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Prevalence and predictors of medication non-adherence among people living with multimorbidity: a systematic review and meta-analysis

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Title: Prevalence and predictors of medication non-adherence among people living with multimorbidity: a systematic review and meta-analysis

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ABSTRACT (299/300)

Objectives This systematic review aimed to describe medication non-adherence in people living with multimorbidity according to the current literature, and synthesise predictors of non-adherence in this population.

Methods A systematic review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses. PubMed, EMBASE, CINAHL and PsycINFO were searched for relevant articles published in English language between 2009 and April 2019. Quantitative studies reporting medication non-adherence and/or predictors of non-adherence among people with two or more chronic conditions were included in the review. A meta-analysis was conducted with a sub-group of studies that used an inclusive definition of multimorbidity to recruit participants, rather than seeking patients with specific conditions. Remaining studies reporting prevalence and predictors of non-adherence were narratively synthesised.

Results The database search produced 10998 records and a further 75 were identified through other sources. Following full-text screening, 177 studies were included in the review. The range of reported non-adherence differed by measurement method, at 76.5% for self-report, 65.6% for pharmacy data and 44.1% for electronic monitoring. A meta-analysis was conducted with 8 studies (n = 8949) that used an inclusive definition of multimorbidity to recruit participants. The pooled prevalence of non-adherence was 42.6% (95% CI = 34.0-51.3%, k = 8, I^2 = 97%, p < 0.01). The overall range of non-adherence was 7.0% to 83.5%. Frequently reported correlates of non-adherence included previous non-adherence and treatment-related beliefs.

Conclusions The review identified a heterogeneous literature in terms of conditions studied, medication adherence measurement and definition of non-adherence. Results suggest that future attempts to improve medication adherence among patients with multimorbidity should determine which patients, conditions and medications require most support. The high levels of medication non-adherence highlight the need for more attention to be paid by healthcare providers to the impact of multimorbidity on chronic disease self-management.

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Strengths and limitations of this study

- This review synthesised evidence from 177 studies conducted in 29 different countries.
- Criteria for inclusion did not specify particular chronic conditions, resulting in the inclusion of a wide variety of disease combinations.
- Studies were not excluded on the basis of design, setting, or method of non-adherence measurement, generating a comprehensive overview of the literature.
- Due to considerable heterogeneity between studies, meta-analysis and synthesis of predictors was restricted to studies that did not seek specific conditions during recruitment of participants.
- Quantitative subgroup analysis was not conducted to determine the influence of method of measurement on non-adherence estimates.

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INTRODUCTION

Multimorbidity is broadly defined as the co-existence of two or more chronic conditions[1] and has been described as the most common chronic condition experienced by adults.[2] In an influential Scottish study of 1.75 million people, the prevalence of multimorbidity was estimated to be 23.3%, increasing to 65% among those aged 65-84 years.[3] People living with multimorbidity attend more medical consultations, experience higher rates of hospital admissions and face increased healthcare costs.[4] Patients are also expected to engage with complex self-management regimens which can include monitoring symptoms, maintaining changes to lifestyle behaviours and adhering to prescribed medications.[5]

Adherence refers to the extent to which a person's behaviours correspond with agreed recommendations from their healthcare provider.[6] Risk of medication non-adherence increases as people are prescribed more medicines[7] – a point of particular relevance considering the close association between multimorbidity and polypharmacy.[8] The extent of non-adherence among people with multimorbidity is poorly understood, attributable to challenges in measuring multiple medication adherence.[9] However, it is recognised that medication non-adherence in multimorbidity can exacerbate the burden experienced by patients through increased morbidity and mortality.[10] Medication adherence can be influenced by many factors, including treatment characteristics, condition-related factors, and illness and medication beliefs.[11] To our knowledge, reported correlates of non-adherence in multimorbidity have not been synthesised.

Much intervention development to improve medication adherence has focused on single-disease populations.[12] Such a focus may produce an artificial underestimation of the complexity of chronic disease self-management, leading to interventions that are not designed with 'the most common chronic condition'[2] in mind. In order to develop interventions to improve medication adherence among people living with multimorbidity, it is first necessary to establish the extent of non-adherence in this population, as well as factors that might influence adherence. This review therefore aims to (1) describe the prevalence of medication non-adherence among people living with multimorbidity, and (2) describe the clinical and psychosocial predictors of non-adherence in this population.

METHODS

A systematic review and meta-analysis was conducted according to the PRISMA statement[13] (supplementary file 1). The review was prospectively registered on PROSPERO (CRD42019133849) and a peer-reviewed protocol was published.[14]

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Inclusion and exclusion criteria

Original peer-reviewed articles and doctoral theses produced in English from 2009-2019 were reviewed. A ten year range was chosen as an agreed definition of multimorbidity[15] and an increase in usage in the literature has arisen over the last decade. Opinion pieces, conference presentations, books, letters, editorials and abstracts were not reviewed. Quantitative studies of people living with two or more chronic conditions reporting medication non-adherence and/or predictors of non-adherence in an extractable format were considered. Studies referring to treatment by medications for distinct conditions were also considered. Presence of multimorbidity was required to be part of study aims and/or inclusion criteria. Studies were not excluded based on design, however interventions that did not report baseline non-adherence were excluded. Studies including any participants under 18 years were excluded.

Search strategy and study selection

An electronic database search of PubMed, EMBASE, CINAHL and PsycINFO from January 2009 to April 2019 was conducted using a predefined search strategy combining terms related to 'multimorbidity' and 'medication adherence' (supplementary file 2). Reference lists of included studies were screened for relevant articles. Records were exported to EndNote X8 to remove duplicates. Two reviewers (LF, RLV) each screened 50% of titles and abstracts in Covidence.[16] Twenty-percent of each reviewer's records were chosen at random for cross-checking by the other reviewer. Studies meeting inclusion criteria were moved forward for full-text screening by two independent reviewers (LF & JL or EG). Where disagreements arose, a third reviewer (GJM or AWM) was consulted.

Data extraction and quality assessment

A data extraction form was developed and used to extract relevant data from each included study (supplementary file 3). Where studies reported medication adherence, this was converted to medication *non*-adherence by subtracting the number of adherent participants from the total sample. One reviewer (LF) extracted data from all included studies, and a second reviewer (JL) cross-checked 20%.

The quality of included studies was assessed using criteria for observational studies[17] previously adapted for medication adherence reviews.[18] Criteria include 11 items related to selection methods, measurement, sources of bias, control for confounding and use of statistics. Each item was answered with 'yes', 'no', 'don't know' or 'not applicable'. All included studies were assessed using these criteria, including intervention studies as only baseline observational data were reviewed. One reviewer

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(LF) assessed the quality of all studies and a second reviewer (EG) cross-checked a random sample of 20%. No studies were excluded based on quality.

Data analysis

Due to between-study heterogeneity relating to study population, design, setting, and measurement and definition of medication non-adherence, a quantitative synthesis involving all studies was not considered appropriate. Meta-analysis was conducted with studies which recruited participants as 'people living with ≥ 2 chronic conditions', and not studies which recruited participants with combinations of specific conditions. This group of studies reflects the most inclusive definition of multimorbidity that can inform future research. Meta-analysis was conducted in R[19] using the meta[20] and metafor[21] packages (supplementary file 4). A random-effects model was used to estimate the pooled prevalence of non-adherence. Between-study heterogeneity was assessed using I^2.[22]

Remaining studies were narratively synthesised to describe medication non-adherence in multimorbidity, informed by Popay and colleagues' guidelines.[23] The study characteristics table was preliminarily analysed to identify and count patterns of results. This was followed by an exploration of relationships within and between results of individual studies. Predictors of non-adherence were also synthesised in this way using studies which recruited participants using an inclusive definition of multimorbidity. Predictors were not extracted from studies which recruited participants with combinations of specific conditions due to potential confounding associated with specific conditions, e.g. a study of people living with bipolar disorder and depression assessing severity of depressive symptoms and psychological disability as predictors of non-adherence.[24]

Patient and public involvement

A panel of people with experience of living with multimorbidity were consulted to advise on appropriate methods for disseminating findings from the review to patients and the public.

RESULTS

Overview of studies

After removing duplicates, 10998 studies were identified from the database search and 75 from reference lists of included studies. Following screening of titles and abstracts, 448 full-texts were reviewed. Of these, 271 were excluded with reasons (figure 1), with 177 studies included in the final synthesis. The study selection process is outlined in the PRISMA flow diagram in figure 1.

Figure 1. PRISMA flow diagram

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Study quality

Results of study quality assessment are summarised in figure 2.

Figure 2. Results of study quality assessment.

Study characteristics

One-hundred and thirty-three studies recruited participants with ≥ 2 specific conditions,[16,24-152] 25 studies recruited participants with one specific condition and ≥ 1 other condition(s),[153-177] and 19 studies recruited participants using an inclusive definition of multimorbidity, e.g. ≥ 2 conditions.[9,178-194] Eighty-two studies measured non-adherence using pharmacy data,[16,25-28,33,34,36,37,43,44,46,48,52,53,55,56,59,62,67,68,73,76,81,83-86,88-90,92-97,100,103,110,111,113,115,118,119,121-123,129-132,134,137,138,141-143,146,151-154,156,158-164,169-171,173,174,176-178,181,195] 79 used self-report,[9,24,28,29,31,32,35,38,39,41,42,47,50,51,57,58,61,63,69-71,74,75,77,78,80,82,87,90,91,96,98,99,101,102,104-109,112,114,116,117,120,124,125,133,139,140,145,148,149,154,155,157,165-168,172,175,179,180,182-188,190-194,196,197] 20 used electronic monitoring,[9,30,32,40,41,45,47,60,65,69,105,126-128,139,140,144,147,150,198] three used pill counts,[49,79,135] two involved reports by a healthcare provider,[64,66] and one did not report the method used.[54] Ten of these studies used more than one method to measure adherence.[32,41,90,96,139,140,149,154,186,187] One hundred and twenty-seven studies reported prevalence of non-adherence,[16,25-29,32,34-46,48,50-53,55,56,58,59,61-68,70,72-79,81-92,95-97,99,101,104-107,109,112-119,121,122,125,129,131-133,135,138,139,141-144,146-150,152-156,158,159,161-167,169,170,172-176,179,182,186,187,192-197,199] 47 reported average non-adherence only,[9,24,29,31,33,47,49,54,57,60,69,71,80,93,94,98,100,102,103,110,111,120,123,124,126-128,130,134,137,140,145,151,157,160,168,171,177,178,180,181,183,185,188,190,191,198] and three did not report non-adherence.[108,184,189] The characteristics of included studies are summarised in supplementary file 5.

Participant characteristics

The number of participants with multimorbidity in each study ranged from 22[102] to 599141.[195] Among 134 studies reporting the mean (SD) age of participants,[9,16,25-32,35,36,38,40-45,47-52,54-61,63-65,67-69,71-74,77-82,84-87,90,92-100,103-106,108-113,115-117,120-124,126,127,129,130,135,138-145,147-150,152,155-158,160,162-164,166,168,170,172,175,177-187,189-195,197-199] the average age ranged from 32.4 (9.6)[82] to 80.8 (9.1) years.[123] Ninety-two (52.9%) study samples were majority female.[24,26-28,31,33,35,36,41-44,48,50,54,56,57,59,60,62,63,67-69,71,73-75,80-85,87-90,94,97,98,106-108,113,115,119,122,123,131-135,137,142,146,147,149,151-153,155-157,159-161,163,164,166,168,169,171,174,176,178,179,181,182,184,186-188,192,193,195,197,198] Three studies did not report sex/gender.[76,92,167]

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Prevalence of medication non-adherence

A meta-analysis was conducted with eight studies that reported prevalence of non-adherence and used an inclusive definition of multimorbidity, i.e. without seeking people with ≥ 1 specific condition(s).[176,179,182,186,187,192-194] The pooled prevalence of non-adherence was 42.6% (95% CI = 34.0-51.3%, $k = 8$, $I^2 = 97\%$, $p < 0.01$) (figure 3). The range of non-adherence among studies in the meta-analysis was 16.4%[182] to 61.4%.[194] The range among all reviewed studies was 7.0%[38] to 83.5%.[175] The remaining studies were narratively synthesised to address the research questions relating to prevalence and predictors of medication non-adherence.

Figure 3. Results of random effects meta-analysis.

Narrative synthesis findings

Medication non-adherence over time

The direction and extent of change in medication non-adherence across time-points was not consistent between studies. Nine studies reported prevalence of non-adherence over time.[25,34,43,52,74,95,125,174,193] Of these, four measured change in non-adherence following a recent diagnosis or treatment for an additional condition,[34,43,74,125] four measured change following medication initiation,[25,52,95,174] and one reported change not related to any specific patient or treatment characteristic.[193] Three studies reported decreased non-adherence,[34,74,125] with the greatest decrease being 9.6% after 3 months.[74] Four studies reported increased non-adherence,[25,95,174,193] with the greatest increase being 41.2% after 12 months.[25] One study reported variation in how non-adherence changed over time depending on the class of medication,[52] while another reported a 13.9% increase in non-adherence after two years, followed by a 14.9% decrease one year later.[43]

Assessment of multiple medications

In studies reporting prevalence of non-adherence for more than one condition, the extent of non-adherence for one condition did not always match the extent of non-adherence for another condition. Twenty-four studies reported non-adherence prevalence for more than one condition, reporting estimates separately for each condition.[27,35,41,50,55,56,59,63,64,67,73,82,86,101,104,105,107,138,142,147,163-165,196] Differences in non-adherence between conditions ranged from 0.8%,[107] to 33.6%.[26] Four studies reported a difference of ≤ 5%,[41,63,107,164] five reported a difference of 5–9.9%,[73,86,101,104,196] 11 reported a difference of 10–19.9%,[35,50,55,59,67,105,138,147,163,165,196] and six reported a difference of ≥ 20%.[26,56,64,82,142,163] Seventeen studies reported a single prevalence estimate of non-adherence accounting for multiple conditions,[28,37,51,66,72,76,81,113,135,155,179,182,186,187,192-194] ranging from 16.4%[182] to

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71.6%.[28] Eighty-six studies reported prevalence of non-adherence for one condition only, preventing comparisons of non-adherence between conditions.[16,27,29,32,34,36,38-40,42-46,48,53,58,61,62,65,68,153-156,158,159,161,162 70,74,75,77-79,81,83-85,87-92,95-97,99,106,109,112,114-119,121,122,125,129,131-133,139,141,143,144,146,148-150,152,166,167,169,170,172-176,195,197,199]

Physical and mental comorbidities

Non-adherence for mental health conditions was neither consistently higher nor consistently lower than non-adherence for physical health conditions. Thirty-four studies reported prevalence of non-adherence in people living with a physical condition and depression and/or anxiety.[16,26,27,36,38-42,46,48,61,62,67,78,81,84,88-91,97,122,132,139,142,144,146,149,150,155,175,197] Four reported non-adherence prevalence for both the physical and mental health condition, with non-adherence higher for depression in three studies,[41,67,142] and higher for the physical condition in one study.[26] Sixteen studies reported prevalence of non-adherence in people living with a psychiatric condition and a physical condition.[29,32,45,56,79,83,86,87,96,101,104,105,115,159,163,164] Seven reported non-adherence prevalence for both conditions, with non-adherence higher for the psychiatric condition in three studies,[104,105,163] and higher for the physical condition in four studies.[56,87,101,164]

Characteristics of multiple medication regimens

Two studies reported prevalence of non-adherence in terms of specific characteristics of multiple medication regimens.[25,72] Starting medications for two conditions simultaneously resulted in lower non-adherence versus not starting simultaneously (24.8% vs 55.0%), but the magnitude of this difference reduced over time (66.0% vs. 69.0% at 12 months).[25] Hussein and colleagues[72] assessed the difference between a single-pill vs. two-pill antihypertensive, lipid-lowering regimen. Among patients naïve to both drugs, the single pill regimen produced a 0.7% lower prevalence of non-adherence. Among patients experienced with both medications, the single pill regimen resulted in a 1.7% lower prevalence of non-adherence.

Method of non-adherence measurement

Different methods of measuring non-adherence produced different ranges of non-adherence estimates, with self-report measures producing the widest range, and observed adherence producing the narrowest range of non-adherence. Among studies reporting prevalence of non-adherence, sixty-three studies used pharmacy data,[16,26-28,34,36,37,43,44,46,48,52,53,55,56,59,62,67-69,72,73,76,81,83-86,88-90,92,95,97,113,115,118,119,121,122,129,131,132,138,141-143,146,152-154,156,158,159,161-164,169,170,173,174,176,195] with non-adherence prevalence ranging from 14.4%[169] to 80.0%;[27] 50 studies used self-

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report,[30,32,35,38,39,41,42,50,51,58,61,66,70,74,75,77,78,82,87,90,91,96,99,101,104,106,107,109,114,116,117,125,133,139,148,149,154,155,165-167,172,175,179,182,192-194,196,197,199] with non-adherence prevalence ranging from 7.0%[38] to 83.5%;[175] 10 studies used MEMS,[32,40,41,45,65,105,139,144,147,150] with non-adherence prevalence ranging from 20.0%[147] to 64.1%.[105] Two studies reported prevalence of non-adherence assessed by a healthcare provider[66] or a mix of healthcare providers, patients and relatives,[64] with non-adherence ranging from 8.6%,[66] to 52.5%.[64] Two studies used observed medication adherence to identify discrepancies between medications taken and medications prescribed,[186,187] with non-adherence prevalence ranging from 37.5%[187] to 44.3%.[186] Two studies used pill counts,[79,135] with non-adherence ranging from 46.3%[79] to 70.0%.[135]

Nine studies reporting prevalence of non-adherence used more than one measure.[32,41,90,96,139,149,154,186,187] Two studies reported lower non-adherence using self-report compared to MEMS,[41,90] one study reported higher non-adherence using self-report compared to MEMS,[139] one study reported a 1% difference using self-report compared to MEMS,[32] and one study reported no difference between self-report and pharmacy data.[154] One study using multiple self-report measures reported a range of 6.4% between the highest and lowest estimate of non-adherence.[149]

Stack and colleagues used self-report to measure both intentional non-adherence (e.g. deliberately choosing not to take medicines) and unintentional non-adherence (e.g. forgetting to take medications) to three medications.[196] For each medication, unintentional non-adherence was more prevalent (62.9%, 47.1% and 46.9%) than intentional non-adherence (13.1%, 19.2% and 18.1%). Bauer and colleagues used pharmacy data to compare primary non-adherence (medication never dispensed) (3.8%), early non-persistence (dispensed once and never refilled) (32.9%) and second-stage non-persistence at 12 months (6.3%).[36]

Studies reporting average non-adherence only

Forty-seven studies reported an average non-adherence estimate only, [9,24,29,31,33,47,49,54,57,60,69,71,80,93,94,98,100,102,103,110,111,120,123,124,126-128,130,134,137,140,145,151,157,160,168,171,177,178,180,181,183,185,188,190,191,198] without reporting a prevalence estimate for any condition studied. Of these, twenty-three studies measured non-adherence using self-report,[9,24,29,31,47,57,69,71,80,98,102,120,124,140,145,157,168,180,183,185,188,190,191,200] seventeen used pharmacy data,[33,93,94,100,103,110,111,123,130,134,137,151,160,171,177,178,181] eight used MEMS,[29,47,60,69,126-128,140,198] one used unannounced pill counts,[49] and one did not report the method used.[54]

Predictors of medication (non-)adherence

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Eleven studies using an inclusive definition of multimorbidity reported predictors of adherence[178-184,189-191,193] (table 1). Nine used self-report to measure adherence,[179,180,182-184,189-191,193] and two used pharmacy data.[178,181] Five studies assessed demographic variables as predictors,[179,184,189,190,193] including age, sex, marital status, education and deprivation. No demographic variables significantly predicted non-adherence.

One study assessed health status, which significantly predicted adherence i.e. poorer health status was associated with stronger motivation to take medications.[184] Other factors assessed included number of prescribed medications,[179,180,189,190,193] number of chronic conditions,[180,181,184,189,190] physical fitness, [193] and functional health status,[189] as well as presence of other disorders,[179] a mental health condition,[181] anxiety/depression,[184] and diabetes.[179] Schüz and colleagues[189] found lower intentional non-adherence among those prescribed more medications and higher intentional non-adherence among those with more conditions. Conversely, Domino and colleagues[181] found lower non-adherence among those with more comorbidities. Depression was associated with lower adherence, while schizophrenia was linked to both higher and lower adherence, depending on other physical conditions present.[181]

A number of studies explored patients' beliefs about their conditions and medications as predictors of adherence. Two studies using the same sample assessed the strength of necessity and concern beliefs, reporting less non-adherence among those with stronger beliefs about the necessity of medications,[189,190] less non-adherence among those with greater concerns about medications,[190] more intentional non-adherence among those with increasing beliefs that medications are harmful,[189] and more unintentional non-adherence among those with increasing beliefs that medications are overused.[189] Another study assessed changes in beliefs over time.[190] Changes in necessity beliefs predicted intentional non-adherence, with increasing beliefs about necessity predicting improved adherence.[190] Changes in beliefs about the overuse of medications predicted unintentional non-adherence, with stronger beliefs associated with higher unintentional non-adherence.[189,190] Two studies assessed beliefs about consequences and beliefs about treatment control,[184,191] with Schüz and colleagues also assessing beliefs about timeline.[191] Kenning and colleagues did not report significant relationships between beliefs and adherence.[184] Schüz and colleagues found higher adherence was predicted by stronger control beliefs, stronger beliefs about negative consequences and beliefs about a shorter illness timeline.[191] One study assessed beliefs about multimorbidity itself, but did not report a significant relationship with adherence.[184]

Three studies found previous unintentional non-adherence significantly predicted later unintentional non-adherence,[189,190,193] while Schüz and colleagues also reported previous intentional

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non-adherence predicted later intentional non-adherence.[189,190] Previous medication motivation predicted adherence, with patients with higher motivation at baseline reporting higher adherence at follow-up.[184] Other factors assessed included attitudes towards medications,[193] medication knowledge,[184] and medication adherence self-efficacy,[180] none of which predicted non-adherence. One study assessed general received social support and medication specific-support,[193] with higher medication-specific social support associated with lower adherence; this relationship was moderated by social conflict. General received social support did not predict adherence. Other factors assessed included chronic disease self-efficacy,[180] medication assistance,[180] health literacy, and functional health literacy,[180] none of which predicted non-adherence.

One study assessed the relationship between quality of life and adherence, with poorer adherence associated with lower physical (r = -.44) and environmental (r = -.57) quality of life.[180] Another study assessed the association between prospective and retrospective memory and adherence.[183] Lower prospective memory performance was related to more omission errors of adherence (r = -.29), while lower retrospective memory performance was related to more commission errors (r = -.24).[183] Other related factors included medical home enrolment, which was positively associated with adherence,[178] and health provider relational quality, which was also positively associated with adherence (r = 0.22).[182] Hassles with healthcare significantly predicted adherence, with fewer hassles during healthcare system encounters predicting greater motivation to adhere.[184] Self-help behaviours did not predict adherence,[184] better self-monitoring and insight predicted higher motivation to adhere to medications,[184] and a more impulsive/careless problem-solving style was associated with lower adherence (r = -.49).[180]

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Study (country)	Population	Predictors assessed	Dependent variable	Significant predictors of and/or associations with medication adherence
Beadles 2015 (USA)[178]	≥ 2 of the following: major depressive disorder, schizophrenia, diabetes mellitus, hypertension, hyperlipidaemia, seizure disorder, asthma, chronic obstructive pulmonary disease	• Medical home enrolment	Proportion Days Covered (PDC) ≥ 0.8	• Medical home enrolment was positively associated with PDC ($p < .001$) across all cohorts (see original paper for breakdown by cohort)
Cicolini 2016 (Italy)[179]	≥ 2 of the following: cardiovascular, neurological, respiratory, musculoskeletal, oncologic diseases or diabetes mellitus	• Age • Sex • Education • Marital status • Number of prescribed drugs • Presence of diabetes • Presence of other disorders (cancer, musculoskeletal, respiratory, neurological, other) • Beliefs about medications (necessity, concerns) • Beliefs about medications (ambivalent, accepting)	8-item Moriskey Medication Adherence Scale (MMAS-8); "medium adherence" vs "low adherence"	• Necessity beliefs: OR (95% CI) = 1.61 (1.21-2.14), $p < .001$ • Concern beliefs: OR (95% CI) = 2.02 (1.64-2.49), $p < .001$ • Ambivalent vs accepting: OR (95% CI) = 0.24 (0.16-0.37), $p < .001$
Daniels 2016 (USA)[180]	≥ 2 chronic conditions	• Health literacy • Medication assistance • Medication adherence self-efficacy • Health-related problem solving • Functional health literacy • Chronic disease self-efficacy • Quality of life • Number of chronic conditions • Number of medications	MMAS-8; higher score = lower adherence	• Health-related problem solving (total score): $r = -.489$, $p = 0.024$) • Health-related problem solving (impulsive subscale): $r = -.521$, $p = 0.0015$ • Quality of life (physical subscale): $r = -.44$, $p < .05$ • Quality of life (environment subscale): $r = -.566$, $p = 0.009$

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Domino 2014 (USA)[181]	≥ 2 of the following: asthma, COPD, diabetes, hypertension, hyperlipidaemia, seizure disorder, depression, schizophrenia	Adherence in medical index condition: • Number of comorbid medical conditions • Mental health conditions Adherence in mental health condition: • Number of comorbidities	PDC (results show percentage point change (standard error) in adherence associated with clinical diagnosis stated)	Persons with diabetes: • Depression: -0.0427 (0.0030), $p < .01$ • Schizophrenia: 0.0458 (0.0061), $p < .01$ • 2 medical comorbidities: 0.0201 (0.0037), $p < .01$ • ≥ 3 medical comorbidities: 0.0678 (0.0037), $p < .01$ Persons with hyperlipidaemia: • Depression: -0.0200 (0.0025), $p < .01$ • Schizophrenia: 0.0359 (0.0050), $p < .01$ • 2 medical comorbidities: 0.0447 (0.0037), $p < .01$ • ≥ 3 medical comorbidities: 0.0979 (0.0037), $p < .01$ Persons with hypertension: • Depression: -0.0276 (0.0021), $p < .01$ • Schizophrenia: -0.0105 (0.0041), $p < .05$ • 2 medical comorbidities: 0.0365 (0.0023), $p < .01$ • ≥ 3 medical comorbidities: 0.1039 (0.0024), $p < .01$ Persons with depression: • 1 medical comorbidity: 0.0193 (0.0029), $p < .01$ • 2 medical comorbidities: 0.0462 (0.0035), $p < 0.01$ • ≥ 3 medical comorbidities: 0.0894 (0.0037), $p < .01$ Persons with schizophrenia: • 1 medical comorbidity: 0.0362 (0.0067), $p < .01$ • 2 medical comorbidities: 0.0752 (0.0078), $p < 0.01$ • ≥ 3 medical comorbidities: 0.1082 (0.0079), $p < .01$
Eton 2017 (USA)[182]	≥ 2 chronic conditions requiring self-management	• Health provider relational quality (HPRQ)	Single-item self-report; “always” vs “usually/sometimes” taking medications	• Health provider relational quality: $t(294) = 3.79$, $p < .001$ (converted to $r = 0.22$, $p < .001$ for consistency with other studies)

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Ihle 2017 (Switzerland)[183]	≥ 2 chronic conditions	• Prospective memory • Retrospective memory	Omission errors of adherence = 1-item self-report Commission errors of adherence = 1-item self-report	• Omission errors of adherence: prospective memory performance: r = -.29, p < .05 • Commission errors of adherence: retrospective memory performance: r = -.24, p < .05
Kenning 2015 (UK)[184]	Any combination of the following: chronic obstructive pulmonary disease, coronary heart disease, diabetes, osteoarthritis and depression.	• Gender • Age • Deprivation (IMD) • Number of conditions • Health status • Anxiety/depression (HADS) • Illness perceptions – consequences (IPQ) • Illness perceptions – treatment control (IPQ) • Self-help behaviours • Self-monitoring and insight (heiQ) • Baseline Moriskey medication motivation • Baseline Moriskey medication knowledge • Hassles with healthcare • Patient experience of multimorbidity (MULTIPleS)	Modified Moriskey Scale – Medication Motivation at follow-up	• Health status (B = -0.119, SE = 0.051; β= -0.132, p < .05) • Self-monitoring and insight (B = 0.093, SE = 0.045; β = 0.100, p < .05) • Moriskey medication motivation at baseline (B = 0.522, SE = 0.044; β = 0.552, p < .01). • Hassles with health care (B = -0.107, SE = 0.050; β = -0.108, p < .05)
Schüz 2011 (Germany)[190]	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	• Unintentional non-adherence Time 1 (T1) • Intentional non-adherence T1 • Sex • Age • Education • Number of illnesses • Number of medicines • Change in specific necessity (T1 to T2) • Change in specific concerns (T1 to T2) • Change in general harm (T1 to T2) • Change in general overuse (T1 to T2)	Two items from Reported Adherence to Medication Scale; Higher score = higher non-adherence.	Intentional non-adherence as dependent variable: • Intentional non-adherence time 1: B = .58, SE = .09, p < .01 • Change in specific necessity beliefs: B = −.19, SE = .06, p < .01 Unintentional non-adherence as dependent variable: • Unintentional non-adherence time 1: B = .70, SE = .09, p < .01 • Change in general overuse beliefs: B step 3 = .26, SE = .10, p < .01
Schüz 2011 (Germany)[189]	≥ 2 chronic conditions from Charlson Comorbidity Index or	• Unintentional non-adherence Time 1 • Intentional non-adherence Time 1 • Sex	Two items from Reported Adherence to Medication Scale;	Unintentional non-adherence as dependent variable: • Unintentional non-adherence time 1: β step 3 = .29, p < 0.01

	Functional Comorbidity Index	• Age • Education • Functional health status • Number of medicines • Number of illnesses • General harm beliefs • General overuse beliefs • Specific necessity beliefs • Specific concern beliefs	Higher score = higher non-adherence	• General overuse beliefs: β step 3 = .17, $p < .01$ Intentional non-adherence as dependent variable: • Intentional non-adherence time 1: β = .26, $p < .01$ • Number of medicines: β = -.15, $p < .05$ • Number of illnesses: β = .16, $p < .05$ • General harm beliefs: β = .15, $p < .05$ • Specific necessity beliefs: β = -.15, $p < .05$
Schüz 2014 (Germany)[191]	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	• Illness perceptions (illness consequences, timeline, control)	One item from Reported Adherence to Medication Scale. Higher score = higher adherence.	Main effects and interaction model: • Timeline of illness 2: β = -.18, $p < .05$ • Control of illness 2: β = .21, $p < .05$ Peak model: • No significant predictors Combined model: • Higher control: β = .20, $p < .01$ • More negative consequences: β = .16, $p < .05$
Warner 2013 (Germany)[193]	≥ 2 chronic physical conditions from Charlson Comorbidity Index or Functional Comorbidity Index	• Sex • Age • Education • Number of medicines • Physical fitness • Attitudes towards medication • General received support • Social conflict • Medication specific social support Time 1 • Medication specific social support Time 2 • Medication adherence Time 1 • Medication adherence Time 2	One item from Reported Adherence to Medication Scale assessing unintentional non-adherence. Higher score = higher non-adherence	Step 1: OR (95% CI) • Adherence Time 1 = 5.99 (3.18-11.29), $p < .001$ Step 2: OR (95% CI) • Adherence Time 1 = 5.37 (2.82-10.24), $p < .001$ • Medication specific social support = 0.69 (0.49-0.98), $p < .05$. Step 3: OR (95% CI) • Adherence Time 1 = 5.62 (2.91-10.85), $p < .001$ • Interaction medication support x social conflict = 0.42 (0.17-0.99), $p < .05$

Table 1 Predictors of medication (non-)adherence

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DISCUSSION

Our systematic review identified a heterogeneous literature reporting medication non-adherence in people living with multimorbidity. Variation in study design, population, and measurement and definitions of both multimorbidity and non-adherence was observed between studies. From eight studies considered suitable for meta-analysis, 43% (95% CI 34%-51%) of patients with multimorbidity were non-adherent to medications. Among studies using an inclusive definition of multimorbidity (k=11), correlates of non-adherence included treatment related beliefs and previous non-adherence.

In studies reporting non-adherence prevalence for more than one condition, results highlighted disparities in non-adherence between medications. It was previously reported that patients with multimorbidity prioritise certain conditions over others,[201,202] with our results suggesting this prioritisation is reflected in self-management behaviours. Other factors known to predict adherence in general may also explain disparities within an individual patient, including differences in regimen complexity and differences in the expected efficacy of medications.[203] In any case, Inauen and colleagues[9] have indicated that adherence within an individual differs considerably between medications and emphasise the need to assess adherence for multiple medications in order to focus interventions accordingly. Despite such calls, a recent systematic review suggested there are no standard criteria for measuring multiple medication adherence, as well as no agreement on how existing measures developed for measuring adherence to monotherapy can be adapted to multiple medications.[204] Among reviewed studies, seventeen reported a single prevalence estimate that required patients to consider non-adherence for all conditions together, potentially resulting in a concealment of those conditions for which non-adherence is highest. This should be considered when interpreting the meta-analysis, as a single estimate of non-adherence across all medications was reported by studies included in the quantitative synthesis.

Among patients with a physical condition and depression, non-adherence was higher for antidepressants in three of four studies. This somewhat supports previous findings that non-adherence to antidepressants is higher than to medications for physical conditions,[205] and extends this to instances where both occur in one individual, however further research is necessary to draw stronger conclusions. Among patients with a psychiatric and physical condition, prevalence of non-adherence to psychiatric medications was higher than non-adherence to medications for physical conditions in three of seven studies. Previous literature has reported that adherence to psychiatric medications is lower than adherence to medications for physical conditions,[205] while others have found antipsychotic adherence to be higher.[206] Such discrepancies further highlight the heterogeneity that is inherent in

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multimorbidity, as well as the importance of considering between-condition medication adherence in individual patients. Moreover, it has previously been reported that the condition patients with multimorbidity prioritise most can differ over time as conditions and treatments change.[202] Considering the cyclical nature of some mental health conditions[207] it may be important to consider whether self-management behaviours, such as medication taking, fluctuate over time as a result of changing priorities in response to symptoms.

Among the variables frequently tested as predictors were number of chronic conditions and number of prescribed medications. Surprisingly, the direction of association with non-adherence varied between studies. It has been suggested previously that frequency of medication administration has a greater influence on adherence than number of tablets,[208] which may have particular relevance in multimorbidity considering some patients take medications for different conditions throughout the day while others take all of their medications together.[209] Indeed, it has been reported that particular combinations of conditions are associated with higher dosing frequencies in multimorbidity, with potential implications for treatment burden,[209] and therefore possibly influencing medication adherence more than number of medications alone. Mode of drug delivery may also be of more relevance than number of medications/conditions alone, with different modes of delivery varying in convenience and, therefore, adherence.[210]

Illness and medication beliefs were also among the most frequently explored predictors of adherence. Beliefs, as well as changes in beliefs over time, were reported as significant predictors. Schüz and colleagues'[191] results suggest medication adherence in the context of multimorbidity may be influenced by perceptions of unique conditions as well as the effect of multiple combined illness perceptions. A scale to measure such multimorbidity illness perceptions has been developed,[211] accounting for domains such as treatment burden, emotional impact and prioritisation of conditions. However when tested within a study reviewed here, this measure did not predict adherence.[184] Importantly, authors note the measure was in early stages of development at the time of administration.[184] Nonetheless, illness perceptions – and how they present in the context of multimorbidity – appear to be of relevance to both intentional and unintentional non-adherence in the context of multimorbidity.[189-191]

Pharmacy data was the most frequently used method to measure non-adherence, followed by self-report and electronic monitoring. Variation in measurement methods contributed to the heterogeneity that prevented a larger meta-analysis; a limiting factor that has previously been identified among systematic reviews of non-adherence to multiple medications.[212] Keeping in mind variance due

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to specific combinations of conditions in individual studies, the range of reported non-adherence differed by measurement method, at 76.5% for self-report, 65.6% for pharmacy data and 44.1% for MEMS. Results from ten studies using multiple methods to report prevalence of non-adherence emphasised this variability further, extending to multimorbidity the recommendation to use subjective and objective measures of non-adherence in combination to allow the strengths of each to compensate for the limitations of others.[213]

The methods used to conduct this systematic review are not without limitation. Firstly, only 20% of titles and abstracts were screened by two reviewers and authors were not contacted for full-text articles or data that was not accessible, both of which increased the risk of excluding relevant studies. Secondly, criteria for inclusion in the review were limited to English language studies produced between 2009 and 2019. While this date range may exclude relevant articles, the review aimed to synthesise evidence produced since an agreed definition of multimorbidity[15] and an increase in usage in the literature has arisen over the last decade, as well as a call for the careful design of adherence research in multimorbidity that uses reliable and consistent adherence measurement and details adherence for multiple conditions.[12] Non-English language articles were not reviewed for feasibility reasons. Thirdly, studies of patients with depression/anxiety were included if the condition was diagnosed, as well as if identified through self-report. As a result, there is a possibility that not all participants have two or more clinical diagnoses. This decision was made to account for the overrepresentation and underdiagnosis of depression among people with chronic conditions,[214] and has been made previously when reviewing the multimorbidity literature.[200] Finally, while the study adopted 'two or more chronic conditions' to define multimorbidity, studies were also included if participants were prescribed medications that could be inferred as treatment for a chronic condition. While this extended the definition of multimorbidity beyond clinical diagnoses, to exclude these studies would have underrepresented the burden that is characteristic of multimorbidity.

CONCLUSION

This review summarises the uneven landscape of non-adherence in multimorbidity, as well as factors that should be considered for the development of interventions to improve adherence for 'the most common chronic condition'. While the review did not identify what specific combinations of conditions produce the highest levels of non-adherence, results do suggest that non-adherence in multimorbidity varies both between and within studies. Although a move towards a multi-disease focus is welcomed, attempts by researchers and clinicians to improve medication adherence should first

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determine which patients, conditions and medications require most support. The use of measures designed to assess multiple medication adherence will be central to achieving this aim.

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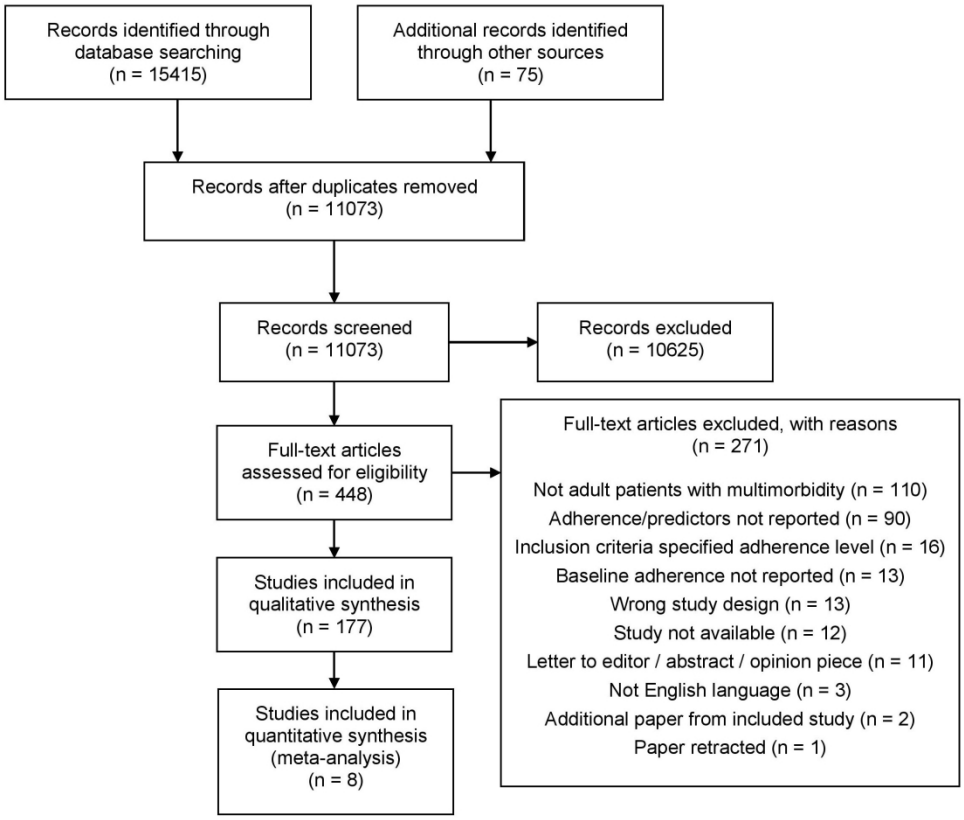


Figure 1. PRISMA flow diagram

633x541mm (96 x 96 DPI)

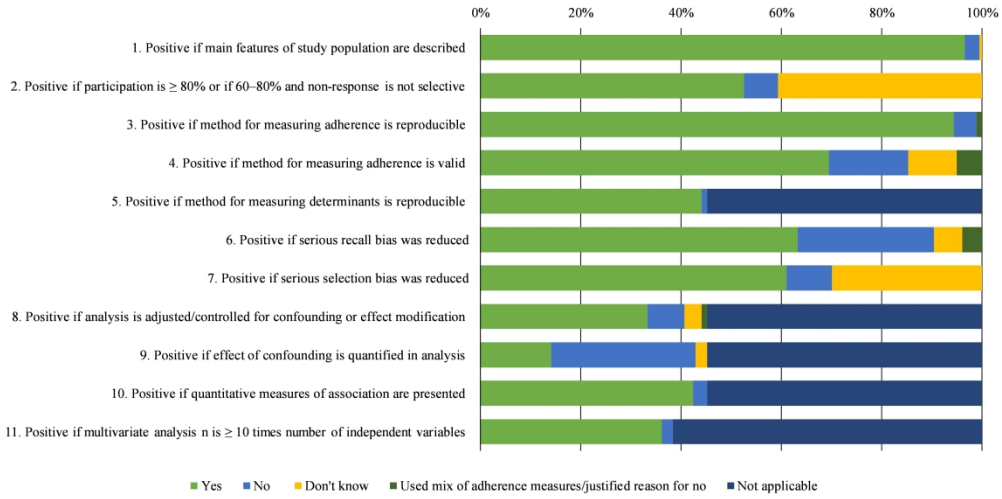


Figure 2. Results of study quality assessment

1730x905mm (72 x 72 DPI)

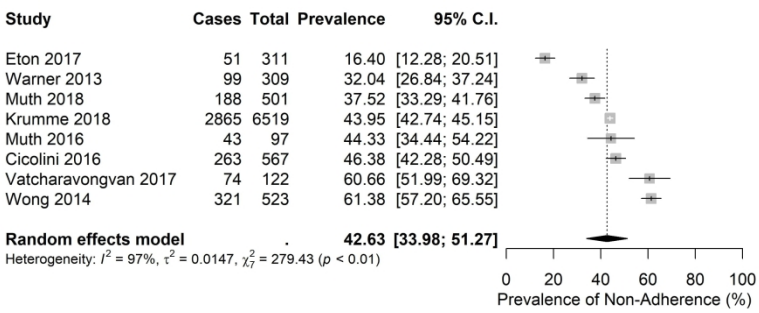


Figure 3. Results of random effects meta-analysis

1411x705mm (72 x 72 DPI)

PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	(i)
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	(ii)
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	1
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	1
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	1
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	2
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	2
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	2+SM
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	2+3
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	2
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	2
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	2+3
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	2
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	3

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	n/r
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	n/r
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	3+4
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	4+SM
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	4
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	5
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	5
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	n/r
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	n/r
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	14-16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	16
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	14-16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	17

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

SM = supplementary material

n/r = not reported

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Search Strategies

PsycINFO:

1. multimorbid*.ab,ti
2. multi-morbid*.ab,ti
3. multidisease*.ab,ti
4. multi-disease*.ab,ti
5. multiple adj2 (ill* or disease* or condition* or syndrom* or disorder*).ab,ti
6. comorbidity/
7. comorbid*.ab,ti
8. co-morbid*.ab,ti
9. chronic* adj2 (disease* or ill* or care or condition* or disorder* or health* or medication* or syndrom* or symptom*).ab,ti
10. polypharma*.ab,ti
11. or/1-10
12. treatment compliance/
13. (medication adherence).ab,ti
14. adheren*.ab,ti
15. nonadheren*.ab,ti
16. non-adheren*.ab,ti
17. (non adheren*).ab,ti
18. complian*.ab,ti
19. noncomplian*.ab,ti
20. non-complian*.ab,ti
21. (non complian*).ab,ti
22. or/12-21
23. S11 AND S22
24. S23 (2009-present)

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CINAHL:

(MH "comorbidity")
OR
TI (multimorbid*)
OR
TI (multi-morbid*)
OR
TI (multidisease*)
OR
TI (multi-disease*)
OR
TI (comorbid*)
OR
TI (co-morbid*)
OR
TI (multiple W2 ill*)
OR
TI (multiple W2 disease*)
OR
TI (multiple W2 condition*)
OR
TI (multiple W2 syndrom*)
OR
TI (multiple W2 disorder*)
OR
TI (chronic* W2 disease*)
OR
TI (chronic* W2 ill*)
OR
TI (chronic* W2 care)
OR
TI (chronic* W2 condition*)
OR
TI (chronic* W2 disorder*)
OR
TI (chronic* W2 health*)
OR
TI (chronic* W2 medication*)
OR
TI (chronic* W2 syndrom*)
OR
TI (chronic* W2 symptom*)
OR
TI (polypharma*)
OR
AB (multimorbid*)
OR
AB (multi-morbid*)
OR
AB (multidisease*)
OR
AB (multi-disease*)

For peer review only

OR
AB (comorbid*)
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AB (multiple W2 ill*)
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AB (multiple W2 disease*)
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AB (chronic* W2 syndrom*)
OR
AB (chronic* W2 symptom*)
OR
AB (polypharma*)

AND

(MH "medication compliance")
OR
TI ("medication adherence")
OR
TI (adheren*)
OR
TI (nonadheren*)
OR
TI (non-adheren*)
OR
TI ("non adheren*")
OR
TI (complian*)
OR
TI (noncomplian*)
OR

For peer review only

TI (non-complian*)
OR
TI ("non complian*")
OR
AB ("medication adherence")
OR
AB (adheren*)
OR
AB (nonadheren*)
OR
AB (non-adheren*)
OR
AB ("non adheren*")
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OR
AB (noncomplian*)
OR
AB (non-complian*)
OR
AB ("non complian*")

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EMBASE:

'multiple chronic conditions'/exp
OR
multimorbid*:ab,ti
OR
multi-morbid*:ab,it
OR
multidisease*:ab,ti
OR
multi-disease*:ab,ti
OR
((multiple) NEXT/2 (ill* OR disease* OR condition* OR syndrom* OR disorder*)):ab,ti
OR
comorbidity/exp
OR
comorbid*:ab,ti
OR
co-morbid*:ab,ti
OR
((chronic) NEXT/2 (disease* OR ill* OR care OR condition* OR disorder* OR health* OR medication* OR syndrom* OR symptom*)):ab,ti
OR
polypharma*:ab,ti

AND

medication compliance/exp
OR
‘medication adherence’:ab,ti
OR
adheren*:ab,ti
OR
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OR
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OR
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OR
complian*:ab,ti
OR
noncomplian*:ab,ti
OR
non-complian*:ab,ti
OR
'non complian*':ab,ti

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PubMed:

multimorbidity[MeSH]
OR
multimorbid*[Title/Abstract]
OR
multi-morbid*[Title/Abstract]
OR
multidisease*[Title/Abstract]
OR
multi-disease*[Title/Abstract]
OR
comorbidity[MeSH]
OR
comorbid*[Title/Abstract]
OR
co-morbid*[Title/Abstract]
OR
"chronic medication*"[Title/Abstract]
OR
polypharma*[Title/Abstract]

AND

"medication adherence"[MeSH]
OR
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OR
nonadheren*[Title/Abstract]
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OR
"non adheren*"[Title/Abstract]
OR
complian*[Title/Abstract]
OR
noncomplian*[Title/Abstract]
OR
non-complian*[Title/Abstract]
OR
"non complian*"[Title/Abstract]

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Data Extraction Form

date form completed (dd/mm/yyyy)
ID of person extracting data (initials)
study ID (author and year)
country of publication
citation (APA format)
study aims
study design
study setting
chronic conditions studied
sample size
age of participants (mean (sd))
gender of participants (% female)
definition of multimorbidity (if applicable)
measure of medication non-adherence
definition of medication non-adherence
proportion of non-adherent participants (or mean (sd) if proportion not reported)
predictors of medication non-adherence (if reported)
key conclusions by study authors

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Meta-analysis of Proportions: R Script

```
library(metafor)
library(meta)
#read in the data file
dat=read.csv("data.csv", header=T, sep=",")
#set-up file without transformation
ies=escalc(xi=cases, ni=total, data=dat, measure = "PR")
#meta-analysis with DerSimonian Laird
pes=rma(yi, vi, data=ies, method = "DL")
#view results
print(pes)
#compute and display confidence intervals of tau-squared and I-squared
confint(pes)
#create simple forest plot
pes.summary=metaprop(cases, total, authoryear, data=dat, sm="PRAW",
method.tau="DL", method.ci="NAsm")
forest(pes.summary, xlim=c(0,100),pscale=100, rightcols = FALSE,
leftcols=c("studlab", "event", "n", "effect", "ci"), leftlabs=c("Study", "Cases",
"Total", "Prevalence", "95% CI.I"), xlab="Prevalence of Non-Adherence", smlab="",
weight.study="random", squaresize=0.5, col.sqaure="black", col.squar.lines="black",
col.diamond="black", col.diamond.lines="black", pooled.totals=FALSE,
comb.fixed=FALSE, fs.hetstat=10, print.tau2=TRUE, print.Q=TRUE, print.pval.Q=TRUE,
print.I2=TRUE, digits=2)
#leave one out to identify outliers
L10=leave1out(pes); print(L10)
#visualise change in summary score with forest plot
l1o=leave1out(pes)
yi=l1o$ estimate; vi=l1o$se^2
forest(yi, vi, slab=paste(dat$author, dat$year, sep=","), refline=pes$b, xlab =
"Summary proportions leaving out each study")
#see which studies are influential using leave-one-out diagnostics
inf=influence(pes)
print(inf)
plot(inf)
```

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Study Characteristics Table

Abbreviations:

COPD = chronic obstructive pulmonary disease
HIV/AIDS = human immunodeficiency virus/acquired immunodeficiency disease
TB = tuberculosis
ACE-I = angiotensin-converting-enzyme inhibitors
HAART = highly active anti-retroviral therapy
ART = antiretroviral therapy
MM = multimorbidity
PDC = proportion of days covered
MPR = medication possession ration
CMG = cumulative mean gap interval
MEMS = medication event monitoring system
VAS = visual analogue scale
MMAS = Morisky Medication Adherence Scale [1,2]
MARS = Medication Adherence Rating Scale[3]
CASE Index = Center for Adherence Support Evaluation [4]
HADS = Hospital Anxiety and Depression Scale[5]
BDI = Beck Depression Inventory [6]
PHQ = Patient Health Questionnaire [7]
MINI = Mini International Neuropsychiatric Interview [8]
CES-D = Centres for Epidemiologic Studies Depression Scale[9]
GDS = Geriatric Depression Scale [10]
SD = standard deviation
IQR = inter quartile range
95% CI = 95% confidence interval
RCT = randomised controlled trial

Note 1: where *n* non-adherent was not reported, *n* was calculated from the reported % non-adherent.
Note 2: where *n* female was not reported, *n* was calculated from the reported % female/male.
Note 3: pooled mean (SD) age was calculated where mean (SD) age was reported for two groups.
Note 4: participants with depression had a clinical diagnosis unless a self-report measure is noted.

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Study (country)	Setting	Design	Chronic Conditions	Sample Size	Adherence Measure	Non-adherence definition	N (%) non-adherent/adherence outcome reported
Agarwal 2009 (USA)[11]	Healthcare database	Retrospective cohort	- Treated with antihypertensive - Treated with lipid-lowering agent	15400	MPR (antihypertensive (AH); lipid-lowering (LL) agent)	MPR < 0.8 for one or more medications	2 months, 6 months, 12 months: Cohort 1 (AH before LL): 40.6%, 67.3%, and 68.7% Cohort 2 (LL before AH): 55.0%, 69.2%, and 69.0% Cohort 3 (AH/LL simultaneously): 24.8%, 65.6%%, and 66.0%
Albrecht 2017 (USA)[12]	Medicare database	Retrospective cohort	- COPD - Depression	16075	PDC (COPD maintenance medications; antidepressants)	PDC < 0.8 (also report PDC = 0)	Person months: COPD: 78.8% Antidepressants: 45.2%
Albrecht 2016 (USA)[13]	Medicare database	Retrospective cohort	- COPD - Depression	31033; MM = 9593	PDC (COPD maintenance medications)	PDC < 0.8 (also report other categories)	7674 (80%)
An 2013 (USA)[14]	Administrative claims and electronic medical records	Retrospective observational	- Hypertension - Type 2 diabetes	2334	PDC (diabetes medications and hypertension medications simultaneously)	PDC < 0.8 for both drugs simultaneously	1671 (71.6%)
Applebaum 2009 (USA)[15]	HIV/AIDS primary care	Baseline data from longitudinal study	- HIV - Substance related disorder - Psychiatric disorder	67	Medication Adherence Questionnaire	Adherence < 95%	20 (29.85%)
Applebaum 2009 (USA)[16]	Healthcare clinics	Cross-sectional (data from RCT screening)	- HIV - History of injection drug use/dependence	108	MEMS (HAART); AIDS Clinical Trials Group Adherence Questionnaire (HAART)	Report mean	Mean MEMS = 68.21% Mean self-report = 93.82%
Axon 2016 (USA)[17]	Veterans health administrative databases	Retrospective cohort	- Type 2 diabetes - Depression	740197; MM = 88895	MPR (diabetes medications)	MPR < 0.8	28143 (31.7%)
Baca-Garcia 2009 (USA)[18]	University hospital	Cross-sectional	- Current bipolar major depressive episode - Alcohol use disorder	97; MM = 39; analysed n = 36	Self-report (antidepressants)	Report mean (SD) median global compliance	Mean (SD) = 90.7 (14.0)
Badiee 2012 (USA)[19]	Ongoing studies at university research center	Cross-sectional	- HIV - Bipolar disorder I or II	74; MM = 43	MEMS (protease inhibitor or related) AIDS Clinical Trials Group (ACTG) 4-day Adherence to	MEMS < 90% ACTG < 95%	MEMS: 22 (51%) ACTG: 6 (13%)

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					Antiretrovirals Questionnaire (antiretroviral medications) Visual Analogue Scale (protease inhibitor)	Visual Analogue Scale < 95%	Visual Analogue Scale: 22 (52%)
Baik 2012 (USA)[20]	Medicare database	Retrospective cohort	- Depression - Heart Failure	552956; MM = 22118; analysed n = 13981	MPR (heart failure medications; antidepressants)	Report mean	No coverage group: Antidepressants: Mean MPR = 0.56 Heart failure medications: Mean MPR = 0.89 Generic coverage group: Antidepressants: Mean MPR = 0.60 Heart failure medications: Mean MPR = 0.90
Banegas 2018 (USA)[21]	Healthcare database	Retrospective cohort	- Treated with statin - Newly diagnosed cancer (breast / prostate / colorectal)	10177	PDC (statins)	Partial adherence = PDC 0.20-0.79 Non-adherence = PDC < 0.2	Year 1: PDC < 0.20 = 837 (8.2%) PDC 0.20-0.79 = 2280 (22.4%) Year 2: PDC < 0.20 = 887 (8.7%) PDC 0.20-0.79 = 2155 (21.2%)
Basak 2014 (USA)[22]	Healthcare database	Longitudinal observational	- Type 2 diabetes - Hypertension and/or dyslipidaemia or angina	6043	PDC	Report mean (SD)	Report according to medication combinations per quarter. Range = 24.43 (39.10) for antianginal medications to 86.40 (23.27) for hypertension medications.
Batchelder 2014 (USA)[23]	Community and hospital	Cross-sectional	- HIV - Type 2 diabetes	62	VAS (antiretrovirals; diabetes medications; insulin injections where appropriate)	VAS < 100%	ARV: 32 (51.6%) Antidiabetes: 39 (63%) Of 32 participants reporting <100 % to HIV medication, 97 % (31) reported < 100 % to diabetes medication
Bauer 2013 (USA)[24]	Healthcare database	Observational new prescription cohort follow-up	- Type 2 diabetes - Depression	1366	Pharmacy utilisation data (antidepressants)	Primary non-adherence: no dispensing of index prescription within 60 days of index prescription date. Non-persistence: dispensing of index	Primary non-adherence: 52 (3.8%) Early non-persistent: 432/1314 (32.9%) 2nd stage non-persistent: 83/1314 (6.3%)

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						prescription once, no additional dispensing within period defined by number of days' supply of medication dispensed plus a grace period equal to 100% of prior days' supply.	
Beadles 2015 (USA)[25]	Medicaid database	Retrospective cohort	≥ 2 of the following: major depressive disorder, schizophrenia, diabetes mellitus, hypertension, hyperlipidaemia, seizure disorder, asthma, chronic obstructive pulmonary disease	depression = 9303; hypertension = 12595; diabetes = 6409; hyperlipidaemia = 9263	PDC (target medication)	Report mean (SD)	Mean (SD): Depression = 40% (43%) Hypertension = 50% (43%) Diabetes = 49% (43%) Hyperlipidaemia = 47% (43%)
Benner 2009 (USA)[26]	Healthcare database	Retrospective claims analysis	- Treated with antihypertensives - Treated with lipid-lowering therapy	5759	PDC (for combination therapy, reported according to prescription burden)	PDC < 80%	By number of other medications: 0 = 59.0% 1 = 65.5% 2 = 69.9% 8 = 79.9% 9 = 74.5% 10/+ = 79.9%
Berg 2018 (Denmark)[27]	Hospital	Cross-sectional	- Cardiac Disease - Depression/anxiety (HADS)	14239; MM (anxiety) = 1978; MM (depression) = 982	Single item self-report (cardiac medication)	Forgetting to take prescribed medication more than once a week among patients who took medication prior to hospital admission	Anxiety group = 113 (7%) Depression group = 70 (8%)
Bifftu 2015 (Ethiopia)[28]	Hospital outpatient	Cross-sectional	- Epilepsy - Depression (BDI-II)	405; MM = 183	Single item self-report (antiepileptic medications)	Not defined (reported as number with difficulties of adherence to antiepileptic drugs)	81 (44.3%)

Blashill 2015 (USA)[29]	HIV primary care clinics	Baseline data of RCT screening	- HIV - Depressive symptoms (PHQ-2)	333	MEMS (antiretrovirals)	MEMS < 80%	191 (57.4%)
Bogner 2014 (USA)[30]	Primary care	Longitudinal	- Type 2 diabetes - Depression	180	MEMS (hypoglycaemic agents and antidepressants) Brief Medication Questionnaire (BMQ) (hypoglycaemic agents and antidepressants)	MEMS < 80% BMQ < 80%	Oral hypoglycaemic agent: MEMS = 103 (57.22%) Self-report = 34 (18.89%) Antidepressants: MEMS = 111 (61.67%) Self-report = 47 (26.11%)
Cabrera-Marroquin 2014 (Mexico)[31]	Tertiary care (ambulatory)	Observational cross-sectional	- Rheumatoid arthritis; - Major depressive disorder	78; MM = 24	Concordance with Therapy self-report measure of adherence and persistence (RA medication)	Based on combinations of item scores	10 (41.7%)
Calip 2013 (USA)[32]	Healthcare database	Retrospective cohort	- Breast cancer survivors - Treated with statins	1393	Adherence: MPR (statins) Persistence: discontinuation rate (statins)	MPR < 0.8 Discontinuation rate (DR): gap ≥ 90 days between end of previous days' supply and subsequent dispensing of statin medication	Cancer treatment period: MPR < 0.8 = 666/1384 (48.1%) Mean (SD) DR = 0.5 (0.6) Year +1: MPR < 0.8 = 572/1129 (50.7%) Mean (SD) DR = 1.2 (1.3) Year +2: MPR < 0.8 = 672/1040 (64.6%) Mean (SD) DR = 1.2 (1.2) Year +3: MPR < 0.8 = 477/959 (49.7%) Mean (SD) DR = 1.2 (1.2)
Calip 2015 (USA)[33]	Healthcare database	Retrospective cohort	- Diabetes mellitus - Breast cancer	509; analysed n = 499	MPR (oral diabetes medications)	MPR < 0.8	376 (75.4%)
Calip 2017 (USA)[34]	Healthcare database	Retrospective cohort	- Breast cancer survivors - Treated with at least one of antihypertensives, oral antidiabetes medications or statins	2308	MPR (antihypertensive or statins or oral antidiabetic medications)	MPR < 0.8	Antihypertensives = 666/1779 (37.4%) Oral diabetes medications = 379/499 (76.0%) Statins = 416/1072 (38.8%)
Cantudo-Cuenca 2014 (Spain)[35]	Hospital pharmacy service	Prospective observational	- HIV - Comorbidity(s)	594; MM = 431	MMAS-4	MMAS-4 < 4 Pharmacy data < 90%	129 (29.9%)

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					Pharmacy dispensing records: Number of medications dispensed/number of medications prescribed x 100		(results show pharmacy data; authors reported no difference in adherence between both measures)
Casaletto 2016 (USA)[36]	Not reported	Cross-sectional	- HIV - Bipolar I or II	50	MEMS (bipolar medications)	MEMS < 90%	28 (56.0%)
Caughey 2013 (Australia)[37]	Veterans' Affairs database	Retrospective cohort	- Diabetes - Depression	29710; MM = 7171	Pharmacy dispensing data – treatment discontinuation (antidiabetic medications)	Gap between two dispensings of more than three times the duration	2868 (40%) Median (95% CI) = 1.81 (1.72-1.94)
Cha 2012 (USA)[38]	Not reported	Cross-sectional using baseline data from intervention	- Heart failure - Diabetes mellitus	116; MM = 46	MMAS-8 (heart failure medications) MEMS (ACE inhibitor and diuretic)	Report mean (SD): MMAS-8 range = 1-12; higher score = more adherent	MMAS-8 mean (SD) = 10.6 (1.6) MEMS: correct doses: ACE-I = 88.0 (16.6) Diuretics = 89.0 (17.0) MEMS: correct days: ACE-I = 81.7 (22.8) Diuretics = 81.1 (22.6)
Chou 2017 (USA)[39]	Healthcare database	Retrospective cohort	- Depression - Breast cancer (newly diagnosed)	2284; MM = 1142	PDC (antidepressants)	PDC < 80%	494 (43.3%)
Cicolini 2016 (Italy)[40]	Primary care	Cross-sectional	≥ 2 chronic condition from cardiovascular, neurological, respiratory, musculoskeletal, oncologic diseases or diabetes mellitus	567	MMAS-8 (Italian Version)	MMAS-8 score between 1-5.75 (defines "low adherence")	263 (46.4%)
Crowley 2013 (USA)[41]	Community practice	Cross-sectional (baseline data from RCT)	- Type 2 diabetes - Hypertension	377	MMAS-4 (hypertension medications; diabetes medications)	Score ≤ 2 on any item	Hypertension = 110 (29.2%) Diabetes = 162 (43.0%)
Crowley 2015 (USA)[42]	Tertiary care health system	Cross-sectional (baseline data from RCT)	- Acute myocardial infarction in previous 3 years - Hypertension	405	MMAS-4 (general adherence)	Response of "strongly agree," "agree," "don't know," or "refused" for any statements	173 (42.7%)

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Cruess 2012 (USA)[43]	Physician outpatient clinics	Prospective cohort	- HIV - Treated with psychotropic medication	324; MM = 106	Monthly unannounced pill counts (HIV medication; psychiatric medication)	Report mean (SD)	HIV = 84.09% (19.26%) Psychiatric = 79.86% (20.88%)
Cziraky 2015 (USA)[44]	Healthcare database	Retrospective cohort	- Acute coronary syndrome or unstable angina - Type 2 diabetes	Year 1 = 85097; MM = 22588 Year 3 = 35540; MM = 9160	MPR (cardiovascular medications) Treatment discontinuation (cardiovascular medications)	MPR: not reported Treatment discontinuation: gap in treatment of 45 days or longer in each medication based on days of supply	Oral diabetes medication: Year 1 = 49.3% Year 3 = 46.8% Insulin Year 1 = 77.4% Year 3 = 78.9% Report other medications in paper.
Daniali 2018 (Iran)[45]	Government healthcare centres	Cross-sectional	- Chronic condition - Depression (PHQ-9)	483; MM = 79	4-item self-report	Not reported	40 (50.6%)
Daniels 2016 (USA)[46]	Primary care	Pilot intervention	≥ 2 chronic conditions	20	MMAS-8	Report mean (SD) MMAS-8 score > 2 (defined as "low adherence"); lower score = lower non-adherence	Mean (SD) = 4.65 (2.37)
Davies 2015 (USA)[47]	Tertiary care Veteran's Affairs Medical Centre (patients receiving primary care)	Retrospective chart review	- HIV/AIDS - Diabetes mellitus	56	MPR (ART)	MPR < 95%	34 (60.7%)
Davis 2011 (USA)[48]	Clinical centres	Data from RCT	- Hypertension - Chronic kidney disease	1065	Not reported (antihypertensives)	Report mean (SD)	Lower BP goal group: Achieved goal: 92.6% (7.7%) Above goal: 83.0% (14.7%) Usual BP goal group: At goal: 91.4% (8.8%) Below goal: 90.8% (13.2%) Above goal: 78.5% (19.1%)
DeLeon 2015 (USA)[49]	Primary care	Retrospective cohort	- Diabetes - Hypertension	20085; MM = 3827	MPR (diabetes medications, hypertension medications, lipid controlling medications)	MPR < 80%	Diabetes: 1056/1679 (62.9%) Hypertension: 1084/1923 (56.4%)

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							Lipid controlling: 928/1295 (71.7%)
Desai 2014 (USA)[50]	Medicaid database	Retrospective analysis	- Treatment by antipsychotic - Treatment by antidiabetic medication	3642; MM = 1821	PDC (oral antidiabetics; oral antipsychotics) Persistence (oral antidiabetics)	PDC < 0.8 Persistence: number of days of continuous access to oral antidiabetics, allowing for a 30-day gap period.	Antidiabetics: 1143 (62.8%) Mean (SD) persistence = 180.8 (124.8) Antipsychotics: 689 (37.8%)
Dhamane 2017 (USA)[51]	Healthcare database	Retrospective cohort	- COPD - Treatment by ≥ 1 of antianxiety, anticoagulants, antidepressants, antihyperglycaemics, antihypertensives, beta-blockers, bisphosphonates, calcium channel blockers, diuretics, insulin, NSAIDs, statins	14117	PDC (COPD medications; non-COPD medications)	PDC < 0.8 Report mean PDC for non-COPD medications	COPD medications: 11177 (79.2%) non-COPD medications mean: antianxiety = 0.39 anticoagulants = 0.55 antidepressants = 0.65 antihyperglycaemics = 0.70 antihypertensives = 0.71 beta-blockers = 0.70 bisphosphonates = 0.54 calcium channel blockers = 0.68 diuretics = 0.65 insulin = 0.53 NSAIDs = 0.33 statins = 0.67
Domino 2014 (USA)[52]	Administrative claims databases	Retrospective secondary data analysis	≥ 2 chronic conditions from asthma, COPD, diabetes, hypertension, hyperlipidaemia, seizure disorder, depression, schizophrenia	188531	PDC	Report mean PDC	Diabetes = 0.29 Hyperlipidaemia = 0.29 Hypertension = 0.43 Depression = 0.33 Schizophrenia = 0.46
Druss 2010 (USA)[53]	Community mental health centre	Pilot RCT	- ≥ 1 chronic medical condition - Severe mental illness	80	MMAS-4 (chronic medical condition)	Report mean (SD) score from 0 (no problems with adherence) to 4 (more problems with adherence)	Mean (SD) = 1.5 (1.3)
DuChane 2014 (USA)[54]	Pharmacy claims database	Retrospective cohort	- HIV - Comorbidity (treated with angiotensin-converting enzyme (ACE) or angiotensin	ACE/ARB group = 2968; MM = 1484; Statins	PDC	ART medications: PDC < 95% Other medications: PDC < 80%	HIV + ACE/ARBs = 1118 (37.7%) HIV + statins = 936 (34.1%) Mean (SD) ART = 83.9 (24.3)

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			receptor blocker (ARB) or statin)	group = 2744; MM = 1372			
Eton 2017 (USA)[55]	Primary care	Cross-sectional	≥ 2 chronic conditions requiring self-management	332 (analysed n = 311)	Single-item self-report	Response "usually /sometimes" vs "always" re adherence to physician-recommended medications	Usually/sometimes = 51 (16.4%)
Farley 2012 (USA)[56]	Medicaid database	Retrospective cohort	- Schizophrenia - Pre-existing cardiometabolic condition(s): hypertension, hyperlipidaemia and/or diabetes	1006	PDC (antipsychotic medications; cardiometabolic medications)	PDC < 80% (unweighted mean of 3 cardiometabolic classes) Report mean (SD)	Antipsychotic (acute phase) = 89 (18.8%) Diabetes medications: acute phase = 0.90; continuation phase = 0.63; maintenance phase = 0.59 Antihypertensives: acute phase = 0.86; continuation phase = 0.71; maintenance phase = 0.66 Statins: acute phase = 0.79; continuation phase = 0.65; maintenance phase = 0.57
Feldman 2016 (USA)[57]	Not reported (recruited from outpatients, emergency room, referrals and flyers)	RCT	- Panic disorder - Asthma	48	MARS (inhaled corticosteroids)	Report mean (SD) Range = 1-5; higher score = lower non-adherence. Score 4.5+ = good adherence.	Mean (SD) = 3.8 (0.4)
Ferguson 2017 (Australia)[58]	Cardiology ward at academic medical centre	Prospective cohort	- Chronic heart failure; - Atrial fibrillation	137; analysed n = 83	MMAS-4	≥ 1 responses to non-adherence items	36 (43.4%)
Foguet-Boreu 2017 (Spain)[59]	Primary care	Cross-sectional	- Hypertension - Diabetes type 1 or type 2	27637	MPR (hypoglycaemic medications; antihypertensive medications; both combined)	MPR < 80% Report MPR ≥ 80%	Diabetes = 10898 (39.8%) Hypertension = 4322 (23.8%) Combined diseases (MPR < 80% for at least one condition) = 13593 (49.2%)

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Gathright 2017 (USA)[60]	Hospital	Secondary analysis of data from longitudinal observational study	- Heart Failure - Depressive Symptoms (PHQ-9)	308; MM = 107	MEMS (up to 4 common heart failure medications)	Report mean (SD) percent of days a participant did not follow their prescribed regimen	Mean (SD) = 65.3% (27.5%)
Gaudiano 2017 (USA)[61]	Psychiatric hospital	Cross-sectional (study of baseline data from clinical trial)	- Bipolar I or II or NOS - Drug and/or alcohol use disorder	39	Medication Compliance Questionnaire (mean rating across all psychotropic medications)	Report mean (SD) Range = 0-4, higher score = higher non-adherence.	Mean (SD) = 1.2 (1.2)
Gaynes 2015 (USA)[62]	Infectious disease clinics	Data from RCT	- HIV - Depression (PHQ-9 and major depressive disorder (MINI)	304	Visual Analogue Scale (HIV/AIDS medications)	VAS score ≤ 80%	78 (25.7%) No psychiatric comorbidity = 13/54 (20.1%) ≥ 1 psychiatric comorbidity = 65/250 (26.0%)
Gentil 2015 (Canada)[63]	Community	Data from longitudinal study	- Type 2 diabetes - Depression/anxiety	301; MM = 47	MPR (oral antidiabetic medication)	MPR < 80%	18 (38.3%)
Gibson (USA) 2018[64]	Primary care	Cross-sectional	- Type 2 diabetes - Hypertension	60	Self-report during semi-structured interview	Self-report missing 1 or more days in previous week	Hypertension: 14 (23.3%) Type 2 diabetes: 16 (26.7%)
Gnanasan 2011 (Malaysia)[65]	Hospital outpatients	Feasibility study (action research)	- Diabetes mellitus - TB	35	Medication-related problems identified by clinical pharmacist (TB medications; non-TB medications)	Not reported	TB treatment: 3 (8.6%) Non-TB treatment: 10 (28.6%)
Gonzalez 2011 (USA)[66]	Methadone maintenance treatment	Cross-sectional	- HIV - History of injection drug use	91	MEMS (HAART)	MEMS < 100% doses as prescribed	73 (80.2%)
Grove 2017 (USA)[67]	Medicaid database	Retrospective cohort	- Schizophrenia - ≥ 1 chronic condition from hypertension, hyperlipidaemia, seizure disorder, chronic obstructive pulmonary disease, diabetes, asthma, or major depression	12662	PDC (antipsychotic medications)	Report mean (SD)	Mean (SD) = 45.4% (37.8%)

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Gundling 2013 (Germany)[68]	Hospital	Retrospective analysis	- Diabetes mellitus - Cirrhosis	285; MM = 87; analysed n = 69	Information about compliance based on doctors', patients' and relatives' reports (report general adherence)	"e.g. concerning regular intake of medication and following the doctor's advice"	36 (52.2%)
Gupte-Singh 2015 (USA)[69]	Medicaid database	Retrospective analysis	- Asthma - Depression	3626; MM = 226	PDC (asthma controller; antidepressant medications)	Asthma controller: PDC < 50% Antidepressant: PDC < 80%	Asthma controller medications: 140 (62.0%) Antidepressants: 168 (74.3%)
Halava 2014 (Finland)[70]	Data from public sector employees	Prospective cohort	- Prescribed statins - Cardiovascular comorbidities	9285; MM = 2827	PDC (statins)	PDC < 80%	1155 (40.9%)
Han 2018 (USA)[71]	Medicare database	Retrospective cohort	- Hypertension - Chronic kidney disease	70201	PDC (antihypertensive medications)	PDC < 80%	27468 (39.1%)
Hansen 2009 (USA)[72]	Ambulatory care practice	Secondary analysis of RCT data	- Heart failure - Depression (GDS)	314; MM = 117	MEMS (heart failure medication) MMAS-4 + one-item screening questionnaire (heart failure medication)	Report mean (95% CI) MMAS-4: Range = 0-1; higher score = better adherence	MMAS-4 = 79% (71-87%) MEMS not reported
Hansen 2018 (USA)[73]	Medicaid database	Retrospective cohort	- Schizophrenia - Cardiometabolic condition (hypertension, hyperlipidaemia and/or diabetes)	4223	PDC (antihypertensive medications; statins; oral antidiabetics; antipsychotic medications)	PDC < 80%	Calculated average %: Antipsychotic = 68.3% Diabetes medication = 48.5% Hypertension medication = 58.6% Statins = 44.9%
Hansen 2012 (USA)[74]	Medicaid database	Cross-sectional time-series	- Schizophrenia - Cardiometabolic condition (hypertension, hyperlipidaemia and/or diabetes)	33230	PDC (oral second-generation antipsychotics; oral antidiabetics; statins; antihypertensive medications)	PDC < 0.8	Antipsychotic: 11565/33230 (34.8%) Antihypertensives: 10069/24843 (40.5%) Antidiabetics: 4869/12349 (39.4%) Antihyperlipidemics: 4192/10909 (38.4%)
Hansen 2014 (USA)[75]	Veterans' Administration Medical Centre	Retrospective cohort	- Cardiometabolic condition (hypertension, hyperlipidaemia and/or diabetes)	7933; Diabetes MM = 1417; Hypertension MM =	CMG (diabetes medications; hypertension medications; dyslipidaemia medications)	CMG ≥ 20%	Diabetes (+ ≥ 1 other): 392 (27.7%) Hypertension (+ ≥ 1 other): 766 (23.1%) Dyslipidaemia (+ ≥ 1 other): 948 (31.5%)

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			- ≥ 1 additional chronic condition(s)	3318; Dyslipidaemia MM = 3009			
Hapangama 2013 (Sri Lanka)[76]	Tertiary care psychiatric unit (inpatient and outpatient)	Cross-sectional	- Mental disorder - Substance use disorder	325; MM = 141	Interview schedule administered by senior registrar in psychiatry	Ceased medication for ≥ 2 days for any reason, without medical advice, during treatment for mental illness	91 (64.5%)
Hawke 2013[77]	Academic medical centres primarily	RCT (secondary data analysis)	- Bipolar I or II - Comorbid anxiety disorder	204; MM = 85	Single-item self-report	Report mean (SD) (possible range not reported)	Mean (SD) = 1.36 (0.53)
Hussein 2010 (USA)[78]	Healthcare database	Retrospective cohort	- Hypertension - Using a two-pill amlodipine + atorvastatin; using a two-pill CCB + statin; or using a single pill amlodipine/ atorvastatin	35430	PDC (statins; calcium channel blocker)	PDC < 80%	Calculated average percent: 43.9%
Ihle 2017 (Switzerland)[79]	Primarily hospital	Cross-sectional	≥ 2 chronic conditions	84	Self-report items adapted from Moriskey, Green, Levine measure (specifically, assesses omission errors and commission errors)	Report mean (SD) from 1-5 (higher score = higher non-adherence)	Omission errors: mean (SD) = 1.39 (0.88) Commission errors: mean (SD) = 1.12 (0.57)
Inauen 2017 (Switzerland)[80]	University hospital (outpatient) and other channels, e.g. online, pharmacy	3-wave panel study	≥ 2 chronic conditions and ≥ 2 prescribed medications	84	Non-Adherence Report, Medication Based (NAR-M) Non-Adherence Report, Diary (NARD) MARS MEMS	Report mean (SD) (lower mean = higher non-adherence)	NAR-M: T1 = 92.8 (11.1); T2 = 92.1 (12.4); T3 = 94.5 (10.5) NARM-D (no MEMS group) = 96.9 (7.9) MEMS group = 98.5 (4.1) MARS T1 = 89.4 (18.3) T2 = 90.1 (18.1) T3 = 94.5 (13.3) MEMS T1-T2 = 95.7 (3.8) T2 – T3 = 93.7 (10.4)

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Iyengar 2016 (USA)[81]	Medicare database	Retrospective claims analysis	- Diabetes - Hypertension - High blood cholesterol	40632	PDC (diabetes medications, cholesterol medications, hypertension medications)	PDC < 80%	Antidiabetics = 13898 (34.2%) Antihypertensives = 13462 (33.1%) Antihyperlipidemics = 10584 (26.1%)
James 2018 (USA)[82]	Hospital	Longitudinal	- Type 2 diabetes - Early stage cancer (recently diagnosed)	56	MMAS-8 (diabetes medications)	MMAS: positive response to ≥ 2 questions	Baseline: 25 (44.6%); 3-months: 20 (35%)
Jaworski 2011 (France)[83]	University hospital psychiatric department (outpatient)	Cross-sectional	- Schizophrenia or bipolar disorder - Substance use disorder	80; MM = 40: schizophrenia + SUD= 20; bipolar + SUD = 20	Self-report number of times forgot to take medication in preceding month (psychotropic medications)	Score < 75%	Bipolar + SUD = 15 (75%); Schizophrenia + SUD = 14 (70%)
Jing 2011 (USA)[84]	Medicare and healthcare databases	Retrospective analysis	- Treatment with antihypertensive - Treatment with antidiabetic	30132	MPR	MPR < 77% for ≥ 1 antihypertensives and/or antidiabetic medications for each standalone calendar quarter	10722 (35.6%)
Joseph 2017 (India [85]	Medical college hospital & dialysis centre (inpatient and outpatient)	Prospective observational	- Hypertension - Chronic kidney disease	120	MMAS-8 (antihypertensives)	Not reported	75 (62.5%)
Kacanek 2010 (USA)[86]	Physician network, newspapers, radio advertisement	Secondary analysis of data from longitudinal cohort	- HIV - Depression symptoms (CES-D and Diagnostic Interview Schedule)	225; MM= 51	For each medication, participants reported the number of days over the last seven days they had missed a dose of that antiretroviral medication.	Missing at least 5% of HAART doses in the seven days prior to study	23 (45.1%)
Kalichman 2015 (USA)[87]	AIDS community services and infectious disease clinics	Prospective	- HIV - Mental health condition	123	Unannounced pill-counts (antiretroviral medications; psychiatric medications)	Antiretrovirals: < 85% Psychiatric medications: report mean (SD)	ART: 57 (46.3%) ART mean (SD) = 77.2 (25.3) Psychiatric mean (SD) = 64.6 (36.4)
Kamal 2018 (Switzerland)[8] [8]	University hospital infectious disease service	Prospective observational cross-sectional	- HIV - Comorbidity (≥ 1 concomitant oral medication from any	105	Swiss HIV Cohort Study Adherence Questionnaire (SHCS-AQ) (ART; co-treatments)	Missing ≥ 1 dose of treatment in previous 4 weeks	ART = 18 (17.1%) Co-treatments = 30 (28.6%)

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			therapeutic class for at least 6 months)				
Katerndahl 2012 (USA)[89]	Primary care	Cross-sectional	- Diabetes - Mild or moderate depression (PHQ-9)	106; MM = 37	Medication subscale from Diabetes Self-Care Activities Measure	Report mean (possible range = 0 – 7; lower score = lower compliance	Mean = 4.89
Katon 2009 (USA)[90]	Primary care	Prospective	- Diabetes type 1 or type 2 - Probably major depression (PHQ-9)	4117; MM = 495	CMG (hyperlipidaemia medications; hypertension medications; hyperglycaemia medications)	CMG ≥ 20%	Only reported in patients with poor glycaemic, blood pressure and lipid control: Poor diabetes control = 47/127 (37.0%) Poor hypertension control = 67/232 (28.9%) Poor lipid control = 17/44 (38.6%)
Kebede 2012 (Ethiopia)[91]	Hospital	Cross-sectional	- HIV/AIDS - TB	24	Self-report tool	Missing a dose for ≥ 1 day per month (ART; TB)	ART: 16 (66.7%) TB: 11 (45.8%)
Kenning 2015 (UK)[92]	General practice	Prospective	≥ 2 from chronic obstructive pulmonary disease, coronary heart disease, diabetes, osteoarthritis and depression (self-reported and/or HADS).	410	Modified Moriskey Scale (6-items)	Not reported	Not reported
Kim 2010 (South Korea)[93]	Pharmacy claims database	Retrospective cohort	- Hypertension - Psychiatric disorder	2454844; MM = 158982	Cumulative Medication Adherence (CMA) (antihypertensives)	CMA < 80%	67297 (42.3%)
Kong 2012 (USA)[94]	Medicaid database	Retrospective cohort	- HIV - Depression	7034; MM = 3286; analysed n = 3202	MPR (antiretrovirals)	Adherence < 90%	2386 (74.5%)
Kostev 2018 (Germany)[95]	General practice	Retrospective cohort	- Type 2 diabetes - Depression	12898; MM = 6449	Persistence (oral antihyperglycaemic medications)	At least 3 months without drug	2096 (32.5%)
Kreyenbuhl 2010 (USA)[96]	Veterans' Affairs administrative database	Retrospective cohort	- Type 2 diabetes - Schizophrenia or schizoaffective disorder	22014; MM = 11454	MPR (oral hyperglycaemic medications; antipsychotic medications)	MPR < 0.8	Oral antihyperglycaemics = 4925 (43%)

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							Antipsychotic medications = 4009 (35%)
Kreyenbuhl 2011 (USA)[97]	Mental health clinics (outpatient)	Cross-sectional	- Type 2 diabetes - Serious mental illness	74; MM = 44	Brief Medication Questionnaire (oral hypoglycaemic medications)	Not taking any hypoglycaemic medications as prescribed over the past week	12 (27%)
Krivoy 2015 (Isreal)[98]	Healthcare database	Retrospective analysis	- Ischaemic heart disease - Depression	63437	MPR with addition of physician prescription data from electronic health record (antidepressants)	MPR < 20% = non-adherent MPR 20% to <50% = poorly adherent MPR 50% - <80% = moderate adherence	Non-adherence = 18212 (28.7%) Poor adherence = 10094 (15.9%) Moderate adherence = 11041 (17.4%) Total < 80% = 39347 (62.03%)
Krivoy 2017 (Israel)[99]	Healthcare database	Retrospective analysis	- Depression - Cerebrovascular accident	32361	MPR plus physician prescription data (antidepressants)	MPR < 20 % = non-adherent MPR 20-50% = poorly adherent MPR 50-80% = moderately adherent	< 20% = 8619 (26.6%) 20-50% = 5108 (15.8%) 50-80% = 5656 (17.5%) Total < 80% = 19383 (59.9%)
Krousel-Wood 2010 (USA)[100]	Managed care organisation	Prospective cohort	- Hypertension - Depression (CEDS)	2180; MM = 284	MMAS-8 (antihypertensives); MPR (antihypertensives)	MMAS-8 score < 6 ("low adherence"); MPR < 0.8	MMAS: 67 (23.6%); MPR = 104 (36.5%)
Krumme 2018 (USA)[101]	Pharmacy	Retrospective cohort	≥ 2 chronic conditions, at least one of which is hypertension, hyperlipidaemia or diabetes	22805; MM = 6519	PDC (cardiovascular medication)	PDC < .80	PDC ≤ 70 = 1303 (20.0%) PDC ≤ 85 = 2865 (44.0%)
Kumar 2009 (USA)[102]	Data from longitudinal study	Cross-sectional	- HIV - Depression	1192; MM = 210	4-item self-report for past week (antiretroviral medication)	Report categories of 100%, 85%, 57-85%, <57.	Adherence < 100% = 88 (41.91%) Adherence < 85% = 67 (31.91%)
Langness 2014 (USA)[103]	Outpatient pharmacy records	Retrospective review	- HIV - Treatment with mental health medication and/or blood pressure medication	Blood pressure group = 269; Mental health group = 295	PDC (blood pressure medication; mental health medication)	PDC < 80%	Blood pressure: 92 (34.2%) Mental health: 108 (36.6%)

Lee 2011 (USA)[104]	Medicaid database	Retrospective cohort	- Serious mental illness (schizophrenia or major affective disorder) - HIV	23413; MM = 788	Continuity of medication use measures (ART and psychotropic medication)	Monthly use of any antiretroviral or psychotropic medications (divide total months patients had prescription filled by follow-up months after indexed prescription)	Percent of continuous time on medication: Psychotropic medication: 72.5% ART: 71.6%
Li 2016 (Hong Kong)[105]	General out-patient clinics	Cross-sectional	- Hypertension - Co-occurring conditions	2445; MM = 1159	MMAS-8 (Chinese version) (antihypertensives)	MMAS-8 ≤ 6	549 (47.4%) + 1 comorbidity = 339 (47.7%) + ≥ 2 comorbidities = 210 (46.9%)
Lin 2012 (USA)[106]	Primary care	RCT	- Depression (PHQ-9) - Diabetes or heart disease	214	CMG (diabetes/heart disease)	Report mean (SD)	Mean (SD): Oral hypoglycaemic = 0.83 (0.19) Antihypertensive = 0.85 (0.18) Lipid lowering = 0.83 (0.20) Antidepressant = 0.80 (0.21)
LoRe 2013 (USA)[107]	Veterans Affairs (VA) database	Retrospective cohort	- HIV - Hepatitis C virus	333	Sum of days' supply of medication dispensed over 12-week interval divided by number of days between first final fills of that interval (antiviral medications)	Pharmacy data < 70% (also report additional intervals above 70%)	Interferon: Interval 1 = 28/267 (10.5%) Interval 2 = 38/189 (20.1%) Interval 3 = 22-131 (16.8%) Interval 4 = 28-94 (29.8%) Ribavirin: Interval 1 = 46/267 (17.2%) Interval 2 = 55/185 (29.7%) Interval 3 = 46/127 (36.2% Interval 4 = 49/91 (53.8%)
Long 2014 (USA)[108]	Veterans Affairs medical facilities (primary care)	Cross-sectional	- Type 2 diabetes - Serious mental illness	363	MPR (diabetes medications including insulin) MMAS-4 (diabetes specific)	MPR: report mean (SD) MMAS-4: report as perfect vs non-perfect adherence	MPR mean (SD) = 0.8 (0.3) MMAS-4 = 178 (49%)
Lunghi 2017 (Canada)[109]	Healthcare database	Retrospective cohort	- Type 2 diabetes - Depression	3106	PDC (antidiabetic medication)	PDC < 90%	1614 (52.0%)
Magidson 2012 (USA)[110]	Substance use treatment centre	Cross-sectional (RCT baseline data)	- HIV/AIDS - Substance use	73	ACTG Adherence to Antiretroviral Medication Questionnaire (ART and other medications)	Report mean adherence over previous 4 days	ART: Yesterday = 100%; 2 days ago = 94.6%; 3 days ago = 97.8% 4 days ago = 97.8%

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							Other medications: Yesterday = 97.3% 2 days ago = 94.9% 3 days ago = 96.5% 4 days ago = 97.4%
Manuel 2011 (USA)[111]	Outpatient mental health centres	Data from RCT	- Psychotic disorder - Active substance use or dependence	198	Self-report missed medication days in past 6 months (psychiatric medications)	≥ 20% missed medication days	30 (15.2%)
Mathew 2013 (India)[112]	Hospital	Non-randomised non-controlled intervention	- Type 2 diabetes - Comorbidity(s) (receiving ≥ 5 drugs)	240	14-item self-report questionnaire (diabetes medication)	Patients were categorised into adherent and non-adherent (criteria for sorting not reported)	116 (48.3%)
May 2010 (USA)[113]	Hospital	Longitudinal observational	- Coronary artery disease - Depression	585; MM = 73	MPR across multiple refills (continuous measure of adherence (CMA) for lipid-lowering medications)	Report mean CMA	Mean adherence: 6 months = 90.9% 1 year = 89.3% 2 years = 85.7%
McCusker 2016 (Canada)[114]	Primary care	Data from RCT	- ≥ 1 chronic physical condition(s) or chronic pain ≥ 6 months - ≥ mild depressive symptoms (PHQ-9)	215; analysed n = 202	MMAS-4 (any medication)	Report mean (SD) MMAS-4 higher score = higher non-adherence (range = 1-4).	Mean (SD) = 0.7 (0.9)
Mellins 2009 (USA)[115]	Community	Cross-sectional (baseline data from cohort study	- HIV - Psychiatric disorder - Substance use disorder	1138 (n prescribed ARV = 542; n prescribed antipsychotics = 501)	Antiretrovirals: Modified version of AACTG Adherence to Antiretroviral Medication Questionnaire Psychiatric medications: 1-item self-report	Self-report < 100% adherence in past 3 days	ARV = 244 (45.0%) Psychiatric medication = 190 (38%)
Miketic 2012 (USA)[116]	Hospital	Prospective descriptive pilot study	- Coronary artery disease - Diabetes mellitus	51; MM = 22	MMAS-4	Report mean (SD)	0.86 (0.94)
Mira 2014 (Spain)[117]	Health centres	RCT	- Multimorbid taking multiple medications	99	MMAS-4	Report mean (SD)	Mean (SD) = 6.9 (1.1)
Monroe 2018 (USA)[118]	HIV clinic	RCT	- HIV - Diabetes and/or hypertension	46	MPR	Report median (IQR)	ART: 92% (71-100%) Other medications: 79% (63-96%)

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							(if patient had hypertension + diabetes, only hypertension adherence recorded)
Moore 2015 (USA)[119]	Ongoing studies at research centre	RCT	- HIV - Bipolar type I or II	50	Visual analogue scale (HIV medication and psychiatric medication)	VAS < 100% = some adherence difficulty in previous 30 days	HIV medication = 12 (47.8%) Psychiatric medication = 14 (54.3%)
Moore 2012 (USA)[120]	Ongoing studies research centre	Cross-sectional	- HIV - Bipolar type I or II	44 (n with psychiatric medication assessed = 39)	MEMS (psychiatric medication and HIV medication)	MEMS <90%	HIV medication = 23/44 (52.3%) Psychiatric medication = 25/39 (64.1%)
Moore 2012 (USA)[121]	Not reported	Cross-sectional	- HIV - Current or lifetime history of methamphetamine use/dependence	127; MM = 75	AIDS Clinical Trials Group (ACTG) four day adherence self-report questionnaire (HIV medications)	Report of any skipped dose in previous 4 days	Total = 16 (21.3%) 12 (18%) with lifetime history METH+ use 4 (50%) with current METH+ use
Muntner 2010 (USA)[122]	Community	Cross-sectional	- Chronic kidney disease - Hypertension	13065; MM = 3936	MMAS-4 (antihypertensive medications)	MMAS-4 = 0: 'better adherence' MMAS-4 ≥ 2: worse adherence	MMAS-4 = 0: 2715 (69.0%) MMAS-4 = 1: 916 (23.3%) MMAS-4 ≥ 2: 305 (7.8%)
Muth 2016 (Germany)[123]	General practice	Pilot cluster-RCT	≥ 3 chronic conditions	100	MMAS-4 Medication Adherence Rating Scale (MARS) Observed medication adherence: (1) drug score (2) dose score (3) the regimen score	MMAS-4: scores 0–4, higher scores = lower adherence MARS: scores 5–25, lower scores = lower adherence Observed medication adherence: Scores outside 0.8–1.2 are divergent	MMAS-4 mean (SD) =0.3 (0.6) MARS mean (SD) = 24.0 (1.4) Observed adherence: Drug score = 43/97 (44.3%) divergent Dose score = 93/94 (98.9%) divergent Regimen score = 93/93 (100%) divergent
Muth 2018 (Germany)[124]	General practice	Pragmatic cluster-RCT	≥ 3 chronic conditions	Total = 505; intervention = 253; control = 252	MMAS-4 Observed medication adherence: (1) drug score (2) dose score (3) the regimen score	Report mean (SD) MMAS-4: scores 0–4, higher scores = lower adherence Observed medication adherence: scores	MMAS-4 =3.7 (0.6) Observed adherence: Drug score = 188/501 (37.5%) divergent Dose score = 259/499 (50.9%) divergent

						outside 0.8–1.2 are divergent	Regimen score = 255/500 (51.0%) divergent
Naidoo 2013 (South Africa)[125]	Primary care	Cross-sectional	- TB - HIV	3107; MM = 1729; analysed n = 757	Anti-TB medication: 1-item self-report of percent medication taken in prior 3–4 weeks HIV medication: 1-item self-report of percent medication taken in prior 4 weeks	Self-report adherence < 90%	TB medication: 315 (41.6%) HIV medication: 321 (42.4%)
Nakimuli-Mpungu 2019 (Uganda)[126]	Primary care	Baseline data from cluster-RCT	- Depression (MINI) - HIV	1140	Missed-dose method: 1-item self-report of days missed taking all doses in prior week (ART)	Self-report adherence< 100%	Not reported
Natarajan 2013 (Canada)[127]	General practice	Cross-sectional	- Diabetes mellitus - Hypertension	527; analysed n = 484	MMAS-4 (antihypertensives)	Answer yes to 1 or more questions = medium or low adherence	Low adherence = 8 (1.7%) Medium adherence = 101 (20.9%) Combined = 109 (22.5%)
Nelson 2011 (USA)[128]	Veterans Affairs medical centre	Retrospective record review	- Hyperlipidaemia - Diabetes - Psychotic disorder (not all sample has psychotic disorder)	total = 124; +psychotic disorder = 62; -psychotic disorder = 62	Cumulative mean gap ratio (CMGR): (both medication classes; for combination therapy, the medication with the greatest non-adherence was used)	Report mean (SD)	Antihyperlipidemic medication: +psychotic disorder mean (SD) = 12.0% (12.2) reflecting approx. 44 days without therapy -psychotic disorder mean (SD) = 16.9% (13.0%) reflecting approx. 62 days without therapy Antipsychotic medication: mean (SD) = 18.3% (16.4%) reflecting approx. 68 days without therapy
Nelson 2011 (USA)[129]	Veteran's Affairs medical centre	Retrospective record review	- Diabetes - Schizophrenia or related psychotic disorder	124; MM = 62	Cumulative mean gap ratio (CMGR) (antihyperglycaemic medication; antipsychotic medication)	Report mean (SD)	Antihyperglycaemics = 15.8% (16.8%) Antipsychotics = 18.3% (16.4%)
Neuget 2016 (USA)[130]	Administrative claims database	Retrospective cohort	- Non-metastatic breast cancer - Chronic condition from hypertension, hyperlipidaemia, GERD, thyroid, diabetes, osteoporosis	21255; MM = 13427	MPR (tamoxifen and/or aromatase inhibitor) Non-adherence after breast cancer diagnosis reported per disease group	MPR < 80%	Hypertension group: 1233/8683 (14.2%) Hyperlipidaemia group: 671/5829 (11.51%) GERD group: 567/4264 (13.97%) Thyroid group: 484/3542 (13.67%) Diabetes group: 267/1742 (15.33%) Osteoporosis group: 147/1384 (10.62%)

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Newville 2014 (USA)[131]	Methadone maintenance clinic	Cross-sectional	- HIV - Treated with methadone maintenance therapy	100	Self-report using questions from AIDS Clinical Trials Group (HAART)	Self-report < 100%	30 (30%)
Nichol 2009 (USA)[132]	Administrative claims database	Retrospective record review	- Hypertension - Treated with lipid-lowering medication	5943	PDC (antihypertensives; lipid-lowering medication)		Year 1 Non-adherent to at least one = 1817 (30.6%) Year 2 Non-adherent to at least one = 2955 (49.7%)
Otenyo 2018 (Kenya)[133]	Hospital renal unit	Cross-sectional	- Chronic kidney disease - Hypertension	144	MMAS-8 (antihypertensives)	MMAS-8 < 6	61 (42.4%)
Owen-Smith 2016 (USA)[134]	Healthcare database	Retrospective analysis	- Schizophrenia or schizoaffective disorder - Treatment by any statin or ACEI/ARB	1420; MM = 710	MPR (cardiovascular medications; if taking both statin and ACEI/ARB, adherence calculated separately for each)	MPR < 0.80	Statin: 350/583 (60.0%) ACEI/ARB: 232/383 (60.6%)
Padman 2017 (India)[135]	Hospital general medicine department; Dialysis centre	Prospective observational	- Chronic renal failure - Diabetes mellitus	107	MMAS-8 (diabetes medication)	Rated MMAS-8 scores as low, moderate and high adherence (criteria for categorising not reported)	Low = 61 (57.0%) Moderate = 36 (33.6%)
Pantalone 2012 (USA)[136]	HIV/AIDS clinics	Cross-sectional	- HIV - Hepatitis C virus	178; MM = 54; analysed n = 45	1-item self-report number of doses missed in prior 30 days (HIV medications)	Self-report adherence < 100%	17 (37.8%)
Parpouchi 2017 (Canada)[137]	Administrative database	Longitudinal (using baseline data, longitudinal data and retrospective data)	- Mental illness - Treated with methadone maintenance	497; MM with adherence to MMT reported = 78	MPR (methadone)	MRP < 0.8	62 (79.5%)
Patel 2016 (USA)[138]	Medicaid database	Retrospective cohort	- Type 2 diabetes - Developmental disability (cerebral palsy, autism, Down's syndrome, cognitive disabilities)	1529	Adherence: MPRm (anti-diabetes medications) Persistence: period from start of medication therapy to	Adherence: MPRm < 0.8. Report mean SD. Persistence: gap > 120 days	Pooled mean (SD) MPRm = 0.90 (0.19) Non-persistence: 464 (30.4%)

					discontinuation (anti-diabetes medications)		
Pence 2015 (USA)[139]	Infectious disease clinics	Pseudo-cluster RCT	- HIV - Major depression (MINI confirmed diagnosis & PHQ-9)	304	Visual analogue scale (ART adherence in prior 30 days)	Report mean (SD)	Mean (SD) = 86.5% (22.7%)
Pentakota 2013 (USA)[140]	Administrative research database	Retrospective cohort	- Diabetes - Comorbidity(s)	79249; MM at some stage (baseline, end, both) = 71334	Adherence: PDC (diabetes medications) Persistence: time from initiation of treatment to first discontinuation in treatment during first 2 years of treatment (diabetes medications)	PDC < 0.8 Persistence: discontinuity > 60 days	Reported according to concordant and discordant comorbidities: Concordant_Concordant = 34442 (44.5%) Discordant_Discordant = 39469 (50.95%) Other combinations reported in full paper.
Poon 2009 (USA)[141]	Veterans Health Administration database	Retrospective cohort	- Dementia - Hypertension	56561	MPR (antihypertensive medications)	MPR < 0.8	Reported by race and drug class: Min = 13.0%; Max = 39.6%
Puyat 2017 (Canada)[142]	Healthcare databases	Retrospective cohort	- Depression - Additional chronic health condition	105476; MM = 13329	PDC (antidepressants)	Report mean (SD) in 180-day period and in 365-day period	Mean (SD) = 0.84 (0.24) Mean (SD) = 0.80 (0.27)
Qian 2014 (USA)[143]	Administrative database	Cross-sectional	- COPD - Depression	74863; MM = 25130; analysed n = 12574	PDC (COPD medications) Discontinuation (COPD medications)	PDC: < 0.80 Discontinuation: binary variable of medication stopped 3 months or more prior to end of observation period or death, whichever first	PDC: 8621 (68.6%) Discontinuation: 3009 (23.9%)
Rattinger 2012 (USA)[144]	Administrative database	Cross-sectional	- Chronic heart failure - Alzheimer's disease or related disorders	9827; MM = 2379	MPR (chronic heart failure medication)	Report mean (SD)	EBM = 0.86 (0.15) Diuretic = 0.81 (0.20) Cardiac glycoside = 0.86 (0.18) Calcium channel blocker = 0.88 (0.17)
Reed 2018 (Australia)[145]	General practice	RCT	≥ 2 chronic conditions from cardiovascular, respiratory, musculoskeletal, psychological, digestive, kidney	254	5-item self-report	Higher scores indicate poorer levels of medication adherence (range of possible scores not reported)	Mean (SD) = 24.14 (0.14)

			disease, diabetes (type 1 or 2), cancer				
Rosenblum 2014 (USA)[146]	Residential and outpatient psychiatric and substance use facilities	RCT	- Mental illness - History of substance use	203	Medication Adherence Rating Scale (MARS) (psychiatric medication)	Report mean (SD) Lower score = higher medication adherence (range of possible scores not reported)	Mean (SD) = 1.55 (0.48)
Roux 2012 (France)[147]	Hospital	Prospective cohort	- HIV - Hepatitis C virus	593 at enrolment; 246 at follow-up	7-item self-report adherence questionnaire (HIV medications)	Self-report adherence < 100% during 4-days prior	Enrolment: 214 (36%) Follow-up: 72 (29.3%)
Saadat 2015 (Iran)[148]	Healthcare centres	Cross-sectional	- Hypertension - ≥ 1 comorbidity	280; MM = 160	MMAS-8 (Persian version) (antihypertensive medication)	MMAS-8 < 6	77 (48.1%)
Safren 2009 (USA)[149]	Community	RCT	- HIV - Depressive mood disorder	45	MEMS (antiretroviral medication)	Report mean (SD)	Control group = 54.41 (30.94) Intervention group = 62.26 (26.82)
Safren 2012 (USA)[150]	Community outreach, hospital HIV clinics, substance use clinics	RCT	- HIV - History of injection drug use (self-endorsed + enrolled in opioid treatment for ≥ 1 month(s)) - Criteria for diagnosis of current or subsyndromal depressive mood disorder	89	MEMS (antiretroviral medication)	Report mean (SD) percent MEMS-based adherence since last visit	Mean (SD) = 69.0% (28.3%)
Safren 2014 (USA)[151]	Primary care	RCT	- Unipolar depression - Type 2 diabetes	87	MEMS (antidiabetic oral medication, insulin or both)	Report mean (SD)	Control group = 85.7 (24.6%) Intervention group = 76.6% (26.0%)
Santorelli 2016 (USA)[152]	Administrative databases	Retrospective cohort	- Breast cancer - Diabetes or hypertension or lipid disorder	9340; MM groups: diabetes = 298; lipid disorders = 508; hypertension = 1062	Adherence: PDC (hypertension; diabetes; lipid disorder medications) Persistence: largest number of consecutive days without coverage by relevant medication class during first 18 months (hypertension;	Adherence: PDC < 80% Persistence: Diabetes = 26 days without medications Hypertension = 26 days without medication	PDC: Diabetes = 78 (26.2%) Hypertension = 151 (14.2%) Lipid disorder = 147 (28.9%) Persistence: Diabetes = 94 (31.5%) Hypertension = 200 (18.8%) Lipid disorder = 123 (24.2%)

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					diabetes; lipid disorder medications)	Lipid-lowering = 62 days without medications	
Schmitt 2010 (USA)[153]	Veteran's Affairs medical centre	Retrospective observational	- Chronic kidney disease - Treatment with antihypertensive	7227	MPR (antihypertensive medication)	MPR < 80%	2360 (32.7%)
Schüz 2011 (a) (Germany)[154]	Community	Longitudinal	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	309	2 self-report items from Reported Adherence to Medication Scale (RAM) measuring: (1) intentional non-adherence (2) unintentional non-adherence	Report mean (SD) Higher values indicate higher degrees of non-adherence (range of possible scores not reported)	Intentional mean (SD) = 1.32 (0.68) Unintentional mean (SD) = 1.40 (0.67)
Schüz 2014 (Germany)[155]	Community	Longitudinal	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	215	1-item self-report intentional non-adherence from Reported Adherence to Medications Scale (RAM)	Report mean (SD) Higher scores indicate better adherence (range of possible scores not reported)	Mean (SD) = 4.68 (0.68)
Schüz 2011 (b) (Germany) [156]	Community	Longitudinal	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	309	2 self-report items from Reported Adherence to Medication Scale (RAM) measuring: (1) intentional non-adherence (2) unintentional non-adherence	Not reported	Not reported
Shi 2010 (USA)[157]	Veteran's Health administration database	Retrospective analysis	- Schizophrenia - Metabolic syndrome	593; MM = 209	Adherence: MPR (antipsychotics) Discontinuation: medication gap (antipsychotics)	Adherence: report mean (SD) MPR Discontinuation: medication gap > 30 consecutive days	Mean (SD) MPR = 0.33 (0.47) Median time to discontinuation = 50 days
Shin 2010 (South Korea) [158]	Pharmacy claims data	Retrospective record review	- Hypertension - Cancer survivor	2455193; MM = 12636	CMA (antihypertensive medication)	CMA < 80%	5762 (45.6%)
Shoval 2017 (Israel)[159]	Healthcare database	Retrospective analysis	- Parkinson's disease - Depression	8553	MPR (antidepressant medication)	MPR < 20% = non-adherent	< 20% = 1566 (18.4%)

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						MPR 20% - 50% = poorly adherent MPR 50% - 80% = moderately adherent	20-50% = 1184 (13.8%) 50-80% = 1584 (18.5%) < 80% = 4334 (50.7%)
Siika 2013 (Kenya)[160]	Ambulatory HIV care program	Retrospective cohort	- HIV - Tuberculosis	21242; MM = 5186; analysis n = 4838	1-item self-report number of pills taken during prior week (ART)	If patient ever answers 'most', 'half', 'few', or 'none' (defined as non-perfect adherence to ART)	986 (20.4%)
Slabbert 2015 (South Africa)[161]	Administrative database	Prospective cohort	- Major depressive disorder - HIV/AIDS	127	MPR (antidepressant medications)	Report mean (SD)	74.43 (32.03)
Soto 2015 (Chile) [162]	Hospital	Prospective	- Type 2 diabetes - Hypertension	50	Pill count (seems like both medications together)	Pill count < 80% or > 110%	35 (70.0%)
Stack 2010 (UK)[163]	Healthcare database (secondary care)	Cross-sectional	- Type 2 diabetes - Hypertension	480	MMAS-4 (separately for antidiabetic medication, antihypertensive medication and statin)	0 = total adherence 4 = non-adherence: presented as no unintentional non-adherence; no intentional non-adherence	Unintentional non-adherence: Diabetes medication = 302 (62.9%) Hypertension medication = 226 (47.1%) Statins = 225 (46.9%) Intentional non-adherence: Diabetes medication = 63 (13.1%) Hypertension medication = 92 (19.2%) Statins = 87 (18.1%)
Stilley 2010 (USA)[164]	Primary care	RCT	≥ 2 from type 2 diabetes, hypertension, hyperlipidaemia	373	MEMS (not clear to what medications)	Report mean (SD): (1) prescribed number of doses (2) days with prescribed dose (3) days with correct inter-dose interval	Prescribed number of doses = 92.6 (17.6) Days with prescribed dose = 82.2 (22.4) Days with correct inter-dose interval = 64.2 (30.1)
Stuart 2015 (USA)[165]	Medicare database	Retrospective cohort	- Diabetes - Cancer	32855; MM = 4348	PDC (diabetes medication)	Report mean PDC	OHA (n = 2248) mean PDC = 73.8% RAAS-I (n = 2618) mean PDC = 70.5% Statins (n = 2272) mean PDC = 69.0%

Taira 2017 (USA)[166]	Healthcare database	Retrospective observational	- Type 2 diabetes - Treatment with antihypertensive, oral antidiabetic and cholesterol lowering medication	5163	MPR (each medication)	MPR < 0.8	Calculated average percent from race/ethnicity groups: Diabetes = 75.2% Hypertension = 78.6% Cholesterol-lowering = 62.4%
Tang 2014 (USA)[167]	Outpatient sites	Secondary analysis of prospective descriptive cohort	- Heart failure - Depression (GDS)	280; MM = 103	MEMS (heart failure medication); Basel Assessment of Adherence Scale (BAAS) (heart failure medication)	MEMS: < 80% BAAS: a positive answer on any question	Self-report = 77 (75%) MEMS = 29 (28%)
Thaler 2015 (USA)[168]	HIV treatment facilities	Not reported	- HIV - neurocognitive impairment and/or substance dependence/use and/or psychiatric illness	204; MM not reported	MEMS (HAART) Self-report (not described) (HAART)	Report mean (SD) per additional risk factor	One risk factor: 64.4% (27.0%) Two risk factors: 62.8% (28.0%)
Vacek 2013 (USA)[169]	Medicaid database	Retrospective analysis	- Developmental disability - Hypertension	280; analysed n = 201	PDC (antihypertensive medication)	PDC < 0.8	90 (45%)
Vatcharavongvan 2017 (Thailand)[170]	Primary care	Cross-sectional	≥ 2 chronic conditions from type 2 diabetes, hyperlipidaemia, hypertension, ischemic heart disease, chronic heart failure, cerebrovascular disease, epilepsy	397; analysed n (polypharmacy group) = 122	Structured questionnaire during face-to-face interview	At least one medication that was (a) taken at wrong time, (b) missed (c) intentionally stopped by patient during 2 weeks prior	74 (60.7%)
Vega 2017 (USA)[171]	Healthcare database	Retrospective cohort	- Major depressive disorder - Type 2 diabetes	1361	Adherence: MPR (depression medication; diabetes medication) Persistence: Length of treatment (LOT) (depression medication)	MPR < 80% LOT ≥ 15 day gap in first 180 days of therapy	Depression medication: Non-adherent = 872 (64.1%) Non-persistent = 926 (68.0%) Non-adherent/non-persistent = 936 (68.8%) Diabetes medication: Non-adherent = 525 (38.6%)
Villar 2009 (Spain)[172]	Ongoing studies, administrative database	Prospective cohort	- Severe obstructive sleep apnoea - Cardiovascular disease	2158; antihypertensives = 659;	MPR (cardiovascular disease medications)	MPR < 80%	Antihypertensives: 118/659 (17.91%) Statins: 59/175 (33.71%)

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				statins = 175; antiplatelets = 109			Antiplatelets: 9/109 (8.26%)
Wagner 2011 (USA)[173]	Pooled data from longitudinal studies	Cross-sectional and longitudinal	- HIV - Depression (BDI-II; BDI (original version); CESD; depression subscale of Brief Symptom Inventory)	1374; MM = 489	MEMS (HIV medication)	MEMS < 90%	Severe depression = 72% (148/205) Mild to moderate depression = 56% (159/284) Combined = 307/489 (62.8%)
Wakefield 2012 (USA)[174]	Primary care	RCT	- Type 2 diabetes - Hypertension	302 (control = 107; low intensity = 102; high-intensity = 93)	1-item self-report adherence	Report mean (SD)	Mean (SD): Control group = 99.5 (4.4) Low-intensity group = 99.7 (1.6) High-intensity group = 99.7 (2.7)
Warner 2013 (Germany)[175]	Community	Longitudinal	≥ 2 chronic physical conditions from Charlson Comorbidity Index or Functional Comorbidity Index	309	1-item self-report unintentional non-adherence from Reported Adherence to Medication Scale (RAM)	1 < RAM < 4	T1 = 83 (27%) T2 = 99 (32%)
Wei 2018 (USA)[176]	Medicare database	Retrospective cohort	- Major depression - Chronic obstructive pulmonary disease	25458; analysed n = 11572	PDC (COPD medications)	PDC < 0.8	7059 (61.0%)
Weiss 2016 (USA)[177]	Ambulatory practices	Longitudinal	- HIV - Hypertension or chronic kidney disease	152; +hypertension = 115 (analysed n = 102); +CKD = 37 (analysed n = 20)	MEMS (1) taking adherence (2) dosing adherence (3) timing adherence (HIV medications and CKD/hypertension medications separately)	MEMS < 95% (HIV medications) MEMS < 80% (other medications)	+hypertension: HIV medication = 40 (39%) Hypertension medication = 20 (20%) +CKD: HIV medication = 7 (35%) CKD medication = 4 (20%)
Wenze 2015 (USA)[178]	Hospital	Pilot RCT	- Bipolar I or II - Substance use disorder	30	Adapted version of Medication Compliance Questionnaire (MCQ) (prescribed mood-stabilizing agent)	MCQ ≥ 2	12 (40.0%)
Wong 2014 (China)[179]	Administrative database	Cross-sectional	≥ 2 chronic conditions	3866; MM = 523	MMAS-8 (antihypertensives, oral hypoglycaemic agents, lipid lowering agents separately)	MMAS-8 ≤ 6 (MMAS-8 sections for each drug should score >6 to be	321 (45.1%)

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						defined as having optimal adherence)	
Wroe 2014 (Rwanda)[180]	Rural health centres	Cross-sectional	- HIV - Depression (Hopkin's Symptom Checklist)	292; MM = 32	Self-report 4-day recall (HIV medications) Self-report 2-day and 7-day treatment interruptions (HIV medications) CASE index (HIV medications)	Self-report 4-day recall: ≥ 1 dose(s) in four days prior Self-report 2-day and 7-day treatment interruptions: non-adherence on sequential days CASE index ≤ 9	4-day recall = 5 (17.0%) 2-day interruptions = 7 (22.2%) CASE index = 5 (15.8%)
Wu 2011 (USA)[181]	Medicaid database	Retrospective observational	- Hyperlipidaemia - Type 2 diabetes	1705	MPR (statins; antidiabetic medication)	MPR < 0.8	Statins = 1079 (63.3%) Mean (SD) MPR (statins) = 0.61 (0.32) Mean (SD) MPR (antidiabetics) = 0.71 (0.30)
Wu 2010 (USA)[182]	Medicaid database	Retrospective cohort	- Major depressive disorder - Anxiety disorder	3083; MM = 776	Adherence: MPR modified (MPRm) (antidepressants) Persistence: Refill gap (antidepressants)	MPRm: report mean (SD) Refill gap: median (IQR)	Mean (SD) MPRm = 0.71 (0.37) Median (IQR) persistence (days) = 237 (216-256)
Wu 2013 (USA)[183]	Inpatient and outpatient cardiology clinics	Secondary analysis of two prospective studies	- Heart failure - Depression (PHQ-9)	216; MM = 55	MEMS (heart failure medication)	MEMS < 88%	30 (54.55%)
Wulaningsih 2018 (Sweden)[184]	Administrative database	Retrospective analysis	- ER-positive breast cancer - Charlson Comorbidity Index score ≥ 1	4645; MM baseline = 715; MM follow-up = 1348	MPR (endocrine therapy)	MPR < 80%	Baseline: 168/715 (23.5%) 5-years: 330/1348 (24.5%)
Xia 2015 (China)[185]	Hospital	Cross-sectional	- ≥ 1 chronic condition from chronic obstructive pulmonary disease, diabetes,	325; +anxiety = 276; +depression = 266	Chinese Medication Adherence Scale (CMAS) (chronic condition)	CMAS <6 to	+depression: 222 (83.5%) +anxiety: 226 (81.9%)

			hyperlipidaemia, hypertension - Depression (PHQ-9) and/or anxiety (GAD-7)				
Xue 2018 (China)[186]	Primary care	Baseline data from RCT	- Hypertension - Depression (PHQ-9)	2362	1-item self-report adherence (antihypertensive medications)	Answer yes to "do you sometimes forget to take your antihypertensive pills"	954 (40.4%)
Yang 2010 (USA)[187]	Medicare database	Retrospective cohort	- Diabetes - Hypertension and/or renal disease	599141	PDC (antihypertensive medication)	PDC < 0.8	276945 (46.22%)

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Title: Prevalence and predictors of medication non-adherence among people living with multimorbidity: a systematic review and meta-analysis

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ABSTRACT (299/300)

Objectives This systematic review aimed to describe medication non-adherence among people living with multimorbidity according to the current literature, and synthesise predictors of non-adherence in this population.

Methods A systematic review was conducted according to the Preferred Reporting Items for Systematic reviews and Meta-Analyses. PubMed, EMBASE, CINAHL and PsycINFO were searched for relevant articles published in English language between January 2009 and April 2019. Quantitative studies reporting medication non-adherence and/or predictors of non-adherence among people with two or more chronic conditions were included in the review. A meta-analysis was conducted with a sub-group of studies that used an inclusive definition of multimorbidity to recruit participants, rather than seeking patients with specific conditions. Remaining studies reporting prevalence and predictors of non-adherence were narratively synthesised.

Results The database search produced 10998 records and a further 75 were identified through other sources. Following full-text screening, 178 studies were included in the review. The range of reported non-adherence differed by measurement method, at 76.5% for self-report, 65.6% for pharmacy data and 44.1% for electronic monitoring. A meta-analysis was conducted with 8 studies (n = 8949) that used an inclusive definition of multimorbidity to recruit participants. The pooled prevalence of non-adherence was 42.6% (95% CI = 34.0-51.3%, k = 8, I^2 = 97%, p < 0.01). The overall range of non-adherence was 7.0% to 83.5%. Frequently reported correlates of non-adherence included previous non-adherence and treatment-related beliefs.

Conclusions The review identified a heterogeneous literature in terms of conditions studied, medication adherence measurement and definition of non-adherence. Results suggest that future attempts to improve medication adherence among people with multimorbidity should determine for which conditions individuals require most support. The high levels of medication non-adherence highlight the need for more attention to be paid by healthcare providers to the impact of multimorbidity on chronic disease self-management.

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Strengths and limitations of this study

- Criteria for inclusion did not specify particular chronic conditions, permitting inclusion of a wide variety of disease combinations.
- Studies were not excluded on the basis of design, setting, or method of non-adherence measurement, generating a comprehensive overview of the literature.
- Due to considerable heterogeneity between studies, meta-analysis and synthesis of predictors was restricted to studies that did not recruit participants with pre-specified combinations of conditions.
- Quantitative subgroup analysis was not conducted to determine the influence of participant or study characteristics on non-adherence estimates.

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INTRODUCTION

Multimorbidity is broadly defined as the co-existence of two or more chronic conditions[1] and has been described as the most common chronic condition experienced by adults.[2] In a Scottish study of 1.75 million people, the prevalence of multimorbidity was estimated to be 23.3%, increasing to 65% among those aged 65-84 years.[3] People living with multimorbidity attend more medical consultations, experience higher rates of hospital admissions and face increased healthcare costs.[4] People with multimorbidity are also expected to engage with complex self-management regimens which can include monitoring symptoms, changing lifestyle behaviours and adhering to prescribed medications.[5]

Adherence refers to the extent to which a person's behaviours correspond with agreed recommendations from their healthcare provider.[6] Risk of medication non-adherence increases as people are prescribed more medicines[7] – a point of relevance considering the close association between multimorbidity and polypharmacy.[8] The extent of non-adherence among people with multimorbidity is poorly understood, attributable to challenges in measuring multiple medication adherence.[9] However, it is recognised that medication non-adherence in multimorbidity can exacerbate the burden experienced by individuals through increased morbidity and mortality.[10] Medication adherence can be influenced by many factors, including treatment characteristics, condition-related factors, and illness and medication beliefs.[11] Identifying significant correlates of non-adherence among people self-managing multiple conditions will contribute to an evidence base for developing behavioural treatments to support adherence in this population.

Much intervention development to improve medication adherence has focused on single-disease populations.[12] Such a focus may produce an artificial underestimation of the complexity of chronic disease self-management when, in reality, most people with a chronic condition have additional morbidity(s).[3] This, in turn, may lead to policies and interventions which are not designed with 'the most common chronic condition'[2] in mind. In order to develop interventions to improve medication adherence among people living with multimorbidity, it is first necessary to understand the extent of non-adherence in this population, as well as factors that might influence adherence. To our knowledge, a synthesis of the prevalence and predictors of medication non-adherence among people with multimorbidity has not been previously reported. This review therefore aims to (1) describe the prevalence of medication non-adherence among people living with multimorbidity, and (2) describe previously identified correlates of non-adherence in this population.

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METHODS

A systematic review and meta-analysis was conducted according to the Preferred Reporting Items for Systematic review and Meta-Analysis statement[13] (supplementary file 1). The review was prospectively registered on PROSPERO (CRD42019133849) and a peer-reviewed protocol was published.[14] Prior to finalising our aims, we searched in PROSPERO[15] to determine if an equivalent review had been conducted or was currently being conducted.

Inclusion and exclusion criteria

Original peer-reviewed articles and doctoral theses produced in English from 2009-2019 were reviewed. A ten year range was chosen to reflect the marked increase in multimorbidity literature in recent years.[16] Opinion pieces, conference presentations, books, letters, editorials and abstracts were not reviewed. Quantitative studies including adults aged 18 years or older, living with two or more chronic conditions (including physical and/or mental health conditions) reporting medication non-adherence and/or predictors of non-adherence in an extractable format were considered. Presence of multimorbidity was required to be part of study aims and/or inclusion criteria. Polypharmacy was not used as a definition of multimorbidity; however studies referring to use of two or more medications that could be inferred as treatment for two or more chronic conditions were considered. Studies were not excluded based on design, however only observational data was extracted for review. Therefore clinical trials that did not report baseline non-adherence were excluded. Studies including any participants under 18 years were excluded.

Search strategy and study selection

An electronic database search of PubMed, EMBASE, CINAHL and PsycINFO from January 2009 to April 2019 was conducted using a predefined search strategy combining terms related to ‘multimorbidity’ and ‘medication adherence’ (supplementary file 2). Reference lists of included studies were screened for relevant articles. Records were exported to EndNote X8 to remove duplicates. Two reviewers (LF, RLV) each screened 50% of titles and abstracts in Covidence.[17] Twenty-percent of each reviewer’s records were chosen at random for cross-checking by the other reviewer. Studies meeting inclusion criteria were moved forward for full-text screening by two independent reviewers (LF & JL or EG). Where disagreements arose, a third reviewer (GJM or AWM) was consulted.

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Data extraction and quality assessment

A data extraction form was developed to extract relevant data from each included study (supplementary file 3). Where studies reported medication adherence, this was converted to medication *non*-adherence by subtracting the number of adherent participants from the total sample. One reviewer (LF) extracted data from all included studies, and a second reviewer (JL) cross-checked 20%.

The quality of included studies was assessed using criteria for observational studies[18] previously adapted for medication adherence reviews.[19] Criteria include 11 items related to selection methods, measurement, sources of bias, control for confounding and use of statistics. Each item was answered with 'yes', 'no', 'don't know' or 'not applicable'. All included studies were assessed using these criteria, including intervention studies as only baseline observational data were reviewed. One reviewer (LF) assessed the quality of all studies and a second reviewer (EG) cross-checked a random sample of 20%. No studies were excluded based on quality.

Data analysis

Due to between-study heterogeneity relating to study population, design, setting, and measurement and definition of medication non-adherence, a quantitative synthesis involving all studies was not considered appropriate. Meta-analysis was conducted with studies which recruited participants as 'people living with ≥ 2 chronic conditions'. This group of studies was considered to be most focussed in addressing a universal experience of multimorbidity – albeit a multifaceted and complex experience across individuals – offering results that can be used in future attempts to address non-adherence in this population. Studies which recruited participants with pre-specified conditions were therefore not included in the meta-analysis. Meta-analysis was conducted in R[20] using the meta[21] and metafor[22] packages (supplementary file 4). A random-effects model was used to estimate the pooled prevalence of non-adherence. Between-study heterogeneity was assessed using I^2.[23]

Remaining studies were narratively synthesised to describe medication non-adherence in multimorbidity, informed by Popay and colleagues' guidelines.[24] The study characteristics table was preliminarily analysed to identify and count patterns of results. This was followed by an exploration of relationships within and between results of individual studies. Predictors of non-adherence were also synthesised in this way using studies which recruited participants using an inclusive definition of multimorbidity. Predictors were not extracted from studies which recruited participants with combinations of specific conditions due to potential confounding associated with specific conditions, e.g. a study of people living with bipolar disorder and depression assessing severity of depressive symptoms and psychological disability as predictors of non-adherence.[25]

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Patient and public involvement

A panel of people with experience of living with multimorbidity were consulted to advise on appropriate methods for disseminating findings from the review to patients and the public.

RESULTS

Overview of studies

After removing duplicates, 10998 studies were identified from the database search and 75 from reference lists of included studies. Following screening of titles and abstracts, 449 full-texts were reviewed. Of these, 271 were excluded with reasons (figure 1), with 178 studies included in the final synthesis.[9,25–201] The study selection process is outlined in the PRISMA flow diagram in figure 1.

Figure 1. PRISMA flow diagram

Study quality

Results of study quality assessment are summarised in figure 2.

Figure 2. Methodological quality graph.

Study characteristics

One-hundred and thirty-three studies included participants with ≥ 2 specific conditions, e.g. Gibson et al. recruited participants self-managing type 2 diabetes and hypertension;[79] 26 studies included participants with one specific condition and ≥ 1 other condition(s), e.g. Puyat et al. recruited participants living with depression and an additional chronic health condition;[156] and 19 studies recruited participants using an inclusive definition of multimorbidity, e.g. Schüz and colleagues[170] recruited people living with ≥ 2 conditions from the Charlson Comorbidity Index[202] or Functional Comorbidity Index.[203]

Eighty-four studies measured non-adherence using pharmacy data, e.g. Kim et al. used pharmacy claims data to assess non-adherence among people living with hypertension and a psychiatric condition;[107] 81 used self-report, e.g. Moore et al. measured adherence to medications for HIV and bipolar disorder using a visual analogue scale;[135] and 20 used electronic monitoring, e.g. Safren et al. assessed non-adherence to antiretroviral medication among people living with HIV and a depressive mood disorder using a Medication Event Monitoring System (MEMS).[163] Three studies used pill counts[59,101,176], two involved reports by a healthcare provider,[80,83] and one did not report the method used.[64] Ten of these studies used more than one method to measure adherence, e.g. Bogner

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et al. used electronic monitoring and self-report to measure non-adherence for diabetes and depression.[45] Forty-seven studies reported average non-adherence only. The characteristics of all included studies are summarised in supplementary files 5 and 6.

Participant characteristics

The number of participants with multimorbidity in each study ranged from 22[130] to 599,141.[201] Among 135 studies reporting the mean (SD) age of participants, the average age ranged from 32.4 (9.6)[105] to 80.8 (9.1) years.[158] Ninety-three (52.2%) study samples were majority women, while three studies did not report gender.[98,117,126]

Prevalence of medication non-adherence

A meta-analysis was conducted with eight studies reporting prevalence of non-adherence and using an inclusive definition of multimorbidity, i.e. without seeking people with ≥ 1 specific condition(s).[56,71,115,137,138,184,189,193] The pooled prevalence of non-adherence was 42.6% (95% CI = 34.0-51.3%, $k = 8$, $I^2 = 97\%$, $p < 0.01$) (figure 3). The range of non-adherence among studies in the meta-analysis was 16.4%[71] to 61.4%.[193] The range among all reviewed studies was 7.0%[42] to 83.5%.[199] Remaining studies were narratively synthesised to describe prevalence and predictors of medication non-adherence.

Figure 3. Results of random effects meta-analysis.

Narrative synthesis findings

Medication non-adherence over time

The direction and extent of change in medication non-adherence over time was variable between studies. Among studies reporting prevalence of non-adherence over time, respective lengths of follow-up were three months,[96] six months,[189] 48 weeks,[121] one year,[26,36] 16 months,[161] two years,[60] 3 years[48] and 5 years.[198] Of these, four studies measured change in non-adherence following a recent diagnosis or treatment for an additional condition,[36,48,96,161] four measured change following medication initiation,[26,60,121,198] and one reported change not related to specific patient or treatment characteristics.[189] Three studies reported decreased non-adherence,[36,96,161] with the greatest decrease being 9.6% after 3 months.[96] Four studies reported increased non-adherence,[26,121,189,198] with the greatest increase being 41.2% after 12 months.[26] One study reported variation in how non-adherence changed over time depending on the class of medication,[60]

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while another reported an initial 13.9% increase after two years, followed by a 14.9% decrease one year later.[48]

Assessment of multiple medications

Within individual studies, the extent of non-adherence for one condition did not always correspond with non-adherence for another condition among people taking medications for multimorbidity. Twenty-six studies reported non-adherence prevalence estimates separately for each condition, e.g. Foguet-Boreu reported that, among people living with both diabetes and hypertension, 39.8% were non-adherent to medication for diabetes, while 23.8% were non-adherent to medications for hypertension.[75] Differences in non-adherence between conditions ranged from 0.8% measured by a 1-item self-report measure in a study of people living with HIV and TB,[139] to 33.6% measured by pharmacy data in a study of people with higher non-adherence to medications for COPD than depression.[27] Four studies reported a difference of ≤ 5%,[45,79,88,139] five reported a difference of 5–9.9%,[95,110,129,135,177] 11 reported a difference of 10–19.9%,[38,57,65,75,84,90,102,133,177,180,191] and six reported a difference of ≥ 20%.[27,66,80,90,105,185]

Physical and mental multimorbidity

Differences in non-adherence between physical and mental health conditions were not consistent across the literature. Among studies of people living with a physical condition and depression and/or anxiety, four reported non-adherence prevalence for both the physical and mental health condition, e.g. Gupte-Singh and colleagues reported non-adherence to medications for asthma (62.0%) and depression (74.3%).[84] Non-adherence was higher for depression in three studies,[45,84,185] and higher for the physical condition in one study.[27]

Among studies of people living with a psychiatric condition and a physical condition, seven reported non-adherence prevalence for both conditions, e.g. Moore et al. reported non-adherence among people living with HIV (52.3%) and bipolar disorder (64.1%).[133] Non-adherence was higher for the psychiatric condition in three studies,[90,133,135] and higher for the physical condition in four studies.[66,88,111,129]

Characteristics of multiple medication regimens

Two studies compared prevalence of non-adherence according to characteristics of multiple medication regimens.[26,93] Starting medications for two conditions simultaneously resulted in lower non-adherence versus not starting simultaneously (24.8% vs 55.0% at 2 months; 66.0% vs. 69.0% at 12

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months).[26] Hussein and colleagues assessed the difference between a single-pill vs. two-pill antihypertensive, lipid-lowering regimen.[93] Results concluded that a single-pill regimen improved adherence compared to a two-pill regimen, and improvements were greater among people with previous experience taking either an antihypertensive or lipid-lowering medication.[93]

Method of non-adherence measurement

Different methods of measuring non-adherence produced differences in the range of non-adherence estimates, with self-report measures producing the widest range, and observed adherence producing the narrowest range of non-adherence. Among sixty-four studies using pharmacy data, non-adherence prevalence ranged from 10.6% among women with breast cancer and osteoporosis[144] to 80.0% among people with COPD and depression;[28] Among 50 studies using self-report, non-adherence prevalence ranged from 7.0% among people with anxiety and cardiac disease[42] to 83.5% among people with depression and an additional chronic condition(s).[199] Among 10 studies using MEMS, non-adherence prevalence ranged from 20.0% among people living with HIV and hypertension or chronic kidney disease[191] to 64.1% among people living with HIV and bipolar disorder.[133] Two studies reported prevalence of non-adherence assessed by a healthcare provider[83] or a mix of healthcare providers, patients and relatives,[80] with non-adherence ranging from 8.6%,[83] to 52.5%.[80] Two studies used observed medication adherence to identify discrepancies between medications taken and medications prescribed,[137,138] with non-adherence prevalence ranging from 37.5%[138] to 44.3%.[137] among people with three or more chronic conditions. Two studies used pill counts, with non-adherence ranging from 46.3%[101] to 70.0%.[176]

Nine studies reporting prevalence of non-adherence used more than one measure.[34,45,51,114,122,137,138,181,194] Among studies using self-report and MEMS, two studies reported lower non-adherence using self-report,[45,114] one study reported higher non-adherence using self-report,[181] and one study reported a 1% difference.[34] Another study reported no difference between self-report and pharmacy data.[51] Stack and colleagues used self-report to measure prevalence of intentional non-adherence (e.g. deliberately choosing not to take medicines) and unintentional non-adherence (e.g. forgetting to take medications) to three medications.[177] For each medication, unintentional non-adherence was more prevalent (62.9%, 47.1%, 46.9%) than intentional non-adherence (13.1%, 19.2%, 18.1%).

Predictors of medication (non-)adherence

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Eleven studies using an inclusive definition of multimorbidity reported predictors of adherence (table 1).[40,56,62,71,94,106,168–170,189] Nine used self-report to measure adherence,[56,62,71,94,106,168–170,189] and two used pharmacy data.[40,68] The mean age of participants was < 65 years in four studies[40,62,68,94] and > 65 in seven studies[56,71,106,168–170,189]. Five studies assessed demographic variables as predictors,[56,106,168,169,189] including age, sex, marital status, education and deprivation. No demographic variables significantly predicted non-adherence.

One study assessed health status, which was associated with adherence i.e. poorer health status was associated with stronger motivation to take medications.[106] Other factors assessed included number of prescribed medications,[56,62,168,169,189] number of chronic conditions,[62,68,106,168,169] physical fitness,[189] and functional health status,[168] as well as presence of other conditions.[56,68,106] Schüz and colleagues[168] found lower intentional non-adherence among people prescribed more medications and higher intentional non-adherence among people with more conditions. Conversely, Domino and colleagues[68] found lower non-adherence among those with more comorbidities. Depression was associated with lower adherence, while schizophrenia was linked to both higher and lower adherence, depending on other physical conditions present.[68]

A number of studies explored beliefs about conditions and medications as predictors of adherence. Two studies using the same sample reported less non-adherence among those with stronger beliefs about the necessity of medications,[168,169] less non-adherence among those with greater concerns about medications,[169] more intentional non-adherence among those with increasing beliefs that medications are harmful,[168] and more unintentional non-adherence among those with increasing beliefs that medications are overused.[168] Changes in necessity beliefs predicted intentional non-adherence, with increasing beliefs about necessity associated with improved adherence.[169] Beliefs about overuse of medications predicted unintentional non-adherence, with stronger beliefs associated with higher unintentional non-adherence.[168,169] Beliefs about consequences, treatment control and timeline were also explored.[106,170] Kenning and colleagues did not report significant relationships between beliefs and adherence.[106] Schüz and colleagues found higher adherence was predicted by stronger control beliefs, stronger beliefs about negative consequences and beliefs about a shorter illness timeline.[170] One study assessed beliefs about multimorbidity itself, reporting no significant relationship with adherence.[106]

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Three studies reported an association between previous and later unintentional non-adherence,[168–170,189] with one study also reporting an association between previous and later intentional non-adherence.[168,169] Previous medication motivation predicted adherence, with participants with higher motivation at baseline reporting higher adherence at follow-up.[106] Other factors assessed included attitudes towards medications,[189] medication knowledge,[106] and medication adherence self-efficacy,[62] none of which predicted non-adherence. One study assessed general received social support and medication specific-support,[189] with higher medication-specific social support associated with lower adherence; this relationship was moderated by social conflict. General received social support did not predict adherence. Other factors assessed included chronic disease self-efficacy, medication assistance, health literacy, and functional health literacy,[62] none of which predicted non-adherence.

Lower adherence associated with lower physical (r = -.44) and environmental (r = -.57) quality of life.[62] Lower prospective memory performance was related to more omission errors of adherence (r = -.29), while lower retrospective memory performance was related to more commission errors (r = -.24).[94] Other factors positively associated with adherence were medical home enrolment,[40] health provider relational quality (r = 0.22)[71] and fewer hassles during healthcare system encounters.[106] Self-help behaviours did not predict adherence,[106] better self-monitoring and insight predicted higher motivation to adhere to medications,[106] and a more impulsive/careless problem-solving style was associated with lower adherence (r = -.49).[62]

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Study (country)	Population	Predictors assessed	Dependent variable	Significant predictors of and/or associations with medication adherence
Beadles 2015 (USA)[40]	≥ 2 of the following: major depressive disorder, schizophrenia, diabetes mellitus, hypertension, hyperlipidaemia, seizure disorder, asthma, chronic obstructive pulmonary disease	• Medical home enrolment	Proportion Days Covered (PDC) ≥ 0.8	• Medical home enrolment was positively associated with PDC ($p < .001$) across all cohorts (see original paper for breakdown by cohort)
Cicolini 2016 (Italy)[56]	≥ 2 of the following: cardiovascular, neurological, respiratory, musculoskeletal, oncologic diseases or diabetes mellitus	• Age • Sex • Education • Marital status • Number of prescribed drugs • Presence of diabetes • Presence of other disorders (cancer, musculoskeletal, respiratory, neurological, other) • Beliefs about medications (necessity, concerns) • Beliefs about medications (ambivalent, accepting)	8-item Moriskey Medication Adherence Scale (MMAS-8); "medium adherence" vs "low adherence"	• Necessity beliefs: OR (95% CI) = 1.61 (1.21-2.14), $p < .001$ • Concern beliefs: OR (95% CI) = 2.02 (1.64-2.49), $p < .001$ • Ambivalent vs accepting: OR (95% CI) = 0.24 (0.16-0.37), $p < .001$
Daniels 2016 (USA)[62]	≥ 2 chronic conditions	• Health literacy • Medication assistance • Medication adherence self-efficacy • Health-related problem solving • Functional health literacy • Chronic disease self-efficacy • Quality of life • Number of chronic conditions • Number of medications	MMAS-8; higher score = lower adherence	• Health-related problem solving (total score): $r = -.489$, $p = 0.024$) • Health-related problem solving (impulsive subscale): $r = -.521$, $p = 0.0015$ • Quality of life (physical subscale): $r = -.44$, $p < .05$ • Quality of life (environment subscale): $r = -.566$, $p = 0.009$

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Domino 2014 (USA)[68]	≥ 2 of the following: asthma, COPD, diabetes, hypertension, hyperlipidaemia, seizure disorder, depression, schizophrenia	Adherence in medical index condition: • Number of comorbid medical conditions • Mental health conditions Adherence in mental health condition: • Number of comorbidities	PDC (results show percentage point change (standard error) in adherence associated with clinical diagnosis stated)	Persons with diabetes: • Depression: -0.0427 (0.0030), $p < .01$ • Schizophrenia: 0.0458 (0.0061), $p < .01$ • 2 medical comorbidities: 0.0201 (0.0037), $p < .01$ • ≥ 3 medical comorbidities: 0.0678 (0.0037), $p < .01$ Persons with hyperlipidaemia: • Depression: -0.0200 (0.0025), $p < .01$ • Schizophrenia: 0.0359 (0.0050), $p < .01$ • 2 medical comorbidities: 0.0447 (0.0037), $p < .01$ • ≥ 3 medical comorbidities: 0.0979 (0.0037), $p < .01$ Persons with hypertension: • Depression: -0.0276 (0.0021), $p < .01$ • Schizophrenia: -0.0105 (0.0041), $p < .05$ • 2 medical comorbidities: 0.0365 (0.0023), $p < .01$ • ≥ 3 medical comorbidities: 0.1039 (0.0024), $p < .01$ Persons with depression: • 1 medical comorbidity: 0.0193 (0.0029), $p < .01$ • 2 medical comorbidities: 0.0462 (0.0035), $p < 0.01$ • ≥ 3 medical comorbidities: 0.0894 (0.0037), $p < .01$ Persons with schizophrenia: • 1 medical comorbidity: 0.0362 (0.0067), $p < .01$ • 2 medical comorbidities: 0.0752 (0.0078), $p < 0.01$ • ≥ 3 medical comorbidities: 0.1082 (0.0079), $p < .01$
Eton 2017 (USA)[71]	≥ 2 chronic conditions requiring self-management	• Health provider relational quality (HPRQ)	Single-item self-report; “always” vs “usually/sometimes” taking medications	• Health provider relational quality: $t(294) = 3.79$, $p < .001$ (converted to $r = 0.22$, $p < .001$ for consistency with other studies)

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Ihle 2017 (Switzerland)[94]	≥ 2 chronic conditions	• Prospective memory • Retrospective memory	Omission errors of adherence = 1-item self-report Commission errors of adherence = 1-item self-report	• Omission errors of adherence: prospective memory performance: r = -.29, p < .05 • Commission errors of adherence: retrospective memory performance: r = -.24, p < .05
Kenning 2015 (UK)[106]	Any combination of the following: chronic obstructive pulmonary disease, coronary heart disease, diabetes, osteoarthritis and depression.	• Gender • Age • Deprivation (IMD) • Number of conditions • Health status • Anxiety/depression (HADS) • Illness perceptions – consequences (IPQ) • Illness perceptions – treatment control (IPQ) • Self-help behaviours • Self-monitoring and insight (heiQ) • Baseline Moriskey medication motivation • Baseline Moriskey medication knowledge • Hassles with healthcare • Patient experience of multimorbidity (MULTIPleS)	Modified Moriskey Scale – Medication Motivation at follow-up	• Health status (B = -0.119, SE = 0.051; β= -0.132, p < .05) • Self-monitoring and insight (B = 0.093, SE = 0.045; β = 0.100, p < .05) • Moriskey medication motivation at baseline (B = 0.522, SE = 0.044; β = 0.552, p < .01). • Hassles with health care (B = -0.107, SE = 0.050; β = -0.108, p < .05)
Schüz 2011 (Germany)[169]	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	• Unintentional non-adherence Time 1 (T1) • Intentional non-adherence T1 • Sex • Age • Education • Number of illnesses • Number of medicines • Change in specific necessity (T1 to T2) • Change in specific concerns (T1 to T2) • Change in general harm (T1 to T2) • Change in general overuse (T1 to T2)	Two items from Reported Adherence to Medication Scale; Higher score = higher non-adherence.	Intentional non-adherence as dependent variable: • Intentional non-adherence time 1: B = .58, SE = .09, p < .01 • Change in specific necessity beliefs: B = −.19, SE = .06, p < .01 Unintentional non-adherence as dependent variable: • Unintentional non-adherence time 1: B = .70, SE = .09, p < .01 • Change in general overuse beliefs: B step 3 = .26, SE = .10, p < .01
Schüz 2011 (Germany)[168]	≥ 2 chronic conditions from Charlson Comorbidity Index or	• Unintentional non-adherence Time 1 • Intentional non-adherence Time 1 • Sex	Two items from Reported Adherence to Medication Scale;	Unintentional non-adherence as dependent variable: • Unintentional non-adherence time 1: β step 3 = .29, p < 0.01

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	Functional Comorbidity Index	• Age • Education • Functional health status • Number of medicines • Number of illnesses • General harm beliefs • General overuse beliefs • Specific necessity beliefs • Specific concern beliefs	Higher score = higher non-adherence	• General overuse beliefs: β step 3 = .17, $p < .01$ Intentional non-adherence as dependent variable: • Intentional non-adherence time 1: β = .26, $p < .01$ • Number of medicines: β = -.15, $p < .05$ • Number of illnesses: β = .16, $p < .05$ • General harm beliefs: β = .15, $p < .05$ • Specific necessity beliefs: β = -.15, $p < .05$
Schüz 2014 (Germany)[170]	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	• Illness perceptions (illness consequences, timeline, control)	One item from Reported Adherence to Medication Scale. Higher score = higher adherence.	Main effects and interaction model: • Timeline of illness 2: β = -.18, $p < .05$ • Control of illness 2: β = .21, $p < .05$ Peak model: • No significant predictors Combined model: • Higher control: β = .20, $p < .01$ • More negative consequences: β = .16, $p < .05$
Warner 2013 (Germany)[189]	≥ 2 chronic physical conditions from Charlson Comorbidity Index or Functional Comorbidity Index	• Sex • Age • Education • Number of medicines • Physical fitness • Attitudes towards medication • General received support • Social conflict • Medication specific social support Time 1 • Medication specific social support Time 2 • Medication adherence Time 1 • Medication adherence Time 2	One item from Reported Adherence to Medication Scale assessing unintentional non-adherence. Higher score = higher non-adherence	Step 1: OR (95% CI) • Adherence Time 1 = 5.99 (3.18-11.29), $p < .001$ Step 2: OR (95% CI) • Adherence Time 1 = 5.37 (2.82-10.24), $p < .001$ • Medication specific social support = 0.69 (0.49-0.98), $p < .05$. Step 3: OR (95% CI) • Adherence Time 1 = 5.62 (2.91-10.85), $p < .001$ • Interaction medication support x social conflict = 0.42 (0.17-0.99), $p < .05$

Table 1 Predictors of medication (non-)adherence

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DISCUSSION

Our systematic review identified a heterogeneous literature reporting medication non-adherence in people living with multimorbidity. Variation in study design, population, and measurement and definitions of both multimorbidity and non-adherence was observed between studies. Among studies using an inclusive definition of multimorbidity (k=11), correlates of non-adherence included treatment related beliefs and previous non-adherence. From eight studies considered suitable for meta-analysis, 43% (95% CI 34%-51%) of participants with multimorbidity were non-adherent to medications. Non-adherence ranged from 16.4%[71] to 61.4%[193], with the lowest estimate arising from a single-item self-report measure,[71] and the highest derived from pharmacy data.[193] One study in the meta-analysis contributed to 76% of the sample.[115] These characteristics of individual studies must be considered when interpreting findings of the quantitative synthesis.

In studies reporting non-adherence prevalence for more than one condition, results highlighted disparities in non-adherence between conditions. It was previously reported that people with multimorbidity prioritise certain conditions over others,[204,205] with our results suggesting this prioritisation may be reflected in self-management behaviours. Other factors known to predict adherence in general may also explain disparities within an individual, including differences in regimen complexity and differences in the expected efficacy of medications.[206] Inauen and colleagues[9] have reported that adherence within an individual differs between medications, and emphasise the need to assess adherence for multiple medications in order to focus interventions accordingly. Despite such calls, a recent systematic review suggested there are no standard criteria for measuring multiple medication adherence, as well as no agreement on how existing measures developed for measuring adherence to monotherapy can be adapted to multiple medications.[207] A number of studies reviewed here, including those in the quantitative synthesis, reported a single prevalence estimate representing average non-adherence across conditions, potentially resulting in a concealment of those conditions for which non-adherence is highest. This should be considered when interpreting the meta-analysis.

Among participants with a physical condition and depression, non-adherence was higher for antidepressants in three of four studies. This somewhat supports previous findings that non-adherence to antidepressants is higher than to medications for physical conditions,[208] and extends this to instances where both occur in one individual; however further research is necessary to draw stronger conclusions. Among participants with a psychiatric and physical condition, prevalence of non-adherence to psychiatric medications was higher than non-adherence to medications for physical conditions in three of seven studies. Previous literature has reported that adherence to antipsychotic

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medications may be lower than adherence to medications for physical conditions,[208] while others have found antipsychotic adherence to be higher.[209] These discrepancies further highlight the heterogeneity that is inherent in multimorbidity, as well as the importance of considering between-condition medication adherence within individuals. Moreover, it has previously been reported that the condition people with multimorbidity prioritise most can differ over time as conditions and treatments change.[205] Considering the cyclical nature of some mental health conditions,[210] it is important to consider whether self-management behaviours, such as medication taking, fluctuate over time as a result of changing priorities in response to symptoms.

Among the variables frequently tested as predictors were number of chronic conditions and number of prescribed medications. Surprisingly, the strength and direction of association with non-adherence was not consistent across studies. A recent systematic review concluded that deprescribing interventions do not consistently improve adherence,[211] suggesting an equivocal relationship between medication burden and non-adherence. It has been suggested previously that frequency of medication administration has a greater influence on adherence than number of tablets,[212] which may have particular relevance in multimorbidity considering some people take medications for different conditions throughout the day while others take all of their medications together.[213] Indeed, it has been reported that particular combinations of conditions are associated with higher dosing frequencies in multimorbidity, with potential implications for treatment burden,[213] and therefore possibly influencing medication adherence. Mode of drug delivery may also be an influencing factor, with different modes of delivery varying in convenience and, therefore, adherence.[214] Due to lack of individual patient data regarding treatment regimens, it is beyond the scope of this review to explore this further.

Illness and medication beliefs were among the most frequently explored predictors of adherence. Schüz and colleagues'[170] results suggest medication adherence in the context of multimorbidity may be influenced by perceptions of unique conditions as well as the effect of multiple combined illness perceptions. A scale to measure multimorbidity illness perceptions has been developed,[215] accounting for domains such as treatment burden, emotional impact and prioritisation. When tested within a study reviewed here, this measure did not predict adherence;[106] however, authors note the measure was in early stages of development at the time of administration.[106] Nonetheless, illness perceptions – and how they present in the context of multimorbidity – appear to be of relevance to both intentional and unintentional non-adherence in the context of multimorbidity.[168–170] This tendency to be influenced by illness beliefs may position intentional non-adherence as a

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rational judgement for some people self-managing chronic conditions; how this transpires as selective non-adherence among people self-managing multiple conditions should be explored in future research.[216]

Pharmacy data was the most frequently used method to measure non-adherence, followed by self-report and electronic monitoring. Variation in measurement methods contributed to the heterogeneity that prevented a larger meta-analysis; a limiting factor that has previously been identified in reviews of non-adherence to multiple medications.[217] Keeping in mind variance due to specific combinations of conditions in individual studies, the range of reported non-adherence differed by measurement method, at 76.5% for self-report, 65.6% for pharmacy data and 44.1% for MEMS. Results from ten studies using multiple methods to report prevalence of non-adherence emphasised this variability further, extending to multimorbidity the recommendation to use subjective and objective measures of non-adherence in combination to allow the strengths of each to compensate for the limitations of others.[218]

We did not stratify our analysis of prevalence by age, limiting the interpretation of findings which may be confounded by age-related experiences. It has been noted that there is no consensus on age as a predictor of non-adherence,[219] and age was not found to be a significant predictor of non-adherence in this review. However, it may be plausible that variables are differentially associated with non-adherence at different life stages. For example, Domino and colleagues[40] reported lower levels of non-adherence among those with more conditions, with a mean age among participants of 43 years; Schüz and colleagues found higher non-adherence among participants with more conditions, with a mean age of 73 years.[168] Exploring how non-adherence is variably predicted by clinical and psychosocial influences across the lifespan could inform intervention development which is appropriately targeted according to life stage.

The quality assessment produced a number of recommendations for future research exploring non-adherence among people with multimorbidity. Of note, while almost all studies (94%) were judged to have used non-adherence measures that were reproducible, 16% of studies used methods that were not considered to be valid. The validity of non-adherence assessment should be a priority for future related investigations. Furthermore, almost 30% of studies were judged not to have reduced serious recall bias. It has been suggested that the appropriate period for retrospective self-report of non-adherence should be balanced between optimising recall and capturing instances of infrequent non-adherence.[220] The optimum recall period for measures of multiple medication non-adherence should be determined.

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The methods used to conduct this systematic review have limitations. Firstly, only 20% of titles and abstracts were screened by two reviewers and authors were not contacted for full-text articles or data that was not accessible, both of which increased the risk of excluding relevant studies. Secondly, criteria for inclusion in the review were limited to English language studies produced between 2009 and 2019. While this date range may exclude relevant articles, the review aimed to synthesise evidence produced since the rise of multimorbidity literature over the last decade,[16] as well as a call for the careful design of adherence research in multimorbidity that uses reliable and consistent adherence measurement and details adherence for multiple conditions.[12] Non-English language articles were not reviewed for feasibility reasons. Thirdly, studies of people with depression or anxiety were eligible for inclusion if the condition was identified through a self-report measure. As a result, it is possible that not all participants had a clinical diagnosis. This decision was made to account for the overrepresentation and underdiagnosis of depression among people with chronic conditions due to incomplete recording in medical records,[221] and has been made previously when reviewing the multimorbidity literature.[222] Finally, while the study adopted 'two or more chronic conditions' to define multimorbidity, studies were also included if participants were prescribed medications that could be inferred as treatment for a chronic condition. While this extended the definition of multimorbidity beyond clinical diagnoses, to exclude these studies would have underrepresented the burden that is characteristic of multimorbidity.

CONCLUSION

This review summarises the complexity of non-adherence in multimorbidity, as well as factors that should be considered for the development of interventions to improve adherence for 'the most common chronic condition'. While the review did not identify specific combinations of conditions producing the highest levels of non-adherence, results suggest that non-adherence in multimorbidity varies both between and within studies. Management of multimorbidity demands integration across conditions and clinical guidelines, requiring practitioners and researchers to focus on individual people as opposed to individual diseases.[2] While the move towards this multi-disease focus is welcomed, attempts by researchers and clinicians to improve medication adherence should determine for which conditions individuals require most support. The use of measures designed to assess multiple medication adherence will be central to achieving this aim.

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Data Sharing: All data relevant to the study are included in the article, uploaded as supplementary information or deposited on Open Science Framework: https://osf.io/nym7v/. Data are available under the terms of the Creative Commons Zero "No rights reserved" data waiver (CC0 1.0 Public domain dedication). Our systematic review produced a large amount of information and the arising database is available for future collaboration on additional analyses. Please contact the corresponding author with any inquiries.

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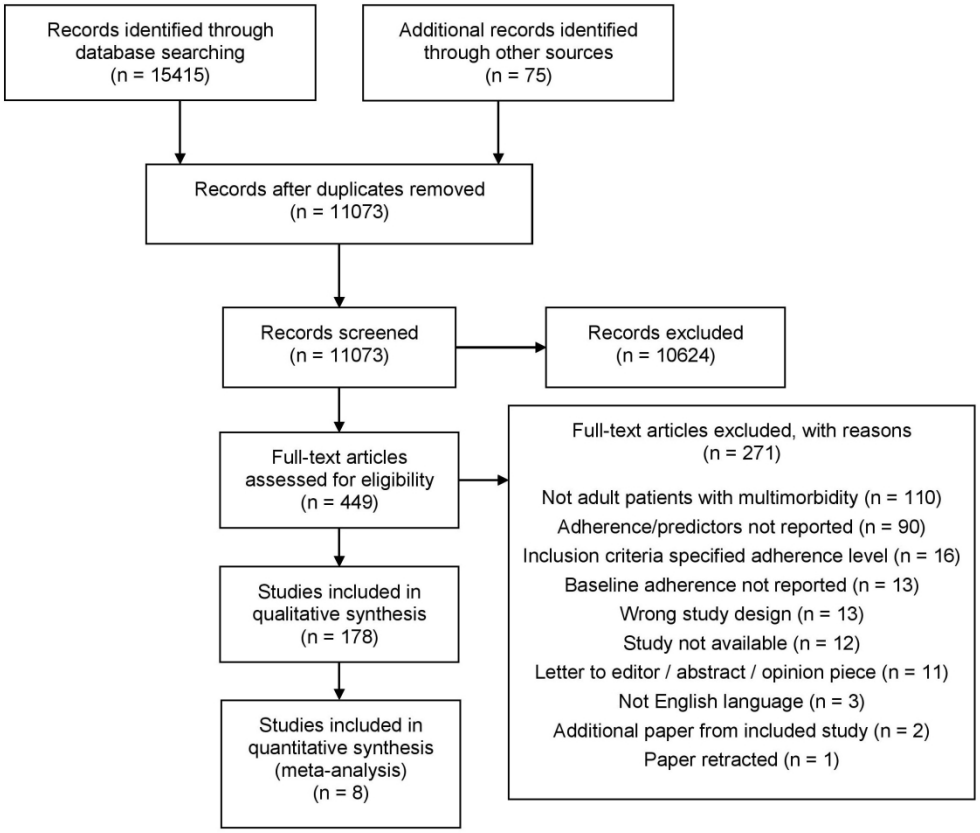


Figure 1. PRISMA flow diagram

199x177mm (300 x 300 DPI)

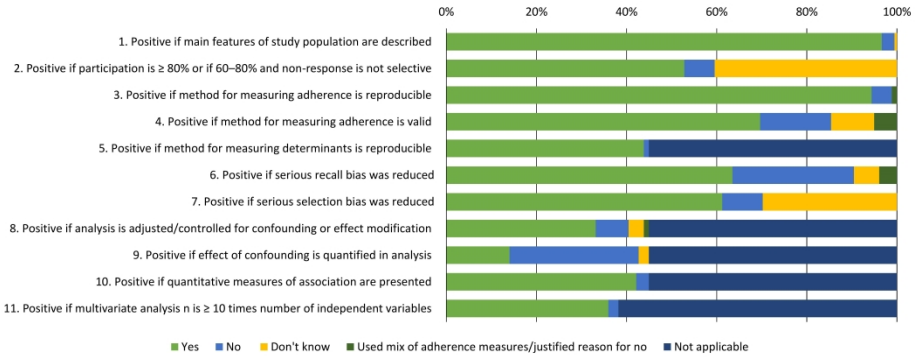


Figure 2. Methodological quality graph

556x218mm (300 x 300 DPI)

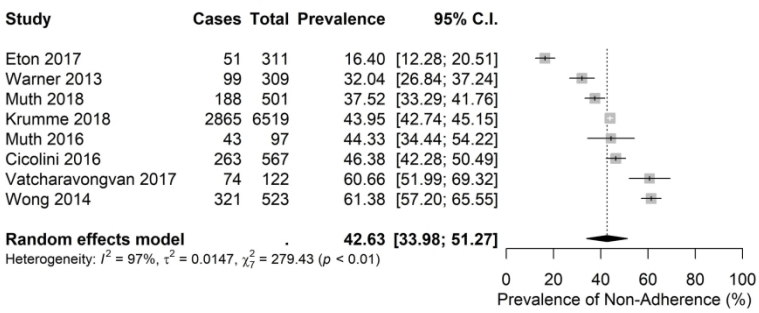


Figure 3. Results of random effects meta-analysis

254x127mm (300 x 300 DPI)

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	(i)
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	(ii)
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	1
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	1
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	2
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	2
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	2
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	2+SM
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	2+3
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	3
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	3
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	3
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	2
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	3

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	n/r
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	n/r
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	4
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	4+SM
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	4
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	5
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	5
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	n/r
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	n/r
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	14-16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	17
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	17

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

SM = supplementary material

n/r = not reported

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Search Strategies

PsycINFO:

1. multimorbid*.ab,ti
2. multi-morbid*.ab,ti
3. multidisease*.ab,ti
4. multi-disease*.ab,ti
5. multiple adj2 (ill* or disease* or condition* or syndrom* or disorder*).ab,ti
6. comorbidity/
7. comorbid*.ab,ti
8. co-morbid*.ab,ti
9. chronic* adj2 (disease* or ill* or care or condition* or disorder* or health* or medication* or syndrom* or symptom*).ab,ti
10. polypharma*.ab,ti
11. or/1-10
12. treatment compliance/
13. (medication adherence).ab,ti
14. adheren*.ab,ti
15. nonadheren*.ab,ti
16. non-adheren*.ab,ti
17. (non adheren*).ab,ti
18. complian*.ab,ti
19. noncomplian*.ab,ti
20. non-complian*.ab,ti
21. (non complian*).ab,ti
22. or/12-21
23. S11 AND S22
24. S23 (2009-present)

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CINAHL:

(MH “comorbidity”)
OR
TI (multimorbid*)
OR
TI (multi-morbid*)
OR
TI (multidisease*)
OR
TI (multi-disease*)
OR
TI (comorbid*)
OR
TI (co-morbid*)
OR
TI (multiple W2 ill*)
OR
TI (multiple W2 disease*)
OR
TI (multiple W2 condition*)
OR
TI (multiple W2 syndrom*)
OR
TI (multiple W2 disorder*)
OR
TI (chronic* W2 disease*)
OR
TI (chronic* W2 ill*)
OR
TI (chronic* W2 care)
OR
TI (chronic* W2 condition*)
OR
TI (chronic* W2 disorder*)
OR
TI (chronic* W2 health*)
OR
TI (chronic* W2 medication*)
OR
TI (chronic* W2 syndrom*)
OR
TI (chronic* W2 symptom*)
OR
TI (polypharma*)
OR
AB (multimorbid*)
OR
AB (multi-morbid*)
OR
AB (multidisease*)
OR
AB (multi-disease*)

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OR
AB (comorbid*)
OR
AB (co-morbid*)
OR
AB (multiple W2 ill*)
OR
AB (multiple W2 disease*)
OR
AB (multiple W2 condition*)
OR
AB (multiple W2 syndrom*)
OR
AB (multiple W2 disorder*)
OR
AB (chronic* W2 disease*)
OR
AB (chronic* W2 ill*)
OR
AB (chronic* W2 care)
OR
AB (chronic* W2 condition*)
OR
AB (chronic* W2 disorder*)
OR
AB (chronic* W2 health*)
OR
AB (chronic* W2 medication*)
OR
AB (chronic* W2 syndrom*)
OR
AB (chronic* W2 symptom*)
OR
AB (polypharma*)

AND

(MH "medication compliance")
OR
TI ("medication adherence")
OR
TI (adheren*)
OR
TI (nonadheren*)
OR
TI (non-adheren*)
OR
TI ("non adheren*")
OR
TI (complian*)
OR
TI (noncomplian*)
OR

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TI (non-complian*)
OR
TI ("non complian*")
OR
AB ("medication adherence")
OR
AB (adheren*)
OR
AB (nonadheren*)
OR
AB (non-adheren*)
OR
AB ("non adheren*")
OR
AB (complian*)
OR
AB (noncomplian*)
OR
AB (non-complian*)
OR
AB ("non complian*")

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EMBASE:

'multiple chronic conditions'/exp
OR
multimorbid*:ab,ti
OR
multi-morbid*:ab,it
OR
multidisease*:ab,ti
OR
multi-disease*:ab,ti
OR
((multiple) NEXT/2 (ill* OR disease* OR condition* OR syndrom* OR disorder*)):ab,ti
OR
comorbidity/exp
OR
comorbid*:ab,ti
OR
co-morbid*:ab,ti
OR
((chronic) NEXT/2 (disease* OR ill* OR care OR condition* OR disorder* OR health* OR medication* OR syndrom* OR symptom*)):ab,ti
OR
polypharma*:ab,ti

AND

medication compliance/exp
OR
‘medication adherence’:ab,ti
OR
adheren*:ab,ti
OR
nonadheren*:ab,ti
OR
non-adheren*:ab,ti
OR
'non adheren*':ab,ti
OR
complian*:ab,ti
OR
noncomplian*:ab,ti
OR
non-complian*:ab,ti
OR
'non complian*':ab,ti

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PubMed:

multimorbidity[MeSH]
OR
multimorbid*[Title/Abstract]
OR
multi-morbid*[Title/Abstract]
OR
multidisease*[Title/Abstract]
OR
multi-disease*[Title/Abstract]
OR
comorbidity[MeSH]
OR
comorbid*[Title/Abstract]
OR
co-morbid*[Title/Abstract]
OR
"chronic medication*"[Title/Abstract]
OR
polypharma*[Title/Abstract]

AND

"medication adherence"[MeSH]
OR
adheren*[Title/Abstract]
OR
nonadheren*[Title/Abstract]
OR
non-adheren*[Title/Abstract]
OR
"non adheren*"[Title/Abstract]
OR
complian*[Title/Abstract]
OR
noncomplian*[Title/Abstract]
OR
non-complian*[Title/Abstract]
OR
"non complian*"[Title/Abstract]

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Data Extraction Form

date form completed (dd/mm/yyyy)
ID of person extracting data (initials)
study ID (author and year)
country of publication
citation (APA format)
study aims
study design
study setting
chronic conditions studied
sample size
age of participants (mean (sd))
gender of participants (% female)
definition of multimorbidity (if applicable)
measure of medication non-adherence
definition of medication non-adherence
proportion of non-adherent participants (or mean (sd) if proportion not reported)
predictors of medication non-adherence (if reported)
key conclusions by study authors

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Meta-analysis of Proportions: R Script

```
library(metafor)
library(meta)
#read in the data file
dat=read.csv("data.csv", header=T, sep=",")
#set-up file without transformation
ies=escalc(xi=cases, ni=total, data=dat, measure = "PR")
#meta-analysis with DerSimonian Laird
pes=rma(yi, vi, data=ies, method = "DL")
#view results
print(pes)
#compute and display confidence intervals of tau-squared and I-squared
confint(pes)
#create simple forest plot
pes.summary=metaprop(cases, total, authoryear, data=dat, sm="PRAW",
method.tau="DL", method.ci="NAsm")
forest(pes.summary, xlim=c(0,100),pscale=100, rightcols = FALSE,
leftcols=c("studlab", "event", "n", "effect", "ci"), leftlabs=c("Study", "Cases",
"Total", "Prevalence", "95% CI.I"), xlab="Prevalence of Non-Adherence", smlab="",
weight.study="random", squaresize=0.5, col.sqaure="black", col.squar.lines="black",
col.diamond="black", col.diamond.lines="black", pooled.totals=FALSE,
comb.fixed=FALSE, fs.hetstat=10, print.tau2=TRUE, print.Q=TRUE, print.pval.Q=TRUE,
print.I2=TRUE, digits=2)
#leave one out to identify outliers
L10=leave1out(pes); print(L10)
#visualise change in summary score with forest plot
l1o=leave1out(pes)
yi=l1o$ estimate; vi=l1o$se^2
forest(yi, vi, slab=paste(dat$author, dat$year, sep=","), refline=pes$b, xlab =
"Summary proportions leaving out each study")
#see which studies are influential using leave-one-out diagnostics
inf=influence(pes)
print(inf)
plot(inf)
```

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Study Characteristics Table

Abbreviations:

COPD = chronic obstructive pulmonary disease
HIV/AIDS = human immunodeficiency virus/acquired immunodeficiency disease
TB = tuberculosis
ACE-I = angiotensin-converting-enzyme inhibitors
HAART = highly active anti-retroviral therapy
ART = antiretroviral therapy
MM = multimorbidity
PDC = proportion of days covered
MPR = medication possession ration
CMG = cumulative mean gap interval
MEMS = medication event monitoring system
VAS = visual analogue scale
MMAS = Morisky Medication Adherence Scale [1,2]
MARS = Medication Adherence Rating Scale[3]
CASE Index = Center for Adherence Support Evaluation [4]
HADS = Hospital Anxiety and Depression Scale[5]
BDI = Beck Depression Inventory [6]
PHQ = Patient Health Questionnaire [7]
MINI = Mini International Neuropsychiatric Interview [8]
CES-D = Centres for Epidemiologic Studies Depression Scale[9]
GDS = Geriatric Depression Scale [10]
SD = standard deviation
IQR = inter quartile range
95% CI = 95% confidence interval
RCT = randomised controlled trial

Note 1: where *n* non-adherent was not reported, *n* was calculated from the reported % non-adherent.
Note 2: where *n* female was not reported, *n* was calculated from the reported % female/male.
Note 3: pooled mean (SD) age was calculated where mean (SD) age was reported for two groups.
Note 4: participants with depression had a clinical diagnosis unless a self-report measure is noted.

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Study (country)	Setting	Design	Chronic Conditions	Sample Size	Adherence Measure	Non-adherence definition	N (%) non-adherent/adherence outcome reported
Agarwal 2009 (USA)[11]	Healthcare database	Retrospective cohort	- Treated with antihypertensive - Treated with lipid-lowering agent	15400	MPR (antihypertensive (AH); lipid-lowering (LL) agent)	MPR < 0.8 for one or more medications	2 months, 6 months, 12 months: Cohort 1 (AH before LL): 40.6%, 67.3%, and 68.7% Cohort 2 (LL before AH): 55.0%, 69.2%, and 69.0% Cohort 3 (AH/LL simultaneously): 24.8%, 65.6%%, and 66.0%
Albrecht 2017 (USA)[12]	Medicare database	Retrospective cohort	- COPD - Depression	16075	PDC (COPD maintenance medications; antidepressants)	PDC < 0.8 (also report PDC = 0)	Person months: COPD: 78.8% Antidepressants: 45.2%
Albrecht 2016 (USA)[13]	Medicare database	Retrospective cohort	- COPD - Depression	31033; MM = 9593	PDC (COPD maintenance medications)	PDC < 0.8 (also report other categories)	7674 (80%)
An 2013 (USA)[14]	Administrative claims and electronic medical records	Retrospective observational	- Hypertension - Type 2 diabetes	2334	PDC (diabetes medications and hypertension medications simultaneously)	PDC < 0.8 for both drugs simultaneously	1671 (71.6%)
Applebaum 2009 (USA)[15]	HIV/AIDS primary care	Baseline data from longitudinal study	- HIV - Substance related disorder - Psychiatric disorder	67	Medication Adherence Questionnaire	Adherence < 95%	20 (29.85%)
Applebaum 2009 (USA)[16]	Healthcare clinics	Cross-sectional (data from RCT screening)	- HIV - History of injection drug use/dependence	108	MEMS (HAART); AIDS Clinical Trials Group Adherence Questionnaire (HAART)	Report mean	Mean MEMS = 68.21% Mean self-report = 93.82%
Axon 2016 (USA)[17]	Veterans health administrative databases	Retrospective cohort	- Type 2 diabetes - Depression	740197; MM = 88895	MPR (diabetes medications)	MPR < 0.8	28143 (31.7%)
Baca-Garcia 2009 (USA)[18]	University hospital	Cross-sectional	- Current bipolar major depressive episode - Alcohol use disorder	97; MM = 39; analysed n = 36	Self-report (antidepressants)	Report mean (SD) median global compliance	Mean (SD) = 90.7 (14.0)
Badiee 2012 (USA)[19]	Ongoing studies at university research centre	Cross-sectional	- HIV - Bipolar disorder I or II	74; MM = 43	MEMS (protease inhibitor or related) AIDS Clinical Trials Group (ACTG) 4-day Adherence to	MEMS < 90% ACTG < 95%	MEMS: 22 (51%) ACTG: 6 (13%)

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					Antiretrovirals Questionnaire (antiretroviral medications) Visual Analogue Scale (protease inhibitor)	Visual Analogue Scale < 95%	Visual Analogue Scale: 22 (52%)
Baik 2012 (USA)[20]	Medicare database	Retrospective cohort	- Depression - Heart Failure	552956; MM = 22118; analysed n = 13981	MPR (heart failure medications; antidepressants)	Report mean	No coverage group: Antidepressants: Mean MPR = 0.56 Heart failure medications: Mean MPR = 0.89 Generic coverage group: Antidepressants: Mean MPR = 0.60 Heart failure medications: Mean MPR = 0.90
Banegas 2018 (USA)[21]	Healthcare database	Retrospective cohort	- Treated with statin - Newly diagnosed cancer (breast / prostate / colorectal)	10177	PDC (statins)	Partial adherence = PDC 0.20-0.79 Non-adherence = PDC < 0.2	Year 1: PDC < 0.20 = 837 (8.2%) PDC 0.20-0.79 = 2280 (22.4%) Year 2: PDC < 0.20 = 887 (8.7%) PDC 0.20-0.79 = 2155 (21.2%)
Basak 2014 (USA)[22]	Healthcare database	Longitudinal observational	- Type 2 diabetes - Hypertension and/or dyslipidaemia or angina	6043	PDC	Report mean (SD)	Report according to medication combinations per quarter. Range = 24.43 (39.10) for antianginal medications to 86.40 (23.27) for hypertension medications.
Batchelder 2014 (USA)[23]	Community and hospital	Cross-sectional	- HIV - Type 2 diabetes	62	VAS (antiretrovirals; diabetes medications; insulin injections where appropriate)	VAS < 100%	ARV: 32 (51.6%) Antidiabetes: 39 (63%) Of 32 participants reporting <100 % to HIV medication, 97 % (31) reported < 100 % to diabetes medication
Bauer 2013 (USA)[24]	Healthcare database	Observational new prescription cohort follow-up	- Type 2 diabetes - Depression	1366	Pharmacy utilisation data (antidepressants)	Primary non-adherence: no dispensing of index prescription within 60 days of index prescription date. Non-persistence: dispensing of index	Primary non-adherence: 52 (3.8%) Early non-persistent: 432/1314 (32.9%) 2nd stage non-persistent: 83/1314 (6.3%)

					prescription once, no additional dispensing within period defined by number of days' supply of medication dispensed plus a grace period equal to 100% of prior days' supply.	
Beadles 2015 (USA)[25]	Medicaid database	Retrospective cohort	≥ 2 of the following: major depressive disorder, schizophrenia, diabetes mellitus, hypertension, hyperlipidaemia, seizure disorder, asthma, chronic obstructive pulmonary disease	depression = 9303; hypertension = 12595; diabetes = 6409; hyperlipidaemia = 9263	PDC (target medication)	Report mean (SD)	Mean (SD): Depression = 40% (43%) Hypertension = 50% (43%) Diabetes = 49% (43%) Hyperlipidaemia = 47% (43%)
Benner 2009 (USA)[26]	Healthcare database	Retrospective claims analysis	- Treated with antihypertensives - Treated with lipid-lowering therapy	5759	PDC (for combination therapy, reported according to prescription burden)	PDC < 80%	By number of other medications: 0 = 59.0% 1 = 65.5% 2 = 69.9% 8 = 79.9% 9 = 74.5% 10/+ = 79.9%
Berg 2018 (Denmark)[27]	Hospital	Cross-sectional	- Cardiac Disease - Depression/anxiety (HADS)	14239; MM (anxiety) = 1978; MM (depression) = 982	Single item self-report (cardiac medication)	Forgetting to take prescribed medication more than once a week among patients who took medication prior to hospital admission	Anxiety group = 113 (7%) Depression group = 70 (8%)
Bifftu 2015 (Ethiopia)[28]	Hospital outpatient	Cross-sectional	- Epilepsy - Depression (BDI-II)	405; MM = 183	Single item self-report (antiepileptic medications)	Not defined (reported as number with difficulties of adherence to antiepileptic drugs)	81 (44.3%)

Blashill 2015 (USA)[29]	HIV primary care clinics	Baseline data of RCT screening	- HIV - Depressive symptoms (PHQ-2)	333	MEMS (antiretrovirals)	MEMS < 80%	191 (57.4%)
Bogner 2014 (USA)[30]	Primary care	Longitudinal	- Type 2 diabetes - Depression	180	MEMS (hypoglycaemic agents and antidepressants) Brief Medication Questionnaire (BMQ) (hypoglycaemic agents and antidepressants)	MEMS < 80% BMQ < 80%	Oral hypoglycaemic agent: MEMS = 103 (57.22%) Self-report = 34 (18.89%) Antidepressants: MEMS = 111 (61.67%) Self-report = 47 (26.11%)
Cabrera-Marroquin 2014 (Mexico)[31]	Tertiary care (ambulatory)	Observational cross-sectional	- Rheumatoid arthritis; - Major depressive disorder	78; MM = 24	Concordance with Therapy self-report measure of adherence and persistence (RA medication)	Based on combinations of item scores	10 (41.7%)
Calderón-Larrañaga 2016 (Spain)[32]	Primary care	Cross-sectional	-Hypertension -Physical and/or mental comorbidity	113,397; MM with physical co-morbidity = 103913; MM with mental comorbidity = 20349	MPR (antihypertensives)	MPR < 80%	+ physical comorbidity = 20868 (20.08%) + mental comorbidity = 4387 (21.56%)
Calip 2013 (USA)[33]	Healthcare database	Retrospective cohort	- Breast cancer survivors - Treated with statins	1393	Adherence: MPR (statins) Persistence: discontinuation rate (statins)	MPR < 0.8 Discontinuation rate (DR): gap ≥ 90 days between end of previous days' supply and subsequent dispensing of statin medication	Cancer treatment period: MPR < 0.8 = 666/1384 (48.1%) Mean (SD) DR = 0.5 (0.6) Year +1: MPR < 0.8 = 572/1129 (50.7%) Mean (SD) DR = 1.2 (1.3) Year +2: MPR < 0.8 = 672/1040 (64.6%) Mean (SD) DR = 1.2 (1.2) Year +3: MPR < 0.8 = 477/959 (49.7%) Mean (SD) DR = 1.2 (1.2)
Calip 2015 (USA)[34]	Healthcare database	Retrospective cohort	- Diabetes mellitus - Breast cancer	509; analysed n = 499	MPR (oral diabetes medications)	MPR < 0.8	376 (75.4%)

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Calip 2017 (USA)[35]	Healthcare database	Retrospective cohort	- Breast cancer survivors - Treated with at least one of antihypertensives, oral antidiabetes medications or statins	2308	MPR (antihypertensive or statins or oral antidiabetic medications)	MPR < 0.8	Antihypertensives = 666/1779 (37.4%) Oral diabetes medications = 379/499 (76.0%) Statins = 416/1072 (38.8%)
Cantudo-Cuenca 2014 (Spain)[36]	Hospital pharmacy service	Prospective observational	- HIV - Comorbidity(s)	594; MM = 431	MMAS-4 Pharmacy dispensing records: Number of medications dispensed/number of medications prescribed x 100	MMAS-4 < 4 Pharmacy data < 90%	129 (29.9%) (results show pharmacy data; authors reported no difference in adherence between both measures)
Casaletto 2016 (USA)[37]	Not reported	Cross-sectional	- HIV - Bipolar I or II	50	MEMS (bipolar medications)	MEMS < 90%	28 (56.0%)
Caughey 2013 (Australia)[38]	Veterans' Affairs database	Retrospective cohort	- Diabetes - Depression	29710; MM = 7171	Pharmacy dispensing data – treatment discontinuation (antidiabetic medications)	Gap between two dispensings of more than three times the duration	2868 (40%) Median (95% CI) = 1.81 (1.72-1.94)
Cha 2012 (USA)[39]	Not reported	Cross-sectional using baseline data from intervention	- Heart failure - Diabetes mellitus	116; MM = 46	MMAS-8 (heart failure medications) MEMS (ACE inhibitor and diuretic)	Report mean (SD): MMAS-8 range = 1-12; higher score = more adherent	MMAS-8 mean (SD) = 10.6 (1.6) MEMS: correct doses: ACE-I = 88.0 (16.6) Diuretics = 89.0 (17.0) MEMS: correct days: ACE-I = 81.7 (22.8) Diuretics = 81.1 (22.6)
Chou 2017 (USA)[40]	Healthcare database	Retrospective cohort	- Depression - Breast cancer (newly diagnosed)	2284; MM = 1142	PDC (antidepressants)	PDC < 80%	494 (43.3%)
Cicolini 2016 (Italy)[41]	Primary care	Cross-sectional	≥ 2 chronic condition from cardiovascular, neurological, respiratory, musculoskeletal, oncologic diseases or diabetes mellitus	567	MMAS-8 (Italian Version)	MMAS-8 score between 1-5.75 (defines "low adherence")	263 (46.4%)

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Crowley 2013 (USA)[42]	Community practice	Cross-sectional (baseline data from RCT)	- Type 2 diabetes - Hypertension	377	MMAS-4 (hypertension medications; diabetes medications)	Score ≤ 2 on any item	Hypertension = 110 (29.2%) Diabetes = 162 (43.0%)
Crowley 2015 (USA)[43]	Tertiary care health system	Cross-sectional (baseline data from RCT)	- Acute myocardial infarction in previous 3 years - Hypertension	405	MMAS-4 (general adherence)	Response of "strongly agree," "agree," "don't know," or "refused" for any statements	173 (42.7%)
Cruess 2012 (USA)[44]	Physician outpatient clinics	Prospective cohort	- HIV - Treated with psychotropic medication	324; MM = 106	Monthly unannounced pill counts (HIV medication; psychiatric medication)	Report mean (SD)	HIV = 84.09% (19.26%) Psychiatric = 79.86% (20.88%)
Cziraky 2015 (USA)[45]	Healthcare database	Retrospective cohort	- Acute coronary syndrome or unstable angina - Type 2 diabetes	Year 1 = 85097; MM = 22588 Year 3 = 35540; MM = 9160	MPR (cardiovascular medications) Treatment discontinuation (cardiovascular medications)	MPR: not reported Treatment discontinuation: gap in treatment of 45 days or longer in each medication based on days of supply	Oral diabetes medication: Year 1 = 49.3% Year 3 = 46.8% Insulin Year 1 = 77.4% Year 3 = 78.9% Additional medications reported in original paper.
Daniali 2018 (Iran)[46]	Government healthcare centres	Cross-sectional	- Chronic condition - Depression (PHQ-9)	483; MM = 79	4-item self-report	Not reported	40 (50.6%)
Daniels 2016 (USA)[47]	Primary care	Pilot intervention	≥ 2 chronic conditions	20	MMAS-8	Report mean (SD) MMAS-8 score > 2 (defined as "low adherence"); lower score = lower non-adherence	Mean (SD) = 4.65 (2.37)
Davies 2015 (USA)[48]	Tertiary care Veterans Affairs Medical Centre (patients receiving primary care)	Retrospective chart review	- HIV/AIDS - Diabetes mellitus	56	MPR (ART)	MPR < 95%	34 (60.7%)
Davis 2011 (USA)[49]	Clinical centres	Data from RCT	- Hypertension	1065	Not reported (antihypertensives)	Report mean (SD)	Lower BP goal group: Achieved goal: 92.6% (7.7%)

			- Chronic kidney disease				Above goal: 83.0% (14.7%) Usual BP goal group: At goal: 91.4% (8.8%) Below goal: 90.8% (13.2%) Above goal: 78.5% (19.1%)
DeLeon 2015 (USA)[50]	Primary care	Retrospective cohort	- Diabetes - Hypertension	20085; MM = 3827	MPR (diabetes medications, hypertension medications, lipid controlling medications)	MPR < 80%	Diabetes: 1056/1679 (62.9%) Hypertension: 1084/1923 (56.4%) Lipid controlling: 928/1295 (71.7%)
Desai 2014 (USA)[51]	Medicaid database	Retrospective analysis	- Treatment by antipsychotic - Treatment by antidiabetic medication	3642; MM = 1821	PDC (oral antidiabetics; oral antipsychotics) Persistence (oral antidiabetics)	PDC < 0.8 Persistence: number of days of continuous access to oral antidiabetics, allowing for a 30-day gap period.	Antidiabetics: 1143 (62.8%) Mean (SD) persistence = 180.8 (124.8) Antipsychotics: 689 (37.8%)
Dhamane 2017 (USA)[52]	Healthcare database	Retrospective cohort	- COPD - Treatment by ≥ 1 of antianxiety, anticoagulants, antidepressants, antihyperglycaemics, antihypertensives, beta-blockers, bisphosphonates, calcium channel blockers, diuretics, insulin, NSAIDs, statins	14117	PDC (COPD medications; non-COPD medications)	PDC < 0.8 Report mean PDC for non-COPD medications	COPD medications: 11177 (79.2%) non-COPD medications mean: antianxiety = 0.39 anticoagulants = 0.55 antidepressants = 0.65 antihyperglycaemics = 0.70 antihypertensives = 0.71 beta-blockers = 0.70 bisphosphonates = 0.54 calcium channel blockers = 0.68 diuretics = 0.65 insulin = 0.53 NSAIDs = 0.33 statins = 0.67
Domino 2014 (USA)[53]	Administrative claims databases	Retrospective secondary data analysis	≥ 2 chronic conditions from asthma, COPD, diabetes, hypertension, hyperlipidaemia, seizure disorder, depression, schizophrenia	188531	PDC	Report mean PDC	Diabetes = 0.29 Hyperlipidaemia = 0.29 Hypertension = 0.43 Depression = 0.33 Schizophrenia = 0.46

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Druss 2010 (USA)[54]	Community mental health centre	Pilot RCT	- ≥ 1 chronic medical condition - Severe mental illness	80	MMAS-4 (chronic medical condition)	Report mean (SD) score from 0 (no problems with adherence) to 4 (more problems with adherence)	Mean (SD) = 1.5 (1.3)
DuChane 2014 (USA)[55]	Pharmacy claims database	Retrospective cohort	- HIV - Comorbidity (treated with angiotensin-converting enzyme (ACE) or angiotensin receptor blocker (ARB) or statin)	ACE/ARB group = 2968; MM = 1484; Statins group = 2744; MM = 1372	PDC	ART medications: PDC < 95% Other medications: PDC < 80%	HIV + ACE/ARBs = 1118 (37.7%) HIV + statins = 936 (34.1%) Mean (SD) ART = 83.9 (24.3)
Eton 2017 (USA)[56]	Primary care	Cross-sectional	≥ 2 chronic conditions requiring self-management	332 (analysed n = 311)	Single-item self-report	Response "usually /sometimes" vs "always" re adherence to physician-recommended medications	Usually/sometimes = 51 (16.4%)
Farley 2012 (USA)[57]	Medicaid database	Retrospective cohort	- Schizophrenia - Pre-existing cardiometabolic condition(s): hypertension, hyperlipidaemia and/or diabetes	1006	PDC (antipsychotic medications; cardiometabolic medications)	PDC < 80% (unweighted mean of 3 cardiometabolic classes) Report mean (SD)	Antipsychotic (acute phase) = 89 (18.8%) Diabetes medications: acute phase = 0.90; continuation phase = 0.63; maintenance phase = 0.59 Antihypertensives: acute phase = 0.86; continuation phase = 0.71; maintenance phase = 0.66 Statins: acute phase = 0.79; continuation phase = 0.65; maintenance phase = 0.57
Feldman 2016 (USA)[58]	Not reported (recruited from outpatients, emergency room, referrals and flyers)	RCT	- Panic disorder - Asthma	48	MARS (inhaled corticosteroids)	Report mean (SD) Range = 1-5; higher score = lower non-adherence. Score 4.5+ = good adherence.	Mean (SD) = 3.8 (0.4)

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Ferguson 2017 (Australia)[59]	Cardiology ward at academic medical centre	Prospective cohort	- Chronic heart failure; - Atrial fibrillation	137; analysed n = 83	MMAS-4	≥ 1 responses to non-adherence items	36 (43.4%)
Foguet-Boreu 2017 (Spain)[60]	Primary care	Cross-sectional	- Hypertension - Diabetes type 1 or type 2	27637	MPR (hypoglycaemic medications; antihypertensive medications; both combined)	MPR < 80% Report MPR ≥ 80%	Diabetes = 10898 (39.8%) Hypertension = 4322 (23.8%) Combined diseases (MPR < 80% for at least one condition) = 13593 (49.2%)
Gathright 2017 (USA)[61]	Hospital	Secondary analysis of data from longitudinal observational study	- Heart Failure - Depressive Symptoms (PHQ-9)	308; MM = 107	MEMS (up to 4 common heart failure medications)	Report mean (SD) percent of days a participant did not follow their prescribed regimen	Mean (SD) = 65.3% (27.5%)
Gaudiano 2017 (USA)[62]	Psychiatric hospital	Cross-sectional (study of baseline data from clinical trial)	- Bipolar I or II or NOS - Drug and/or alcohol use disorder	39	Medication Compliance Questionnaire (mean rating across all psychotropic medications)	Report mean (SD) Range = 0-4, higher score = higher non-adherence.	Mean (SD) = 1.2 (1.2)
Gaynes 2015 (USA)[63]	Infectious disease clinics	Data from RCT	- HIV - Depression (PHQ-9 and major depressive disorder (MINI)	304	Visual Analogue Scale (HIV/AIDS medications)	VAS score ≤ 80%	78 (25.7%) No psychiatric comorbidity = 13/54 (20.1%) ≥ 1 psychiatric comorbidity = 65/250 (26.0%)
Gentil 2015 (Canada)[64]	Community	Data from longitudinal study	- Type 2 diabetes - Depression/anxiety	301; MM = 47	MPR (oral antidiabetic medication)	MPR < 80%	18 (38.3%)
Gibson (USA) 2018[65]	Primary care	Cross-sectional	- Type 2 diabetes - Hypertension	60	Self-report during semi-structured interview	Self-report missing 1 or more days in previous week	Hypertension: 14 (23.3%) Type 2 diabetes: 16 (26.7%)
Gnanasan 2011 (Malaysia)[66]	Hospital outpatients	Feasibility study (action research)	- Diabetes mellitus - TB	35	Medication-related problems identified by clinical pharmacist (TB medications; non-TB medications)	Not reported	TB treatment: 3 (8.6%) Non-TB treatment: 10 (28.6%)
Gonzalez 2011 (USA)[67]	Methadone maintenance treatment	Cross-sectional	- HIV - History of injection drug use	91	MEMS (HAART)	MEMS < 100% doses as prescribed	73 (80.2%)

Grove 2017 (USA)[68]	Medicaid database	Retrospective cohort	- Schizophrenia - ≥ 1 chronic condition from hypertension, hyperlipidaemia, seizure disorder, chronic obstructive pulmonary disease, diabetes, asthma, or major depression	12662	PDC (antipsychotic medications)	Report mean (SD)	Mean (SD) = 45.4% (37.8%)
Gundling 2013 (Germany)[69]	Hospital	Retrospective analysis	- Diabetes mellitus - Cirrhosis	285; MM = 87; analysed n = 69	Information about compliance based on doctors', patients' and relatives' reports (report general adherence)	"e.g. concerning regular intake of medication and following the doctor's advice"	36 (52.2%)
Gupte-Singh 2015 (USA)[70]	Medicaid database	Retrospective analysis	- Asthma - Depression	3626; MM = 226	PDC (asthma controller; antidepressant medications)	Asthma controller: PDC < 50% Antidepressant: PDC < 80%	Asthma controller medications: 140 (62.0%) Antidepressants: 168 (74.3%)
Halava 2014 (Finland)[71]	Data from public sector employees	Prospective cohort	- Prescribed statins - Cardiovascular comorbidities	9285; MM = 2827	PDC (statins)	PDC < 80%	1155 (40.9%)
Han 2018 (USA)[72]	Medicare database	Retrospective cohort	- Hypertension - Chronic kidney disease	70201	PDC (antihypertensive medications)	PDC < 80%	27468 (39.1%)
Hansen 2009 (USA)[73]	Ambulatory care practice	Secondary analysis of RCT data	- Heart failure - Depression (GDS)	314; MM = 117	MEMS (heart failure medication) MMAS-4 + one-item screening questionnaire (heart failure medication)	Report mean (95% CI) MMAS-4: Range = 0-1; higher score = better adherence	MMAS-4 = 79% (71-87%) MEMS not reported
Hansen 2018 (USA)[74]	Medicaid database	Retrospective cohort	- Schizophrenia - Cardiometabolic condition (hypertension, hyperlipidaemia and/or diabetes)	4223	PDC (antihypertensive medications; statins; oral antidiabetics; antipsychotic medications)	PDC < 80%	Range of non-adherence reported: Antipsychotic = 24.4% - 44.5% Antidiabetic = 39.4% - 61.5% Antihypertensive = 32.5% - 52.6% Statins = 44.5 – 63.5% Full results reported according to number of primary care prescribers

							and psychiatric specialist prescribers in original paper.
Hansen 2012 (USA)[75]	Medicaid database	Cross-sectional time-series	- Schizophrenia - Cardiometabolic condition (hypertension, hyperlipidaemia and/or diabetes)	33230	PDC (oral second-generation antipsychotics; oral antidiabetics; statins; antihypertensive medications)	PDC < 0.8	Antipsychotic: 11565/33230 (34.8%) Antihypertensives: 10069/24843 (40.5%) Antidiabetics: 4869/12349 (39.4%) Antihyperlipidemics: 4192/10909 (38.4%)
Hansen 2014 (USA)[76]	Veterans' Administration Medical Centre	Retrospective cohort	- Cardiometabolic condition (hypertension, hyperlipidaemia and/or diabetes) - ≥ 1 additional chronic condition(s)	7933; Diabetes MM = 1417; Hypertension MM = 3318; Dyslipidaemia MM = 3009	CMG (diabetes medications; hypertension medications; dyslipidaemia medications)	CMG ≥ 20%	Diabetes (+ ≥ 1 other): 392 (27.7%) Hypertension (+ ≥ 1 other): 766 (23.1%) Dyslipidaemia (+ ≥ 1 other): 948 (31.5%)
Hapangama 2013 (Sri Lanka)[77]	Tertiary care psychiatric unit (inpatient and outpatient)	Cross-sectional	- Mental disorder - Substance use disorder	325; MM = 141	Interview schedule administered by senior registrar in psychiatry	Ceased medication for ≥ 2 days for any reason, without medical advice, during treatment for mental illness	91 (64.5%)
Hawke 2013[78]	Academic medical centres primarily	RCT (secondary data analysis)	- Bipolar I or II - Comorbid anxiety disorder	204; MM = 85	Single-item self-report	Report mean (SD) (possible range not reported)	Mean (SD) = 1.36 (0.53)
Hussein 2010 (USA)[79]	Healthcare database	Retrospective cohort	- Hypertension - Using a two-pill amlodipine + atorvastatin; using a two-pill CCB + statin; or using a single pill amlodipine/ atorvastatin	35430	PDC (statins; calcium channel blocker)	PDC < 80% (both medications together)	Range of non-adherence reported: 39.1 - 73.4% Full results according to status of experienced/naïve with respective medications reported in original paper.
Ihle 2017 (Switzerland)[80]	Primarily hospital	Cross-sectional	≥ 2 chronic conditions	84	Self-report items adapted from Moriskey, Green, Levine measure (specifically, assesses	Report mean (SD) from 1-5 (higher score = higher non-adherence)	Omission errors: mean (SD) = 1.39 (0.88)

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					omission errors and commission errors)		Commission errors: mean (SD) = 1.12 (0.57)
Inauen 2017 (Switzerland) [81]	University hospital (outpatient) and other channels, e.g. online, pharmacy	3-wave panel study	≥ 2 chronic conditions and ≥ 2 prescribed medications	84	Non-Adherence Report, Medication Based (NAR-M) Non-Adherence Report, Diary (NARD) MARS MEMS	Report mean (SD) (lower mean = higher non-adherence)	NAR-M: T1 = 92.8 (11.1); T2 = 92.1 (12.4); T3 = 94.5 (10.5) NARM-D (no MEMS group) = 96.9 (7.9) MEMS group = 98.5 (4.1) MARS T1 = 89.4 (18.3) T2 = 90.1 (18.1) T3 = 94.5 (13.3) MEMS T1-T2 = 95.7 (3.8) T2 – T3 = 93.7 (10.4)
Iyengar 2016 (USA)[82]	Medicare database	Retrospective claims analysis	- Diabetes - Hypertension - High blood cholesterol	40632	PDC (diabetes medications, cholesterol medications, hypertension medications)	PDC < 80%	Antidiabetics = 13898 (34.2%) Antihypertensives = 13462 (33.1%) Antihyperlipidemics = 10584 (26.1%)
James 2018 (USA)[83]	Hospital	Longitudinal	- Type 2 diabetes - Early stage cancer (recently diagnosed)	56	MMAS-8 (diabetes medications)	MMAS: positive response to ≥ 2 questions	Baseline: 25 (44.6%); 3-months: 20 (35%)
Jaworski 2011 (France)[84]	University hospital psychiatric department (outpatient)	Cross-sectional	- Schizophrenia or bipolar disorder - Substance use disorder	80; MM = 40: schizophrenia + SUD= 20; bipolar + SUD = 20	Self-report number of times forgot to take medication in preceding month (psychotropic medications)	Score < 75%	Bipolar + SUD = 15 (75%); Schizophrenia + SUD = 14 (70%)
Jing 2011 (USA)[85]	Medicare and healthcare databases	Retrospective analysis	- Treatment with antihypertensive - Treatment with antidiabetic	30132	MPR	MPR < 77% for ≥ 1 antihypertensives and/or antidiabetic medications for each standalone calendar quarter	10722 (35.6%)
Joseph 2017 (India [86]	Medical college hospital & dialysis centre	Prospective observational	- Hypertension - Chronic kidney disease	120	MMAS-8 (antihypertensives)	Not reported	75 (62.5%)

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	(inpatient and outpatient)						
Kacanek 2010 (USA)[87]	Physician network, newspapers, radio advertisement	Secondary analysis of data from longitudinal cohort	- HIV - Depression symptoms (CES-D and Diagnostic Interview Schedule)	225; MM= 51	For each medication, participants reported the number of days over the last seven days they had missed a dose of that antiretroviral medication.	Missing at least 5% of HAART doses in the seven days prior to study	23 (45.1%)
Kalichman 2015 (USA)[88]	AIDS community services and infectious disease clinics	Prospective	- HIV - Mental health condition	123	Unannounced pill-counts (antiretroviral medications; psychiatric medications)	Antiretrovirals: < 85% Psychiatric medications: report mean (SD)	ART: 57 (46.3%) ART mean (SD) = 77.2 (25.3) Psychiatric mean (SD) = 64.6 (36.4)
Kamal 2018 (Switzerland)[89]	University hospital infectious disease service	Prospective observational cross-sectional	- HIV - Comorbidity (≥ 1 concomitant oral medication from any therapeutic class for at least 6 months)	105	Swiss HIV Cohort Study Adherence Questionnaire (SHCS-AQ) (ART; co-treatments)	Missing ≥ 1 dose of treatment in previous 4 weeks	ART = 18 (17.1%) Co-treatments = 30 (28.6%)
Katerndahl 2012 (USA)[90]	Primary care	Cross-sectional	- Diabetes - Mild or moderate depression (PHQ-9)	106; MM = 37	Medication subscale from Diabetes Self-Care Activities Measure	Report mean (possible range = 0 – 7; lower score = lower compliance	Mean = 4.89
Katon 2009 (USA)[91]	Primary care	Prospective	- Diabetes type 1 or type 2 - Probably major depression (PHQ-9)	4117; MM = 495	CMG (hyperlipidaemia medications; hypertension medications; hyperglycaemia medications)	CMG ≥ 20%	Only reported in patients with poor glycaemic, blood pressure and lipid control: Poor diabetes control = 47/127 (37.0%) Poor hypertension control = 67/232 (28.9%) Poor lipid control = 17/44 (38.6%)
Kebede 2012 (Ethiopia)[92]	Hospital	Cross-sectional	- HIV/AIDS - TB	24	Self-report tool	Missing a dose for ≥ 1 day per month (ART; TB)	ART: 16 (66.7%) TB: 11 (45.8%)
Kenning 2015 (UK)[93]	General practice	Prospective	≥ 2 from chronic obstructive pulmonary disease, coronary heart disease, diabetes, osteoarthritis and	410	Modified Moriskey Scale (6-items)	Not reported	Not reported

			depression (self-reported and/or HADS).				
Kim 2010 (South Korea)[94]	Pharmacy claims database	Retrospective cohort	- Hypertension - Psychiatric disorder	2454844; MM = 158982	Cumulative Medication Adherence (CMA) (antihypertensives)	CMA < 80%	67297 (42.3%)
Kong 2012 (USA)[95]	Medicaid database	Retrospective cohort	- HIV - Depression	7034; MM = 3286; analysed n = 3202	MPR (antiretrovirals)	Adherence < 90%	2386 (74.5%)
Kostev 2018 (Germany)[96]	General practice	Retrospective cohort	- Type 2 diabetes - Depression	12898; MM = 6449	Persistence (oral antihyperglycaemic medications)	At least 3 months without drug	2096 (32.5%)
Kreyenbuhl 2010 (USA)[97]	Veterans' Affairs administrative database	Retrospective cohort	- Type 2 diabetes - Schizophrenia or schizoaffective disorder	22014; MM = 11454	MPR (oral hyperglycaemic medications; antipsychotic medications)	MPR < 0.8	Oral antihyperglycaemics = 4925 (43%) Antipsychotic medications = 4009 (35%)
Kreyenbuhl 2011 (USA)[98]	Mental health clinics (outpatient)	Cross-sectional	- Type 2 diabetes - Serious mental illness	74; MM = 44	Brief Medication Questionnaire (oral hypoglycaemic medications)	Not taking any hypoglycaemic medications as prescribed over the past week	12 (27%)
Krivoy 2015 (Isreal)[99]	Healthcare database	Retrospective analysis	- Ischaemic heart disease - Depression	63437	MPR with addition of physician prescription data from electronic health record (antidepressants)	MPR < 20% = non-adherent MPR 20% to <50% = poorly adherent MPR 50% - <80% = moderate adherence	Non-adherence = 18212 (28.7%) Poor adherence = 10094 (15.9%) Moderate adherence = 11041 (17.4%) Total < 80% = 39347 (62.03%)
Krivoy 2017 (Israel)[100]	Healthcare database	Retrospective analysis	- Depression - Cerebrovascular accident	32361	MPR plus physician prescription data (antidepressants)	MPR < 20 % = non-adherent MPR 20-50% = poorly adherent MPR 50-80% = moderately adherent	< 20% = 8619 (26.6%) 20-50% = 5108 (15.8%) 50-80% = 5656 (17.5%) Total < 80% = 19383 (59.9%)
Krousel-Wood 2010 (USA)[101]	Managed care organisation	Prospective cohort	- Hypertension - Depression (CEDS)	2180; MM = 284	MMAS-8 (antihypertensives); MPR (antihypertensives)	MMAS-8 score < 6 ("low adherence"); MPR < 0.8	MMAS: 67 (23.6%); MPR = 104 (36.5%)

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Krumme 2018 (USA)[102]	Medicare database	Retrospective cohort	≥ 2 chronic conditions, at least one of which is hypertension, hyperlipidaemia or diabetes	22805; MM = 6519	PDC (cardiovascular medication)	PDC < .80	PDC ≤ 70 = 1303 (20.0%) PDC ≤ 85 = 2865 (44.0%)
Kumar 2009 (USA)[103]	Data from longitudinal study	Cross-sectional	- HIV - Depression	1192; MM = 210	4-item self-report for past week (antiretroviral medication)	Report categories of 100%, 85%, 57-85%, <57.	Adherence < 100% = 88 (41.91%) Adherence < 85% = 67 (31.91%)
Langness 2014 (USA)[104]	Pharmacy records	Retrospective review	- HIV - Treatment with mental health medication and/or blood pressure medication	Blood pressure group = 269; Mental health group = 295	PDC (blood pressure medication; mental health medication)	PDC < 80%	Blood pressure: 92 (34.2%) Mental health: 108 (36.6%)
Lee 2011 (USA)[105]	Medicaid database	Retrospective cohort	- Serious mental illness (schizophrenia or major affective disorder) - HIV	23413; MM = 788	Continuity of medication use measures (ART and psychotropic medication)	Monthly use of any antiretroviral or psychotropic medications (divide total months patients had prescription filled by follow-up months after indexed prescription)	Percent of continuous time on medication: Psychotropic medication: 72.5% ART: 71.6%
Li 2016 (Hong Kong)[106]	Primary care	Cross-sectional	- Hypertension - Co-occurring conditions	2445; MM = 1159	MMAS-8 (Chinese version) (antihypertensives)	MMAS-8 ≤ 6	549 (47.4%) + 1 comorbidity = 339 (47.7%) + ≥ 2 comorbidities = 210 (46.9%)
Lin 2012 (USA)[107]	Primary care	RCT	- Depression (PHQ-9) - Diabetes or heart disease	214	CMG (diabetes/heart disease)	Report mean (SD)	Mean (SD): Oral hypoglycaemic = 0.83 (0.19) Antihypertensive = 0.85 (0.18) Lipid lowering = 0.83 (0.20) Antidepressant = 0.80 (0.21)
LoRe 2013 (USA)[108]	Veterans Affairs (VA) database	Retrospective cohort	- HIV - Hepatitis C virus	333	Sum of days' supply of medication dispensed over 12-week interval divided by number of days between first final fills of that interval (antiviral medications)	Pharmacy data < 70% (also report additional intervals above 70%)	Interferon: Interval 1 = 28/267 (10.5%) Interval 2 = 38/189 (20.1%) Interval 3 = 22-131 (16.8%) Interval 4 = 28-94 (29.8%) Ribavirin: Interval 1 = 46/267 (17.2%)

							Interval 2 = 55/185 (29.7%) Interval 3 = 46/127 (36.2% Interval 4 = 49/91 (53.8%)
Long 2014 (USA)[109]	Veterans Affairs medical facilities (primary care)	Cross-sectional	- Type 2 diabetes - Serious mental illness	363	MPR (diabetes medications including insulin) MMAS-4 (diabetes specific)	MPR: report mean (SD) MMAS-4: report as perfect vs non-perfect adherence	MPR mean (SD) = 0.8 (0.3) MMAS-4 = 178 (49%)
Lunghi 2017 (Canada)[110]	Healthcare database	Retrospective cohort	- Type 2 diabetes - Depression	3106	PDC (antidiabetic medication)	PDC < 90%	1614 (52.0%)
Magidson 2012 (USA)[111]	Substance use treatment centre	Cross-sectional (RCT baseline data)	- HIV/AIDS - Substance use	73	ACTG Adherence to Antiretroviral Medication Questionnaire (ART and other medications)	Report mean adherence over previous 4 days	ART: Yesterday = 100%; 2 days ago = 94.6%; 3 days ago = 97.8% 4 days ago = 97.8% Other medications: Yesterday = 97.3% 2 days ago = 94.9% 3 days ago = 96.5% 4 days ago = 97.4%
Manuel 2011 (USA)[112]	Outpatient community mental health centres	Data from RCT	- Psychotic disorder - Active substance use or dependence	198	Self-report missed medication days in past 6 months (psychiatric medications)	≥ 20% missed medication days	30 (15.2%)
Mathew 2013 (India)[113]	Hospital	Non-randomised non-controlled intervention	- Type 2 diabetes - Comorbidity(s) (receiving ≥ 5 drugs)	240	14-item self-report questionnaire (diabetes medication)	Patients were categorised into adherent and non-adherent (criteria for sorting not reported)	116 (48.3%)
May 2010 (USA)[114]	Hospital	Longitudinal observational	- Coronary artery disease - Depression	585; MM = 73	MPR across multiple refills (continuous measure of adherence (CMA) for lipid-lowering medications)	Report mean CMA	Mean adherence: 6 months = 90.9% 1 year = 89.3% 2 years = 85.7%
McCusker 2016 (Canada)[115]	Primary care	Data from RCT	- ≥ 1 chronic physical condition(s) or chronic pain ≥ 6 months - ≥ mild depressive symptoms (PHQ-9)	215; analysed n = 202	MMAS-4 (any medication)	Report mean (SD) MMAS-4 higher score = higher non-adherence (range = 1-4).	Mean (SD) = 0.7 (0.9)

Mellins 2009 (USA)[116]	Community	Cross-sectional (baseline data from cohort study	- HIV - Psychiatric disorder - Substance use disorder	1138 (n prescribed ARV = 542; n prescribed antipsychotics = 501)	Antiretrovirals: Modified version of AACTG Adherence to Antiretroviral Medication Questionnaire Psychiatric medications: 1-item self-report	Self-report < 100% adherence in past 3 days	ARV = 244 (45.0%) Psychiatric medication = 190 (38%)
Miketic 2012 (USA)[117]	Hospital	Prospective descriptive pilot study	- Coronary artery disease - Diabetes mellitus	51; MM = 22	MMAS-4	Report mean (SD)	Mean (SD) = 0.86 (0.94)
Mira 2014 (Spain)[118]	Health centres	RCT	- Multimorbid taking multiple medications	99	MMAS-4	Report mean (SD)	Mean (SD) = 6.9 (1.1)
Monroe 2018 (USA)[119]	HIV clinic	RCT	- HIV - Diabetes and/or hypertension	46	MPR	Report median (IQR)	ART: 92% (71-100%) Other medications: 79% (63-96%) (if patient had hypertension + diabetes, only hypertension adherence recorded)
Moore 2015 (USA)[120]	Ongoing studies at research centre	RCT	- HIV - Bipolar type I or II	50	Visual analogue scale (HIV medication and psychiatric medication)	VAS < 100% = some adherence difficulty in previous 30 days	HIV medication = 12 (47.8%) Psychiatric medication = 14 (54.3%)
Moore 2012 (USA)[121]	Ongoing studies at research centre	Cross-sectional	- HIV - Bipolar type I or II	44 (n with psychiatric medication assessed = 39)	MEMS (psychiatric medication and HIV medication)	MEMS <90%	HIV medication = 23/44 (52.3%) Psychiatric medication = 25/39 (64.1%)
Moore 2012 (USA)[122]	Ongoing studies at research centre	Cross-sectional	- HIV - Current or lifetime history of methamphetamine use/dependence	127; MM = 75	AIDS Clinical Trials Group (ACTG) four day adherence self-report questionnaire (HIV medications)	Report of any skipped dose in previous 4 days	Total = 16 (21.3%) 12 (18%) with lifetime history METH+ use 4 (50%) with current METH+ use
Muntner 2010 (USA)[123]	Community	Cross-sectional	- Chronic kidney disease - Hypertension	13065; MM = 3936	MMAS-4 (antihypertensive medications)	MMAS-4 = 0: 'better adherence' MMAS-4 ≥ 2: worse adherence	MMAS-4 = 0: 2715 (69.0%) MMAS-4 = 1: 916 (23.3%) MMAS-4 ≥ 2: 305 (7.8%)
Muth 2016 (Germany)[124]	General practice	Pilot cluster-RCT	≥ 3 chronic conditions	100	MMAS-4 Medication Adherence Rating Scale (MARS)	MMAS-4: scores 0–4, higher scores = lower adherence	MMAS-4 mean (SD) =0.3 (0.6) MARS mean (SD) = 24.0 (1.4)

					Observed medication adherence: (1) drug score (2) dose score (3) the regimen score	MARS: scores 5–25, lower scores = lower adherence Observed medication adherence: Scores outside 0.8–1.2 are divergent	Observed adherence: Drug score = 43/97 (44.3%) divergent Dose score = 93/94 (98.9%) divergent Regimen score = 93/93 (100%) divergent
Muth 2018 (Germany)[125]	General practice	Pragmatic cluster-RCT	≥ 3 chronic conditions	Total = 505; intervention = 253; control = 252	MMAS-4 Observed medication adherence: (1) drug score (2) dose score (3) the regimen score	Report mean (SD) MMAS-4: scores 0–4, higher scores = lower adherence Observed medication adherence: scores outside 0.8–1.2 are divergent	MMAS-4 =3.7 (0.6) Observed adherence: Drug score = 188/501 (37.5%) divergent Dose score = 259/499 (50.9%) divergent Regimen score = 255/500 (51.0%) divergent
Naidoo 2013 (South Africa)[126]	Primary care	Cross-sectional	- TB - HIV	3107; MM = 1729; analysed n = 757	Anti-TB medication: 1-item self-report of percent medication taken in prior 3–4 weeks HIV medication: 1-item self-report of percent medication taken in prior 4 weeks	Self-report adherence < 90%	TB medication: 315 (41.6%) HIV medication: 321 (42.4%)
Nakimuli-Mpungu 2019 (Uganda)[127]	Primary care	Baseline data from cluster-RCT	- Depression (MINI) - HIV	1140	Missed-dose method: 1-item self-report of days missed taking all doses in prior week (ART)	Self-report adherence< 100%	Not reported
Natarajan 2013 (Canada)[128]	General practice	Cross-sectional	- Diabetes mellitus - Hypertension	527; analysed n = 484	MMAS-4 (antihypertensives)	Answer yes to 1 or more questions = medium or low adherence	Low adherence = 8 (1.7%) Medium adherence = 101 (20.9%) Combined = 109 (22.5%)
Nelson 2011 (USA)[129]	Veterans Affairs medical centre	Retrospective record review	- Hyperlipidaemia - Diabetes - Psychotic disorder (not all sample has psychotic disorder)	total = 124; +psychotic disorder = 62; -psychotic	Cumulative mean gap ratio (CMGR): (both medication classes; for combination therapy, the medication with the greatest non-adherence was used)	Report mean (SD)	Antihyperlipidemic medication: +psychotic disorder mean (SD) = 12.0% (12.2) reflecting approx. 44 days without therapy -psychotic disorder mean (SD) = 16.9% (13.0%) reflecting approx. 62 days without therapy

				disorder = 62			Antipsychotic medication: mean (SD) = 18.3% (16.4%) reflecting approx. 68 days without therapy
Nelson 2011 (USA)[130]	Veterans Affairs medical centre	Retrospective record review	- Diabetes - Schizophrenia or related psychotic disorder	124; MM = 62	Cumulative mean gap ratio (CMGR) (antihyperglycaemic medication; antipsychotic medication)	Report mean (SD)	Antihyperglycaemics = 15.8% (16.8%) Antipsychotics = 18.3% (16.4%)
Neuget 2016 (USA)[131]	Administrative claims database	Retrospective cohort	- Non-metastatic breast cancer - Chronic condition from hypertension, hyperlipidaemia, GERD, thyroid, diabetes, osteoporosis	21255; MM = 13427	MPR (tamoxifen and/or aromatase inhibitor) Non-adherence after breast cancer diagnosis reported per disease group	MPR < 80%	Hypertension group: 1233/8683 (14.2%) Hyperlipidaemia group: 671/5829 (11.51%) GERD group: 567/4264 (13.97%) Thyroid group: 484/3542 (13.67%) Diabetes group: 267/1742 (15.33%) Osteoporosis group: 147/1384 (10.62%)
Newville 2014 (USA)[132]	Methadone maintenance clinic	Cross-sectional	- HIV - Treated with methadone maintenance therapy	100	Self-report using questions from AIDS Clinical Trials Group (HAART)	Self-report < 100%	30 (30%)
Nichol 2009 (USA)[133]	Administrative claims database	Retrospective record review	- Hypertension - Treated with lipid-lowering medication	5943	PDC (antihypertensives; lipid-lowering medication)		Year 1 Non-adherent to at least one = 1817 (30.6%) Year 2 Non-adherent to at least one = 2955 (49.7%)
Otenyo 2018 (Kenya)[134]	Hospital renal unit	Cross-sectional	- Chronic kidney disease - Hypertension	144	MMAS-8 (antihypertensives)	MMAS-8 < 6	61 (42.4%)
Owen-Smith 2016 (USA)[135]	Healthcare database	Retrospective analysis	- Schizophrenia or schizoaffective disorder - Treatment by any statin or ACEI/ARB	1420; MM = 710	MPR (cardiovascular medications; if taking both statin and ACEI/ARB, adherence calculated separately for each)	MPR < 0.80	Statin: 350/583 (60.0%) ACEI/ARB: 232/383 (60.6%)
Padman 2017 (India)[136]	Hospital general medicine department; Dialysis centre	Prospective observational	- Chronic renal failure - Diabetes mellitus	107	MMAS-8 (diabetes medication)	Rated MMAS-8 scores as low, moderate and high adherence (criteria for categorising not reported)	Low = 61 (57.0%) Moderate = 36 (33.6%)

Pantalone 2012 (USA)[137]	HIV/AIDS clinics	Cross-sectional	- HIV - Hepatitis C virus	178; MM = 54; analysed n = 45	1-item self-report number of doses missed in prior 30 days (HIV medications)	Self-report adherence < 100%	17 (37.8%)
Parpouchi 2017 (Canada)[138]	Administrative database	Longitudinal (using baseline data, longitudinal data and retrospective data)	- Mental illness - Treated with methadone maintenance	497; MM with adherence to MMT reported = 78	MPR (methadone)	MRP < 0.8	62 (79.5%)
Patel 2016 (USA)[139]	Medicaid database	Retrospective cohort	- Type 2 diabetes - Developmental disability (cerebral palsy, autism, Down's syndrome, cognitive disabilities)	1529	Adherence: MPRm (anti-diabetes medications) Persistence: period from start of medication therapy to discontinuation (anti-diabetes medications)	Adherence: MPRm < 0.8. Report mean SD. Persistence: gap > 120 days	Pooled mean (SD) MPRm = 0.90 (0.19) Non-persistence: 464 (30.4%)
Pence 2015 (USA)[140]	Infectious disease clinics	Pseudo-cluster RCT	- HIV - Major depression (MINI confirmed diagnosis & PHQ-9)	304	Visual analogue scale (ART adherence in prior 30 days)	Report mean (SD)	Mean (SD) = 86.5% (22.7%)
Pentakota 2013 (USA)[141]	Administrative research database	Retrospective cohort	- Diabetes - Comorbidity(s)	79249; MM at some stage (baseline, end, both) = 71334	Adherence: PDC (diabetes medications) Persistence: time from initiation of treatment to first discontinuation in treatment during first 2 years of treatment (diabetes medications)	PDC < 0.8 Persistence: discontinuity > 60 days	Reported according to concordant and discordant comorbidities: Concordant_Concordant = 34442 (44.5%) Discordant_Discordant = 39469 (50.95%) Additional combinations reported in original paper.
Poon 2009 (USA)[142]	Veterans Health Administration database	Retrospective cohort	- Dementia - Hypertension	56561	MPR (antihypertensive medications)	MPR < 0.8	Reported by race and drug class: Min = 13.0%; Max = 39.6%
Puyat 2017 (Canada)[143]	Healthcare databases	Retrospective cohort	- Depression - Additional chronic health condition	105476; MM = 13329	PDC (antidepressants)	Report mean (SD) in 180-day period and in 365-day period	Mean (SD) = 0.84 (0.24) Mean (SD) = 0.80 (0.27)
Qian 2014 (USA)[144]	Administrative database	Cross-sectional	- COPD - Depression	74863; MM = 25130; analysed n = 12574	PDC (COPD medications) Discontinuation (COPD medications)	PDC: < 0.80 Discontinuation: binary variable of medication stopped	PDC: 8621 (68.6%) Discontinuation: 3009 (23.9%)

						3 months or more prior to end of observation period or death, whichever first	
Rattinger 2012 (USA)[145]	Administrative database	Cross-sectional	- Chronic heart failure - Alzheimer's disease or related disorders	9827; MM = 2379	MPR (chronic heart failure medication)	Report mean (SD)	EBM = 0.86 (0.15) Diuretic = 0.81 (0.20) Cardiac glycoside = 0.86 (0.18) Calcium channel blocker = 0.88 (0.17)
Reed 2018 (Australia)[146]	General practice	RCT	≥ 2 chronic conditions from cardiovascular, respiratory, musculoskeletal, psychological, digestive, kidney disease, diabetes (type 1 or 2), cancer	254	5-item self-report	Higher scores indicate poorer levels of medication adherence (range of possible scores not reported)	Mean (SD) = 24.14 (0.14)
Rosenblum 2014 (USA)[147]	Residential and outpatient psychiatric and substance use facilities	RCT	- Mental illness - History of substance use	203	Medication Adherence Rating Scale (MARS) (psychiatric medication)	Report mean (SD) Lower score = higher medication adherence (range of possible scores not reported)	Mean (SD) = 1.55 (0.48)
Roux 2012 (France)[148]	Hospital	Prospective cohort	- HIV - Hepatitis C virus	593 at enrolment; 246 at follow-up	7-item self-report adherence questionnaire (HIV medications)	Self-report adherence < 100% during 4-days prior	Enrolment: 214 (36%) Follow-up: 72 (29.3%)
Saadat 2015 (Iran)[149]	Healthcare centres	Cross-sectional	- Hypertension - ≥ 1 comorbidity	280; MM = 160	MMAS-8 (Persian version) (antihypertensive medication)	MMAS-8 < 6	77 (48.1%)
Safren 2009 (USA)[150]	Community	RCT	- HIV - Depressive mood disorder	45	MEMS (antiretroviral medication)	Report mean (SD)	Control group = 54.41 (30.94) Intervention group = 62.26 (26.82)
Safren 2012 (USA)[151]	Community outreach, hospital HIV clinics, substance use clinics	RCT	- HIV - History of injection drug use (self-endorsed + enrolled in opioid treatment for ≥ 1 month(s)) - Criteria for diagnosis of current or subsyndromal	89	MEMS (antiretroviral medication)	Report mean (SD) percent MEMS-based adherence since last visit	Mean (SD) = 69.0% (28.3%)

			depressive mood disorder				
Safren 2014 (USA)[152]	Primary care	RCT	- Unipolar depression - Type 2 diabetes	87	MEMS (antidiabetic oral medication, insulin or both)	Report mean (SD)	Control group = 85.7 (24.6%) Intervention group = 76.6% (26.0%)
Santorelli 2016 (USA)[153]	Administrative databases	Retrospective cohort	- Breast cancer - Diabetes or hypertension or lipid disorder	9340; MM groups: diabetes = 298; lipid disorders = 508; hypertension = 1062	Adherence: PDC (hypertension; diabetes; lipid disorder medications) Persistence: largest number of consecutive days without coverage by relevant medication class during first 18 months (hypertension; diabetes; lipid disorder medications)	Adherence: PDC < 80% Persistence: Diabetes = 26 days without medications Hypertension = 26 days without medication Lipid-lowering = 62 days without medications	PDC: Diabetes = 78 (26.2%) Hypertension = 151 (14.2%) Lipid disorder = 147 (28.9%) Persistence: Diabetes = 94 (31.5%) Hypertension = 200 (18.8%) Lipid disorder = 123 (24.2%)
Schmitt 2010 (USA)[154]	Veterans Affairs medical centre	Retrospective observational	- Chronic kidney disease - Treatment with antihypertensive	7227	MPR (antihypertensive medication)	MPR < 80%	2360 (32.7%)
Schüz 2011 (a) (Germany)[155]	Community	Longitudinal	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	309	2 self-report items from Reported Adherence to Medication Scale (RAM) measuring: (1) intentional non-adherence (2) unintentional non-adherence	Report mean (SD) Higher values indicate higher degrees of non-adherence (range of possible scores not reported)	Intentional mean (SD) = 1.32 (0.68) Unintentional mean (SD) = 1.40 (0.67)
Schüz 2013 (Germany)[156]	Community	Longitudinal	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	215	1-item self-report intentional non-adherence from Reported Adherence to Medications Scale (RAM)	Report mean (SD) Higher scores indicate better adherence (range of possible scores not reported)	Mean (SD) = 4.68 (0.68)
Schüz 2011 (b) (Germany) [157]	Community	Longitudinal	≥ 2 chronic conditions from Charlson Comorbidity Index or	309	2 self-report items from Reported Adherence to Medication Scale (RAM) measuring:	Not reported	Not reported

			Functional Comorbidity Index		(1) intentional non-adherence (2) unintentional non-adherence		
Shi 2010 (USA)[158]	Veterans Health administration database	Retrospective analysis	- Schizophrenia - Metabolic syndrome	593; MM = 209	Adherence: MPR (antipsychotics) Discontinuation: medication gap (antipsychotics)	Adherence: report mean (SD) MPR Discontinuation: medication gap > 30 consecutive days	Mean (SD) MPR = 0.33 (0.47) Median time to discontinuation = 50 days
Shin 2010 (South Korea) [159]	Pharmacy claims data	Retrospective record review	- Hypertension - Cancer survivor	2455193; MM = 12636	CMA (antihypertensive medication)	CMA < 80%	5762 (45.6%)
Shoval 2017 (Israel)[160]	Healthcare database	Retrospective analysis	- Parkinson's disease - Depression	8553	MPR (antidepressant medication)	MPR < 20% = non-adherent MPR 20% - 50% = poorly adherent MPR 50% - 80% = moderately adherent	< 20% = 1566 (18.4%) 20-50% = 1184 (13.8%) 50-80% = 1584 (18.5%) < 80% = 4334 (50.7%)
Siika 2013 (Kenya)[161]	Ambulatory HIV care program	Retrospective cohort	- HIV - Tuberculosis	21242; MM = 5186; analysis n = 4838	1-item self-report number of pills taken during prior week (ART)	If patient ever answers 'most', 'half', 'few', or 'none' (defined as non-perfect adherence to ART)	986 (20.4%)
Slabbert 2015 (South Africa)[162]	Administrative database	Prospective cohort	- Major depressive disorder - HIV/AIDS	127	MPR (antidepressant medications)	Report mean (SD)	74.43 (32.03)
Soto 2015 (Chile) [163]	Hospital	Prospective	- Type 2 diabetes - Hypertension	50	Pill count (seems like both medications together)	Pill count < 80% or > 110%	35 (70.0%)
Stack 2010 (UK)[164]	Healthcare database (secondary care)	Cross-sectional	- Type 2 diabetes - Hypertension	480	MMAS-4 (separately for antidiabetic medication, antihypertensive medication and statin)	0 = total adherence 4 = non-adherence: presented as no unintentional non-adherence; no intentional non-adherence	Unintentional non-adherence: Diabetes medication = 302 (62.9%) Hypertension medication = 226 (47.1%) Statins = 225 (46.9%) Intentional non-adherence: Diabetes medication = 63 (13.1%)

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							Hypertension medication = 92 (19.2%) Statins = 87 (18.1%)
Stilley 2010 (USA)[165]	Primary care	RCT	≥ 2 from type 2 diabetes, hypertension, hyperlipidaemia	373	MEMS (not clear to what medications)	Report mean (SD): (1) prescribed number of doses (2) days with prescribed dose (3) days with correct inter-dose interval	Prescribed number of doses = 92.6 (17.6) Days with prescribed dose = 82.2 (22.4) Days with correct inter-dose interval = 64.2 (30.1)
Stuart 2015 (USA)[166]	Medicare database	Retrospective cohort	- Diabetes - Cancer	32855; MM = 4348	PDC (diabetes medication)	Report mean PDC	OHA (n = 2248) mean PDC = 73.8% RAAS-I (n = 2618) mean PDC = 70.5% Statins (n = 2272) mean PDC = 69.0%
Taira 2017 (USA)[167]	Healthcare database	Retrospective observational	- Type 2 diabetes - Treatment with antihypertensive, oral antidiabetic and cholesterol lowering medication	5163	MPR (each medication)	MPR < 0.8	Range of non-adherence reported: 16% - 48.2% Non-adherence for each medication according to ethnicity reported in original paper.
Tang 2014 (USA)[168]	Outpatient sites	Secondary analysis of prospective descriptive cohort	- Heart failure - Depression (GDS)	280; MM = 103	MEMS (heart failure medication); Basel Assessment of Adherence Scale (BAAS) (heart failure medication)	MEMS: < 80% BAAS: a positive answer on any question	Self-report = 77 (75%) MEMS = 29 (28%)
Thaler 2015 (USA)[169]	HIV treatment facilities	Not reported	- HIV - neurocognitive impairment and/or substance dependence/use and/or psychiatric illness	204; MM not reported	MEMS (HAART) Self-report (not described) (HAART)	Report mean (SD) per additional risk factor	One risk factor: 64.4% (27.0%) Two risk factors: 62.8% (28.0%)
Vacek 2013 (USA)[170]	Medicaid database	Retrospective analysis	- Developmental disability - Hypertension	280; analysed n = 201	PDC (antihypertensive medication)	PDC < 0.8	90 (45%)
Vatcharavongvan 2017 (Thailand)[171]	Primary care	Cross-sectional	≥ 2 chronic conditions from type 2 diabetes, hyperlipidaemia,	397; analysed n (polyphar	Structured questionnaire during face-to-face interview	At least one medication that was (a) taken at wrong	74 (60.7%)

			hypertension, ischemic heart disease, chronic heart failure, cerebrovascular disease, epilepsy	macy group) = 122		time, (b) missed (c) intentionally stopped by patient during 2 weeks prior	
Vega 2017 (USA)[172]	Healthcare database	Retrospective cohort	- Major depressive disorder - Type 2 diabetes	1361	Adherence: MPR (depression medication; diabetes medication) Persistence: Length of treatment (LOT) (depression medication)	MPR < 80% LOT ≥ 15 day gap in first 180 days of therapy	Depression medication: Non-adherent = 872 (64.1%) Non-persistent = 926 (68.0%) Non-adherent/non-persistent = 936 (68.8%) Diabetes medication: Non-adherent = 525 (38.6%)
Villar 2009 (Spain)[173]	Ongoing studies, administrative database	Prospective cohort	- Severe obstructive sleep apnoea - Cardiovascular disease	2158; antihypertensives = 659; statins = 175; antiplatelets = 109	MPR (cardiovascular disease medications)	MPR < 80%	Antihypertensives: 118/659 (17.91%) Statins: 59/175 (33.71%) Antiplatelets: 9/109 (8.26%)
Wagner 2011 (USA)[174]	Pooled data from longitudinal studies	Cross-sectional and longitudinal	- HIV - Depression (BDI-II; BDI (original version); CESD; depression subscale of Brief Symptom Inventory)	1374; MM = 489	MEMS (HIV medication)	MEMS < 90%	Severe depression = 72% (148/205) Mild to moderate depression = 56% (159/284) Combined = 307/489 (62.8%)
Wakefield 2012 (USA)[175]	Primary care	RCT	- Type 2 diabetes - Hypertension	302 (control = 107; low intensity = 102; high-intensity = 93)	1-item self-report adherence	Report mean (SD)	Mean (SD): Control group = 99.5 (4.4) Low-intensity group = 99.7 (1.6) High-intensity group = 99.7 (2.7)
Warner 2013 (Germany)[176]	Community	Longitudinal	≥ 2 chronic physical conditions from Charlson Comorbidity Index or Functional Comorbidity Index	309	1-item self-report unintentional non-adherence from Reported Adherence to Medication Scale (RAM)	1 < RAM < 4	T1 = 83 (27%) T2 = 99 (32%)
Wei 2018 (USA)[177]	Medicare database	Retrospective cohort	- Major depression - Chronic obstructive pulmonary disease	25458; analysed n = 11572	PDC (COPD medications)	PDC < 0.8	7059 (61.0%)

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Weiss 2016 (USA)[178]	Ambulatory practices	Longitudinal	- HIV - Hypertension or chronic kidney disease	152; +hypertension = 115 (analysed n = 102); +CKD = 37 (analysed n = 20)	MEMS (1) taking adherence (2) dosing adherence (3) timing adherence (HIV medications and CKD/hypertension medications separately)	MEMS < 95% (HIV medications) MEMS < 80% (other medications)	+hypertension: HIV medication = 40 (39%) Hypertension medication = 20 (20%) +CKD: HIV medication = 7 (35%) CKD medication = 4 (20%)
Wenze 2015 (USA)[179]	Hospital	Pilot RCT	- Bipolar I or II - Substance use disorder	30	Adapted version of Medication Compliance Questionnaire (MCQ) (prescribed mood-stabilizing agent)	MCQ ≥ 2	12 (40.0%)
Wong 2014 (China)[180]	Administrative database	Cross-sectional	≥ 2 chronic conditions	3866; MM = 523	MMAS-8 (antihypertensives, oral hypoglycaemic agents, lipid lowering agents separately)	MMAS-8 ≤ 6 (MMAS-8 sections for each drug should score >6 to be defined as having optimal adherence)	321 (45.1%)
Wroe 2014 (Rwanda)[181]	Rural health centres	Cross-sectional	- HIV - Depression (Hopkin's Symptom Checklist)	292; MM = 32	Self-report 4-day recall (HIV medications) Self-report 2-day and 7-day treatment interruptions (HIV medications) CASE index (HIV medications)	Self-report 4-day recall: ≥ 1 dose(s) in four days prior Self-report 2-day and 7-day treatment interruptions: non-adherence on sequential days CASE index ≤ 9	4-day recall = 5 (17.0%) 2-day interruptions = 7 (22.2%) CASE index = 5 (15.8%)
Wu 2011 (USA)[182]	Medicaid database	Retrospective observational	- Hyperlipidaemia - Type 2 diabetes	1705	MPR (statins; antidiabetic medication)	MPR < 0.8	Statins = 1079 (63.3%) Mean (SD) MPR (statins) = 0.61 (0.32) Mean (SD) MPR (antidiabetics) = 0.71 (0.30)
Wu 2010 (USA)[183]	Medicaid database	Retrospective cohort	- Major depressive disorder - Anxiety disorder	3083; MM = 776	Adherence: MPR modified (MPRm) (antidepressants) Persistence: Refill gap (antidepressants)	MPRm: report mean (SD) Refill gap: median (IQR)	Mean (SD) MPRm = 0.71 (0.37) Median (IQR) persistence (days) = 237 (216-256)

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Wu 2013 (USA)[184]	Inpatient and outpatient cardiology clinics	Secondary analysis of two prospective studies	- Heart failure - Depression (PHQ-9)	216; MM = 55	MEMS (heart failure medication)	MEMS < 88%	30 (54.55%)
Wulaningsih 2018 (Sweden)[185]	Administrative database	Retrospective analysis	- ER-positive breast cancer - Charlson Comorbidity Index score ≥ 1	4645; MM baseline = 715; MM follow-up = 1348	MPR (endocrine therapy)	MPR < 80%	Baseline: 168/715 (23.5%) 5-years: 330/1348 (24.5%)
Xia 2015 (China)[186]	Hospital	Cross-sectional	- ≥ 1 chronic condition from chronic obstructive pulmonary disease, diabetes, hyperlipidaemia, hypertension - Depression (PHQ-9) and/or anxiety (GAD-7)	325; +anxiety = 276; +depression = 266	Chinese Medication Adherence Scale (CMAS) (chronic condition)	CMAS <6 to	+depression: 222 (83.5%) +anxiety: 226 (81.9%)
Xue 2018 (China)[187]	Primary care	Baseline data from RCT	- Hypertension - Depression (PHQ-9)	2362	1-item self-report adherence (antihypertensive medications)	Answer yes to "do you sometimes forget to take your antihypertensive pills"	954 (40.4%)
Yang 2010 (USA)[188]	Medicare database	Retrospective cohort	- Diabetes - Hypertension and/or renal disease	599141	PDC (antihypertensive medication)	PDC < 0.8	276945 (46.22%)

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References to studies included in analyses

Multimorbidity sample	
≥ 2 specific conditions	[1–134]
1 specified condition and ≥ 1 other condition(s)	[135–160]
Inclusive definition of multimorbidity, e.g. ≥ 2 conditions	[161–178]

Adherence measurement	
Pharmacy data	[1–4,7,10,11,13,14,20,21,23,25,29,30,32,33,36,40,45–47,51,54,59,61–64,66–68,70–75,78,81,89,90,92,94,97,98,100–102,108–111,113,117,118,121–123,126,131,132,135–151,161,133,160,164,50]
Self-report	[5,6,8,9,12,15,16,18,19,24,27,28,34,35,38,39,41,47–49,52,53,55,56,58,60,65,68,69,74,76,77,79,80,82,83,85–88,91,93,93,95,96,99,103,104,112,115,119,120,125,128,129,134,137,152–159,162,163,165–169,171,170,172–174,178,176,177,175,43]
Electronic monitoring	[5,9,17,18,22,24,37,43,47,84,105–107,116,119,120,124,127,130,167]
Pill counts	[26,57,114]
Healthcare provider report	[42,44]

Reporting of non-adherence	
Prevalence of non-adherence	[1–4,6,7,9,11–23,25,27–30,32,33,35,36,39–46,48,50–57,59–70,73–75,77,79,82–86,88,91–98,100,101,104,108,110–112,114,115,118,119,121–124,126–131,133,134,136–140,142–148,150–152,154–156,158–160,162,165,170,171,176–178]
Average non-adherence	[5,8,10,24,26,31,34,37,38,47,49,58,71,72,76,78,80,81,89,90,99,102,103,105–107,109,113,116,117,120,125,132,135,141,149,153,157,161,163,164,166,167,169,172,173,175]

Study setting	
Primary care / community / general practice	[6,17,18,26,27,30,32,36,40,41,57–59,63,72,74,77,79,85–88,99,105,107,116,125,129,134,153,155,157,160,162,163,165,168,170–177]
Secondary care / hospital /tertiary care	[8,15,16,19,28,35,37,38,42,44,48,52,53,55,60,65,76,78,80,93,95,103,104,114,115,128,130,137,144,152,154,156,159,166]
Administrative database	[1–4,7,10,11,13,14,20,21,23,25,29,33,45,46,50,51,54,61,62,64,66,67,70,71,73,75,92,94,97,98,100–102,109–111,113,117,118,121–123,126,131–133,135,136,138–143,145–151,161,164,178]
Other / not clearly reported / multiple sources	[152,5,9,12,22,24,31,34,39,43,47,49,167,56,68,69,169,83,84,89–91,96,158,106,108,112,119,120,124,127,81,82]

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	(i)
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	(ii)
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	1
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	1
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	2
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	2
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	2
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	2+SM
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	2+3
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	3
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	3
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	3
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	2
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	3

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	n/r
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	n/r
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	4
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	4+SM
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	4
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	5
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	5
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	n/r
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	n/r
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	14-16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	17
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	17

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SM = supplementary material

n/r = not reported

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Title: Prevalence and predictors of medication non-adherence among people living with multimorbidity: a systematic review and meta-analysis

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ABSTRACT (298/300)

Objectives This systematic review aimed to describe medication non-adherence among people living with multimorbidity according to the current literature, and synthesise predictors of non-adherence in this population.

Methods A systematic review was conducted according to the Preferred Reporting Items for Systematic reviews and Meta-Analyses. PubMed, EMBASE, CINAHL and PsycINFO were searched for relevant articles published in English language between January 2009 and April 2019. Quantitative studies reporting medication non-adherence and/or predictors of non-adherence among people with two or more chronic conditions were included in the review. A meta-analysis was conducted with a sub-group of studies that used an inclusive definition of multimorbidity to recruit participants, rather than seeking people with specific conditions. Remaining studies reporting prevalence and predictors of non-adherence were narratively synthesised.

Results The database search produced 10998 records and a further 75 were identified through other sources. Following full-text screening, 178 studies were included in the review. The range of reported non-adherence differed by measurement method, at 76.5% for self-report, 65.6% for pharmacy data, and 44.1% for electronic monitoring. A meta-analysis was conducted with 8 studies (n = 8949) that used an inclusive definition of multimorbidity to recruit participants. The pooled prevalence of non-adherence was 42.6% (95% CI = 34.0-51.3%, k = 8, I^2 = 97%, p < 0.01). The overall range of non-adherence was 7.0% to 83.5%. Frequently reported correlates of non-adherence included previous non-adherence and treatment-related beliefs.

Conclusions The review identified a heterogeneous literature in terms of conditions studied, medication adherence measurement, and definition of non-adherence. Results suggest that future attempts to improve adherence among people with multimorbidity should determine for which conditions individuals require most support. The variable levels of medication non-adherence highlight the need for more attention to be paid by healthcare providers to the impact of multimorbidity on chronic disease self-management.

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Strengths and limitations of this study

- Criteria for inclusion did not specify particular chronic conditions, permitting inclusion of a wide variety of disease combinations.
- Studies were not excluded on the basis of design, setting, or method of non-adherence measurement, generating a comprehensive overview of the literature.
- Due to considerable heterogeneity between studies, meta-analysis and synthesis of predictors was restricted to studies that did not recruit participants with pre-specified combinations of conditions.
- Quantitative subgroup analysis was not conducted to determine the influence of participant or study characteristics on non-adherence estimates.

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INTRODUCTION

Multimorbidity is broadly defined as the co-existence of two or more chronic conditions[1] and has been described as the most common chronic condition experienced by adults.[2] In a Scottish study of 1.75 million people, the prevalence of multimorbidity was estimated to be 23.3%, increasing to 65% among those aged 65-84 years.[3] People living with multimorbidity attend more medical consultations, experience higher rates of hospital admissions, and face increased healthcare costs.[4] People with multimorbidity are also expected to engage with complex self-management regimens which can include monitoring symptoms, changing lifestyle behaviours and adhering to prescribed medications.[5]

Adherence refers to the extent to which a person's behaviours correspond with agreed recommendations from their healthcare provider.[6] Multimorbidity is closely associated with polypharmacy,[7] and it has been reported previously that the risk of medication non-adherence can increase as people are prescribed more medicines.[8] The extent of non-adherence among people with multimorbidity is poorly understood, attributable to challenges in measuring multiple medication adherence.[9] However, it is recognised that medication non-adherence in multimorbidity can exacerbate the burden experienced by individuals through increased morbidity and mortality.[10] Medication adherence can be influenced by many factors, including treatment characteristics, condition-related factors, and illness and medication beliefs.[11] Identifying significant correlates of non-adherence among people self-managing multiple conditions will contribute to an evidence base for developing behavioural treatments which support adherence in the context of multimorbidity.

Much intervention development to improve medication adherence has focused on single-disease populations.[12] Such a focus may produce an artificial underestimation of the complexity of chronic disease self-management when, in reality, most people with a chronic condition have additional morbidity(s).[3] This, in turn, may lead to policies and interventions which are not designed with 'the most common chronic condition'[2] in mind. In order to develop interventions to improve medication adherence among people living with multimorbidity, it is first necessary to understand the extent of non-adherence in this population, as well as factors that might influence adherence. To our knowledge, a synthesis of the prevalence and predictors of medication non-adherence among people with multimorbidity has not been previously reported. This review therefore aims to (1) describe the prevalence of medication non-adherence among people living with multimorbidity, and (2) describe previously identified correlates of non-adherence in this population.

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METHODS

A systematic review and meta-analysis was conducted according to the Preferred Reporting Items for Systematic review and Meta-Analysis statement[13] (supplementary file 1). The review was prospectively registered on PROSPERO (CRD42019133849) and a peer-reviewed protocol was published.[14] Prior to finalising our aims, we searched in PROSPERO[15] to determine if an equivalent review had been conducted or was currently being conducted.

Inclusion and exclusion criteria

Original peer-reviewed articles and doctoral theses produced in English from 2009-2019 were reviewed. A ten year range was chosen to reflect the marked increase in multimorbidity literature in recent years.[16] Opinion pieces, conference presentations, books, letters, editorials and abstracts were not reviewed. Quantitative studies including adults aged 18 years or older, living with two or more chronic conditions (including physical and/or mental health conditions) reporting medication non-adherence and/or predictors of non-adherence in an extractable format were considered. Presence of multimorbidity was required to be part of study aims and/or inclusion criteria. Polypharmacy was not used as a definition of multimorbidity; however studies referring to use of two or more medications that could be inferred as treatment for two or more chronic conditions were considered. Studies were not excluded based on design, however only observational data was extracted for review. Therefore clinical trials that did not report baseline non-adherence were excluded. Studies including any participants under 18 years were excluded.

Search strategy and study selection

An electronic database search of PubMed, EMBASE, CINAHL and PsycINFO from January 2009 to April 2019 was conducted using a predefined search strategy combining terms related to ‘multimorbidity’ and ‘medication adherence’ (supplementary file 2). Reference lists of included studies were screened for relevant articles. Records were exported to EndNote X8 to remove duplicates. Two reviewers (LF, RLV) each screened 50% of titles and abstracts in Covidence.[17] Twenty-percent of each reviewer’s records were chosen at random for cross-checking by the other reviewer. Studies meeting inclusion criteria were moved forward for full-text screening by two independent reviewers (LF & JL or EG). Where disagreements arose, a third reviewer (GJM or AWM) was consulted.

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Data extraction and quality assessment

A data extraction form was developed to extract relevant data from each included study (supplementary file 3). Where studies reported medication adherence, this was converted to medication *non*-adherence by subtracting the number of adherent participants from the total sample. One reviewer (LF) extracted data from all included studies, and a second reviewer (JL) cross-checked 20%.

The quality of included studies was assessed using criteria for observational studies[18] previously adapted for medication adherence reviews.[19] Criteria include 11 items related to selection methods, measurement, sources of bias, control for confounding and use of statistics. Each item was answered with 'yes', 'no', 'don't know' or 'not applicable'. All included studies were assessed using these criteria, including intervention studies as only baseline observational data were reviewed. One reviewer (LF) assessed the quality of all studies and a second reviewer (EG) cross-checked a random sample of 20%. No studies were excluded based on quality.

Data analysis

Due to between-study heterogeneity relating to study population, design, setting, and measurement and definition of medication non-adherence, a quantitative synthesis involving all studies was not considered appropriate. Meta-analysis was conducted with studies which recruited participants as 'people living with ≥ 2 chronic conditions'. This group of studies was considered to be most focussed in addressing a universal experience of multimorbidity – albeit a multifaceted and complex experience across individuals – offering results that can be used in future attempts to address non-adherence in this population. Studies which recruited participants with pre-specified conditions were therefore not included in the meta-analysis. Meta-analysis was conducted in R[20] using the meta[21] and metafor[22] packages (supplementary file 4). A random-effects model was used to estimate the pooled prevalence of non-adherence. Between-study heterogeneity was assessed using I^2.[23]

Remaining studies were narratively synthesised to describe medication non-adherence in multimorbidity, informed by Popay and colleagues' guidelines.[24] The study characteristics table was preliminarily analysed to identify and count patterns of results. This was followed by an exploration of relationships within and between results of individual studies. Predictors of non-adherence were also synthesised in this way using studies which recruited participants using an inclusive definition of multimorbidity. Predictors were not extracted from studies which recruited participants with combinations of specific conditions due to potential confounding associated with specific conditions, e.g. a study of people living with bipolar disorder and depression assessing severity of depressive symptoms and psychological disability as predictors of non-adherence.[25]

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Patient and public involvement

A panel of people with experience of living with multimorbidity were consulted to advise on appropriate methods for disseminating findings from the review to patients and the public.

RESULTS

Overview of studies

After removing duplicates, 10998 studies were identified from the database search and 75 from reference lists of included studies. Following screening of titles and abstracts, 449 full-texts were reviewed. Of these, 271 were excluded with reasons (figure 1), with 178 studies included in the final synthesis.[9,25–201] The study selection process is outlined in the PRISMA flow diagram in figure 1.

Figure 1. PRISMA flow diagram

Study quality

Results of study quality assessment are summarised in figure 2.

Figure 2. Methodological quality graph.

Study characteristics

One-hundred and thirty-three studies included participants with ≥ 2 specific conditions, e.g. Gibson et al. recruited participants self-managing type 2 diabetes and hypertension;[79] 26 studies included participants with one specific condition and ≥ 1 other condition(s), e.g. Puyat et al. recruited participants living with depression and an additional chronic health condition;[156] and 19 studies recruited participants using an inclusive definition of multimorbidity, e.g. Schüz and colleagues[170] recruited people living with ≥ 2 conditions from the Charlson Comorbidity Index[202] or Functional Comorbidity Index.[203]

Eighty-four studies measured non-adherence using pharmacy data, e.g. Kim et al. used pharmacy claims data to assess non-adherence among people living with hypertension and a psychiatric condition;[107] 81 used self-report, e.g. Moore et al. measured adherence to medications for HIV and bipolar disorder using a visual analogue scale;[135] and 20 used electronic monitoring, e.g. Safren et al. assessed non-adherence to antiretroviral medication among people living with HIV and a depressive mood disorder using a Medication Event Monitoring System (MEMS).[163] Three studies used pill counts,[59,101,176] two involved reports by a healthcare provider,[80,83] and one did not report the method used.[64] Ten of these studies used more than one method to measure adherence, e.g. Bogner

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et al. used electronic monitoring and self-report to measure non-adherence for diabetes and depression.[45] Forty-seven studies reported average non-adherence only. The characteristics of all included studies are summarised in supplementary files 5 and 6.

Participant characteristics

The number of participants with multimorbidity in each study ranged from 22[130] to 599,141.[201] Among 135 studies reporting the mean (SD) age of participants, the average age ranged from 32.4 (9.6)[105] to 80.8 (9.1) years.[158] Ninety-three (52.2%) study samples were majority female, while three studies did not report gender.[98,117,126]

Prevalence of medication non-adherence

A meta-analysis was conducted with eight studies reporting prevalence of non-adherence and using an inclusive definition of multimorbidity, i.e. without seeking people with ≥ 1 specific condition(s).[56,71,115,137,138,184,189,193] The pooled prevalence of non-adherence was 42.6% (95% CI = 34.0-51.3%, $k = 8$, $I^2 = 97\%$, $p < 0.01$) (figure 3). The range of non-adherence among studies in the meta-analysis was 16.4%[71] to 61.4%.[193] The range among all reviewed studies was 7.0%[42] to 83.5%.[199] Remaining studies were narratively synthesised to describe prevalence and predictors of medication non-adherence.

Figure 3. Results of random effects meta-analysis.

Narrative synthesis findings

Medication non-adherence over time

The direction and extent of change in medication non-adherence over time was variable between studies. Among studies reporting prevalence of non-adherence over time, respective lengths of follow-up were three months,[96] six months,[189] 48 weeks,[121] one year,[26,36] 16 months,[161] two years,[60] 3 years[48] and 5 years.[198] Of these, four studies measured change in non-adherence following a recent diagnosis or treatment for an additional condition,[36,48,96,161] four measured change following medication initiation,[26,60,121,198] and one reported change not related to specific patient or treatment characteristics.[189] Three studies reported decreased non-adherence,[36,96,161] with the greatest decrease being 9.6% after 3 months.[96] Four studies reported increased non-adherence,[26,121,189,198] with the greatest increase being 41.2% after 12 months.[26] One study reported variation in how non-adherence changed over time depending on the class of medication,[60]

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while another reported an initial 13.9% increase after two years, followed by a 14.9% decrease one year later.[48]

Assessment of multiple medications

Within individual studies, the extent of non-adherence for one condition did not always correspond with non-adherence for another condition among people taking medications for multimorbidity. Twenty-six studies reported non-adherence prevalence estimates separately for each condition, e.g. Foguet-Boreu reported that, among people living with both diabetes and hypertension, 39.8% were non-adherent to medication for diabetes, while 23.8% were non-adherent to medications for hypertension.[75] Differences in non-adherence between conditions ranged from 0.8% measured by a 1-item self-report measure in a study of people living with HIV and TB,[139] to 33.6% measured by pharmacy data in a study of people with higher non-adherence to medications for COPD than depression.[27] Four studies reported a difference of ≤ 5%,[45,79,88,139] five reported a difference of 5–9.9%,[95,110,129,135,177] 11 reported a difference of 10–19.9%,[38,57,65,75,84,90,102,133,177,180,191] and six reported a difference of ≥ 20%.[27,66,80,90,105,185]

Physical and mental multimorbidity

Differences in non-adherence between physical and mental health conditions were not consistent across the literature. Among studies of people living with a physical condition and depression and/or anxiety, four reported non-adherence prevalence for both the physical and mental health condition, e.g. Gupte-Singh and colleagues reported non-adherence to medications for asthma (62.0%) and depression (74.3%).[84] Non-adherence was higher for depression in three studies,[45,84,185] and higher for the physical condition in one study.[27]

Among studies of people living with a psychiatric condition and a physical condition, seven reported non-adherence prevalence for both conditions, e.g. Moore et al. reported non-adherence among people living with HIV (52.3%) and bipolar disorder (64.1%).[133] Non-adherence was higher for the psychiatric condition in three studies,[90,133,135] and higher for the physical condition in four studies.[66,88,111,129]

Characteristics of multiple medication regimens

Two studies compared prevalence of non-adherence according to characteristics of multiple medication regimens.[26,93] Starting medications for two conditions simultaneously resulted in lower non-adherence versus not starting simultaneously (24.8% vs 55.0% at 2 months; 66.0% vs. 69.0% at 12

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months).[26] Hussein and colleagues assessed the difference between a single-pill vs. two-pill antihypertensive, lipid-lowering regimen.[93] Results concluded that a single-pill regimen improved adherence compared to a two-pill regimen, and improvements were greater among people with previous experience taking either an antihypertensive or lipid-lowering medication.[93]

Method of non-adherence measurement

Different methods of measuring non-adherence produced differences in the range of non-adherence estimates, with self-report measures producing the widest range, and observed adherence producing the narrowest range. Among sixty-four studies using pharmacy data, non-adherence prevalence ranged from 10.6% among women with breast cancer and osteoporosis[144] to 80.0% among people with COPD and depression;[28] Among 50 studies using self-report, non-adherence prevalence ranged from 7.0% among people with anxiety and cardiac disease[42] to 83.5% among people with depression and an additional chronic condition(s).[199] Among 10 studies using MEMS, non-adherence prevalence ranged from 20.0% among people living with HIV and hypertension or chronic kidney disease[191] to 64.1% among people living with HIV and bipolar disorder.[133] Two studies reported prevalence of non-adherence assessed by a healthcare provider[83] or a mix of healthcare providers, patients and relatives,[80] with non-adherence ranging from 8.6%,[83] to 52.5%.[80] Two studies used observed medication adherence to identify discrepancies between medications taken and medications prescribed,[137,138] with non-adherence prevalence ranging from 37.5%[138] to 44.3%.[137] among people with three or more chronic conditions. Two studies used pill counts, with non-adherence ranging from 46.3%[101] to 70.0%.[176]

Nine studies reporting prevalence of non-adherence used more than one measure.[34,45,51,114,122,137,138,181,194] Among studies using self-report and MEMS, two studies reported lower non-adherence using self-report,[45,114] one study reported higher non-adherence using self-report,[181] and one study reported a 1% difference.[34] Another study reported no difference between self-report and pharmacy data.[51] Stack and colleagues used self-report to measure prevalence of intentional non-adherence (e.g. deliberately choosing not to take medicines) and unintentional non-adherence (e.g. forgetting to take medications) to three medications.[177] For each medication, unintentional non-adherence was more prevalent (62.9%, 47.1%, 46.9%) than intentional non-adherence (13.1%, 19.2%, 18.1%).

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Predictors of medication (non-)adherence

Eleven studies using an inclusive definition of multimorbidity reported predictors of adherence (see supplementary file 7).[40,56,62,71,94,106,168–170,189] Nine used self-report to measure adherence,[56,62,71,94,106,168–170,189] and two used pharmacy data.[40,68] The mean age of participants was < 65 years in four studies[40,62,68,94] and > 65 in seven studies[56,71,106,168–170,189]. Five studies assessed demographic variables as predictors,[56,106,168,169,189] including age, sex, marital status, education and deprivation. No demographic variables significantly predicted non-adherence.

One study assessed health status, which was associated with adherence i.e. poorer health status was associated with stronger motivation to take medications.[106] Other factors assessed included number of prescribed medications,[56,62,168,169,189] number of chronic conditions,[62,68,106,168,169] physical fitness,[189] and functional health status,[168] as well as presence of other conditions.[56,68,106] Schüz and colleagues[168] found lower intentional non-adherence among people prescribed more medications and higher intentional non-adherence among people with more conditions; number of conditions and medications did not predict unintentional non-adherence.[168] Domino and colleagues[68] reported lower non-adherence among those with more comorbidities. Depression was associated with lower adherence, while schizophrenia was linked to both higher and lower adherence, depending on other physical conditions present.[68]

A number of studies explored beliefs about conditions and medications as predictors of adherence. Two studies using the same sample reported less non-adherence among those with stronger beliefs about the necessity of medications,[168,169] less non-adherence among those with greater concerns about medications,[169] more intentional non-adherence among those with increasing beliefs that medications are harmful,[168] and more unintentional non-adherence among those with increasing beliefs that medications are overused.[168] Changes in necessity beliefs predicted intentional non-adherence, with increasing beliefs about necessity associated with improved adherence.[169] Beliefs about overuse of medications predicted unintentional non-adherence, with stronger beliefs associated with higher unintentional non-adherence.[168,169] Beliefs about consequences, treatment control and timeline were also explored.[106,170] Kenning and colleagues did not report significant relationships between beliefs and adherence.[106] Schüz and colleagues found higher adherence was predicted by stronger control beliefs, stronger beliefs about negative consequences, and beliefs about a shorter illness timeline.[170] One study assessed beliefs about multimorbidity itself, reporting no significant relationship with adherence.[106]

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Three studies using the same sample reported an association between previous and later unintentional non-adherence,[168,169,189] two of which also reported an association between previous and later intentional non-adherence.[168,169] Previous medication motivation predicted adherence, with participants with higher motivation at baseline reporting higher adherence at follow-up.[106] Other factors assessed included attitudes towards medications,[189] medication knowledge,[106] and medication adherence self-efficacy,[62] none of which predicted non-adherence. One study assessed general received social support and medication specific-support,[189] with higher medication-specific social support associated with lower adherence; this relationship was moderated by social conflict. General received social support did not predict adherence. Other factors assessed included chronic disease self-efficacy, medication assistance, health literacy, and functional health literacy,[62] none of which predicted non-adherence.

Lower adherence was associated with lower physical (r = -.44) and environmental (r = -.57) quality of life.[62] Lower prospective memory performance was related to more omission errors of adherence (r = -.29), while lower retrospective memory performance was related to more commission errors (r = -.24).[94] Other factors positively associated with adherence were medical home enrolment,[40] better health provider relational quality (r = 0.22)[71] and fewer hassles during healthcare system encounters.[106] Self-help behaviours did not predict adherence,[106] better self-monitoring and insight predicted higher motivation to adhere to medications,[106] and a more impulsive/careless problem-solving style was associated with lower adherence (r = -.49).[62]

DISCUSSION

Our systematic review identified a heterogeneous literature reporting medication non-adherence in people living with multimorbidity. Variation in study design, population, and measurement and definitions of both multimorbidity and non-adherence was observed between studies. Among studies using an inclusive definition of multimorbidity (k=11), correlates of non-adherence included treatment related beliefs and previous non-adherence. From eight studies considered suitable for meta-analysis, 43% (95% CI 34%-51%) of participants with multimorbidity were non-adherent to medications. Non-adherence ranged from 16.4%[71] to 61.4%[193], with the lowest estimate arising from a single-item self-report measure,[71] and the highest derived from pharmacy data.[193] One study in the meta-analysis contributed to 76% of the total sample.[115] These characteristics of individual studies should be considered when interpreting findings of the quantitative synthesis.

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Among studies reporting non-adherence prevalence for more than one condition, results highlighted disparities in non-adherence between conditions. It was previously reported that people with multimorbidity prioritise certain conditions over others,[204,205] with our results suggesting this prioritisation may be reflected in self-management behaviours. Other factors known to predict adherence in general may also explain disparities within an individual, including differences in regimen complexity and differences in the expected efficacy of medications.[206] Inauen and colleagues[9] have reported that adherence within an individual can differ between medications, and emphasise the need to assess adherence for multiple medications separately in order to focus interventions appropriately. Despite such calls, a recent systematic review suggested there are no standard criteria for measuring multiple medication adherence, as well as no agreement on how existing measures developed for measuring adherence to monotherapy can be adapted to multiple medications.[207] A number of studies reviewed here, including those in the quantitative synthesis, reported a single prevalence estimate representing average non-adherence across conditions, potentially resulting in a concealment of those conditions for which non-adherence is highest. This should be considered when interpreting the meta-analysis.

Among participants with a physical condition and depression, non-adherence was higher for antidepressants in three of four studies. This somewhat supports previous findings that non-adherence to antidepressants is higher than to medications for physical conditions,[208] and extends this to instances where both occur in one individual; however further research is necessary to draw stronger conclusions. Among participants with a psychiatric and physical condition, prevalence of non-adherence to psychiatric medications was higher than non-adherence to medications for physical conditions in three of seven studies. Previous literature has reported that adherence to antipsychotic medications may be lower than adherence to medications for physical conditions,[208] while others have found antipsychotic adherence to be higher.[209] These discrepancies further highlight the heterogeneity that is inherent in multimorbidity, as well as the importance of considering between-condition medication adherence within individuals. Moreover, it has previously been reported that the condition people with multimorbidity prioritise most can differ over time as conditions and treatments change.[205] Considering the cyclical nature of some mental health conditions,[210] it is important to consider whether self-management behaviours, such as medication taking, fluctuate over time as a result of changing priorities in response to symptoms.

Among the variables frequently tested as predictors were number of chronic conditions and number of prescribed medications. Surprisingly, the strength and direction of association with non-

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adherence was not consistent across studies. A recent systematic review concluded that deprescribing interventions do not consistently improve adherence,[211] suggesting an equivocal relationship between medication burden and non-adherence. This highlights the clinical dilemma faced by clinicians where deprescribing may be a necessary but not sufficient therapeutic response. It has been suggested previously that the frequency of medication administration has a greater influence on adherence than number of tablets,[212] which may have particular relevance in multimorbidity considering some people take medications for different conditions throughout the day while others take multiple medications together.[213] Indeed, it has been reported that particular combinations of conditions are associated with higher dosing frequencies in multimorbidity, with potential implications for treatment burden,[213] and therefore possibly influencing medication adherence. Mode of drug delivery may also be an influencing factor, with different modes of delivery varying in convenience and, therefore, adherence.[214] Due to lack of individual patient data regarding treatment regimens, it is beyond the scope of this review to explore this further.

Illness and medication beliefs were among the most frequently explored predictors of adherence. Schüz and colleagues'[170] results suggest medication adherence in the context of multimorbidity may be influenced by perceptions of unique conditions as well as the effect of multiple combined illness perceptions. A scale to measure multimorbidity illness perceptions has been developed,[215] accounting for domains such as treatment burden, emotional impact and prioritisation. When tested within a study reviewed here, this measure did not predict adherence;[106] however, authors note the measure was in early stages of development at the time of administration.[106] Nonetheless, illness perceptions – and how they present in the context of multimorbidity – appear to be of relevance to both intentional and unintentional non-adherence in the context of multimorbidity.[168–170] This tendency to be influenced by illness beliefs may position intentional non-adherence as a rational judgement for some people self-managing chronic conditions; how this transpires as selective non-adherence among people self-managing multiple conditions should be explored in future research.[216]

Pharmacy data was the most frequently used method to measure non-adherence, followed by self-report and electronic monitoring. Variation in measurement methods contributed to the heterogeneity that prevented a larger meta-analysis – a limiting factor that has previously been identified in reviews of non-adherence to multiple medications.[217] While also noting variance due to different combinations of conditions within individual studies, the range of reported non-adherence differed by measurement method (77% for self-report, 66% for pharmacy data and 44% for MEMS).

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Results from ten studies using multiple methods to report prevalence of non-adherence emphasised this variability further, extending to multimorbidity the recommendation to use subjective and objective measures of non-adherence in combination to allow the strengths of each to compensate for the limitations of others.[218]

We did not stratify our analysis of prevalence by age, potentially limiting interpretation of our findings. Notably, age was not found to be a significant correlate in our synthesis of predictors, and it has been suggested previously that there is no consensus on age as a predictor of non-adherence.[219] Nevertheless, it may be plausible that variables are differentially associated with non-adherence at different life stages. For example, among reviewed studies, Domino and colleagues[40] reported lower levels of non-adherence in people with more conditions, with a mean age among participants of 43 years; Schüz and colleagues found higher intentional non-adherence among participants with more conditions, with a mean age of 73 years.[168] Exploring how non-adherence is variably predicted by clinical and psychosocial influences across the lifespan could inform intervention development which is appropriately targeted according to life stage.

The quality assessment produced a number of recommendations for future research exploring non-adherence among people with multimorbidity. Of note, while almost all studies (94%) were judged to have used non-adherence measures that were reproducible, 16% of studies used methods that were not considered to be valid. The validity of non-adherence assessment methods should be a priority for future related investigations. Furthermore, almost 30% of studies were judged not to have reduced serious recall bias. It has been suggested that the appropriate period for retrospective self-report of non-adherence should be balanced between optimising recall and capturing instances of infrequent non-adherence.[220] The optimum recall period for measures of multiple medication non-adherence should be determined.

The methods used to conduct this systematic review have limitations. Firstly, only 20% of titles and abstracts were screened by two reviewers and authors were not contacted for full-text articles or data that was not accessible, both of which increased the risk of excluding relevant studies. Secondly, criteria for inclusion in the review were limited to English language studies produced between 2009 and 2019. While this date range may exclude relevant articles, the review aimed to synthesise evidence produced since the expansion of multimorbidity literature over the last decade,[16] as well as a call for the careful design of adherence research in multimorbidity that uses reliable and consistent adherence measurement and details adherence for multiple conditions.[12] Non-English language articles were not reviewed for feasibility reasons. Thirdly, studies of people with depression or anxiety were eligible

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for inclusion if the condition was identified through a self-report measure. As a result, it is possible that not all participants had a clinical diagnosis. This decision was made to account for the overrepresentation and underdiagnosis of depression among people with chronic conditions due to incomplete recording in medical records,[221] and has been made previously when reviewing the multimorbidity literature.[222] Finally, while the study adopted 'two or more chronic conditions' to define multimorbidity, studies were also included if participants were prescribed medications that could be inferred as treatment for a chronic condition. While this extended the definition of multimorbidity beyond clinical diagnoses, to exclude these studies would have underrepresented the burden that is characteristic of multimorbidity.

CONCLUSION

This review summarises the complexity of non-adherence in multimorbidity, as well as factors that should be considered for the development of interventions to improve adherence for 'the most common chronic condition'. While the review did not identify specific combinations of conditions producing the highest levels of non-adherence, results suggest that non-adherence in multimorbidity varies both between and within studies. Management of multimorbidity demands integration across conditions and clinical guidelines, requiring practitioners and researchers to focus on individual people as opposed to individual diseases.[2] While the move towards this multi-disease focus is welcomed, attempts by researchers and clinicians to improve medication adherence should determine for which conditions individuals require most support. The use of measures designed to assess multiple medication adherence will be central to achieving this aim.

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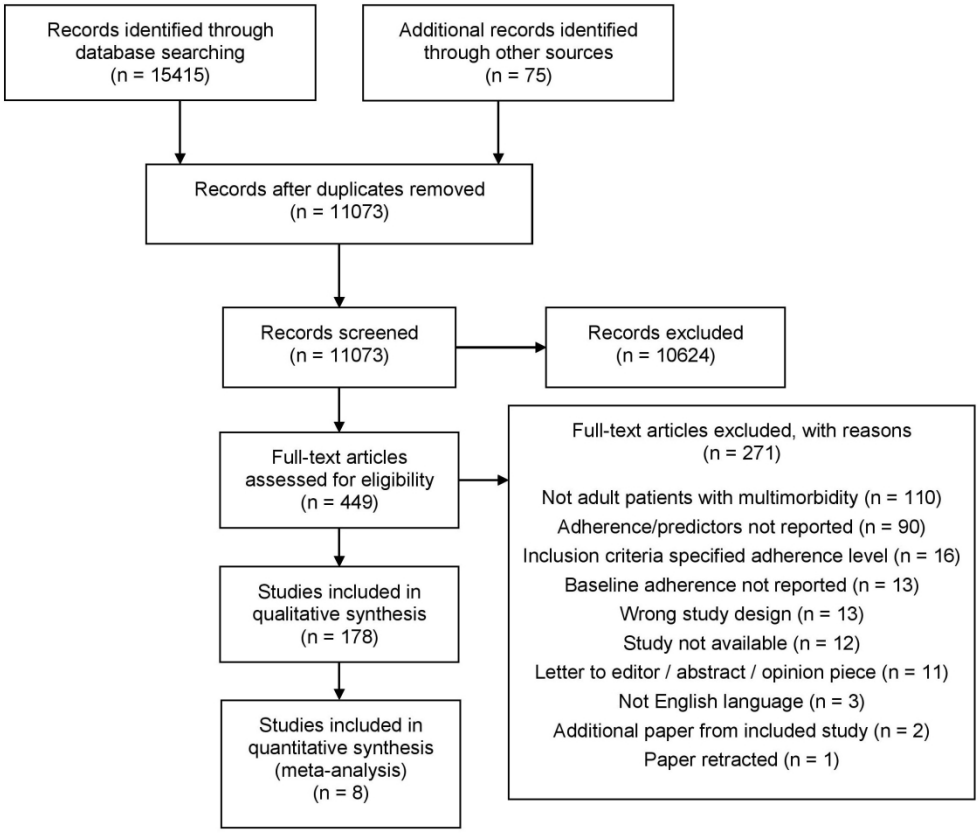


Figure 1. PRISMA flow diagram

199x177mm (300 x 300 DPI)

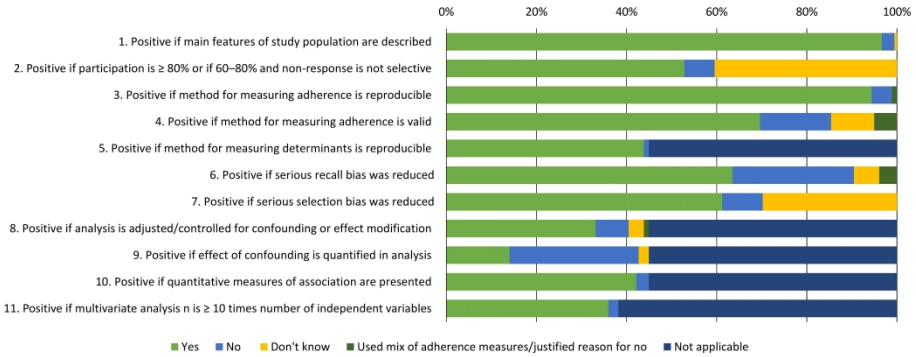


Figure 2. Methodological quality graph

556x218mm (300 x 300 DPI)

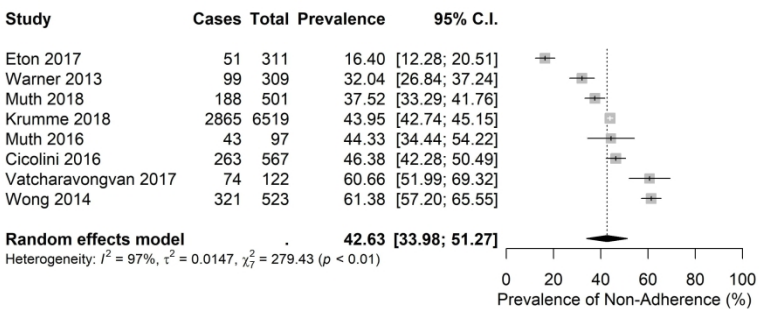


Figure 3. Results of random effects meta-analysis

254x127mm (400 x 400 DPI)

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	(i)
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	(ii)
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	1
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	1
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	2
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	2
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	2
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	2+SM
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	2+3
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	3
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	3
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	3
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	2
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	3

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	n/r
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	n/r
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	4
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	4+SM
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	4
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	5
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	5
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	n/r
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	n/r
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	14-16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	17
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	17

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

SM = supplementary material

n/r = not reported

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Search Strategies

PsycINFO:

1. multimorbid*.ab,ti
2. multi-morbid*.ab,ti
3. multidisease*.ab,ti
4. multi-disease*.ab,ti
5. multiple adj2 (ill* or disease* or condition* or syndrom* or disorder*).ab,ti
6. comorbidity/
7. comorbid*.ab,ti
8. co-morbid*.ab,ti
9. chronic* adj2 (disease* or ill* or care or condition* or disorder* or health* or medication* or syndrom* or symptom*).ab,ti
10. polypharma*.ab,ti
11. or/1-10
12. treatment compliance/
13. (medication adherence).ab,ti
14. adheren*.ab,ti
15. nonadheren*.ab,ti
16. non-adheren*.ab,ti
17. (non adheren*).ab,ti
18. complian*.ab,ti
19. noncomplian*.ab,ti
20. non-complian*.ab,ti
21. (non complian*).ab,ti
22. or/12-21
23. S11 AND S22
24. S23 (2009-present)

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CINAHL:

(MH “comorbidity”)
OR
TI (multimorbid*)
OR
TI (multi-morbid*)
OR
TI (multidisease*)
OR
TI (multi-disease*)
OR
TI (comorbid*)
OR
TI (co-morbid*)
OR
TI (multiple W2 ill*)
OR
TI (multiple W2 disease*)
OR
TI (multiple W2 condition*)
OR
TI (multiple W2 syndrom*)
OR
TI (multiple W2 disorder*)
OR
TI (chronic* W2 disease*)
OR
TI (chronic* W2 ill*)
OR
TI (chronic* W2 care)
OR
TI (chronic* W2 condition*)
OR
TI (chronic* W2 disorder*)
OR
TI (chronic* W2 health*)
OR
TI (chronic* W2 medication*)
OR
TI (chronic* W2 syndrom*)
OR
TI (chronic* W2 symptom*)
OR
TI (polypharma*)
OR
AB (multimorbid*)
OR
AB (multi-morbid*)
OR
AB (multidisease*)
OR
AB (multi-disease*)

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OR
AB (comorbid*)
OR
AB (co-morbid*)
OR
AB (multiple W2 ill*)
OR
AB (multiple W2 disease*)
OR
AB (multiple W2 condition*)
OR
AB (multiple W2 syndrom*)
OR
AB (multiple W2 disorder*)
OR
AB (chronic* W2 disease*)
OR
AB (chronic* W2 ill*)
OR
AB (chronic* W2 care)
OR
AB (chronic* W2 condition*)
OR
AB (chronic* W2 disorder*)
OR
AB (chronic* W2 health*)
OR
AB (chronic* W2 medication*)
OR
AB (chronic* W2 syndrom*)
OR
AB (chronic* W2 symptom*)
OR
AB (polypharma*)

AND

(MH "medication compliance")
OR
TI ("medication adherence")
OR
TI (adheren*)
OR
TI (nonadheren*)
OR
TI (non-adheren*)
OR
TI ("non adheren*")
OR
TI (complian*)
OR
TI (noncomplian*)
OR

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TI (non-complian*)
OR
TI ("non complian*")
OR
AB ("medication adherence")
OR
AB (adheren*)
OR
AB (nonadheren*)
OR
AB (non-adheren*)
OR
AB ("non adheren*")
OR
AB (complian*)
OR
AB (noncomplian*)
OR
AB (non-complian*)
OR
AB ("non complian*")

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EMBASE:

'multiple chronic conditions'/exp
OR
multimorbid*:ab,ti
OR
multi-morbid*:ab,it
OR
multidisease*:ab,ti
OR
multi-disease*:ab,ti
OR
((multiple) NEXT/2 (ill* OR disease* OR condition* OR syndrom* OR disorder*)):ab,ti
OR
comorbidity/exp
OR
comorbid*:ab,ti
OR
co-morbid*:ab,ti
OR
((chronic) NEXT/2 (disease* OR ill* OR care OR condition* OR disorder* OR health* OR medication* OR syndrom* OR symptom*)):ab,ti
OR
polypharma*:ab,ti

AND

medication compliance/exp
OR
‘medication adherence’:ab,ti
OR
adheren*:ab,ti
OR
nonadheren*:ab,ti
OR
non-adheren*:ab,ti
OR
'non adheren*':ab,ti
OR
complian*:ab,ti
OR
noncomplian*:ab,ti
OR
non-complian*:ab,ti
OR
'non complian*':ab,ti

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PubMed:

multimorbidity[MeSH]
OR
multimorbid*[Title/Abstract]
OR
multi-morbid*[Title/Abstract]
OR
multidisease*[Title/Abstract]
OR
multi-disease*[Title/Abstract]
OR
comorbidity[MeSH]
OR
comorbid*[Title/Abstract]
OR
co-morbid*[Title/Abstract]
OR
"chronic medication*"[Title/Abstract]
OR
polypharma*[Title/Abstract]

AND

"medication adherence"[MeSH]
OR
adheren*[Title/Abstract]
OR
nonadheren*[Title/Abstract]
OR
non-adheren*[Title/Abstract]
OR
"non adheren*"[Title/Abstract]
OR
complian*[Title/Abstract]
OR
noncomplian*[Title/Abstract]
OR
non-complian*[Title/Abstract]
OR
"non complian*"[Title/Abstract]

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Data Extraction Form

date form completed (dd/mm/yyyy)
ID of person extracting data (initials)
study ID (author and year)
country of publication
citation (APA format)
study aims
study design
study setting
chronic conditions studied
sample size
age of participants (mean (sd))
gender of participants (% female)
definition of multimorbidity (if applicable)
measure of medication non-adherence
definition of medication non-adherence
proportion of non-adherent participants (or mean (sd) if proportion not reported)
predictors of medication non-adherence (if reported)
key conclusions by study authors

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Meta-analysis of Proportions: R Script

```
library(metafor)
library(meta)
#read in the data file
dat=read.csv("data.csv", header=T, sep=",")
#set-up file without transformation
ies=escalc(xi=cases, ni=total, data=dat, measure = "PR")
#meta-analysis with DerSimonian Laird
pes=rma(yi, vi, data=ies, method = "DL")
#view results
print(pes)
#compute and display confidence intervals of tau-squared and I-squared
confint(pes)
#create simple forest plot
pes.summary=metaprop(cases, total, authoryear, data=dat, sm="PRAW",
method.tau="DL", method.ci="NAsm")
forest(pes.summary, xlim=c(0,100),pscale=100, rightcols = FALSE,
leftcols=c("studlab", "event", "n", "effect", "ci"), leftlabs=c("Study", "Cases",
"Total", "Prevalence", "95% CI.I"), xlab="Prevalence of Non-Adherence", smlab="",
weight.study="random", squaresize=0.5, col.sqaure="black", col.squar.lines="black",
col.diamond="black", col.diamond.lines="black", pooled.totals=FALSE,
comb.fixed=FALSE, fs.hetstat=10, print.tau2=TRUE, print.Q=TRUE, print.pval.Q=TRUE,
print.I2=TRUE, digits=2)
#leave one out to identify outliers
L10=leave1out(pes); print(L10)
#visualise change in summary score with forest plot
l1o=leave1out(pes)
yi=l1o$ estimate; vi=l1o$se^2
forest(yi, vi, slab=paste(dat$author, dat$year, sep=","), refline=pes$b, xlab =
"Summary proportions leaving out each study")
#see which studies are influential using leave-one-out diagnostics
inf=influence(pes)
print(inf)
plot(inf)
```

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Study Characteristics Table

Abbreviations:

COPD = chronic obstructive pulmonary disease
HIV/AIDS = human immunodeficiency virus/acquired immunodeficiency disease
TB = tuberculosis
ACE-I = angiotensin-converting-enzyme inhibitors
HAART = highly active anti-retroviral therapy
ART = antiretroviral therapy
MM = multimorbidity
PDC = proportion of days covered
MPR = medication possession ration
CMG = cumulative mean gap interval
MEMS = medication event monitoring system
VAS = visual analogue scale
MMAS = Morisky Medication Adherence Scale [1,2]
MARS = Medication Adherence Rating Scale[3]
CASE Index = Center for Adherence Support Evaluation [4]
HADS = Hospital Anxiety and Depression Scale[5]
BDI = Beck Depression Inventory [6]
PHQ = Patient Health Questionnaire [7]
MINI = Mini International Neuropsychiatric Interview [8]
CES-D = Centres for Epidemiologic Studies Depression Scale[9]
GDS = Geriatric Depression Scale [10]
SD = standard deviation
IQR = inter quartile range
95% CI = 95% confidence interval
RCT = randomised controlled trial

Note 1: where *n* non-adherent was not reported, *n* was calculated from the reported % non-adherent.
Note 2: where *n* female was not reported, *n* was calculated from the reported % female/male.
Note 3: pooled mean (SD) age was calculated where mean (SD) age was reported for two groups.
Note 4: participants with depression had a clinical diagnosis unless a self-report measure is noted.

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Study (country)	Setting	Design	Chronic Conditions	Sample Size	Adherence Measure	Non-adherence definition	N (%) non-adherent/adherence outcome reported
Agarwal 2009 (USA)[11]	Healthcare database	Retrospective cohort	- Treated with antihypertensive - Treated with lipid-lowering agent	15400	MPR (antihypertensive (AH); lipid-lowering (LL) agent)	MPR < 0.8 for one or more medications	2 months, 6 months, 12 months: Cohort 1 (AH before LL): 40.6%, 67.3%, and 68.7% Cohort 2 (LL before AH): 55.0%, 69.2%, and 69.0% Cohort 3 (AH/LL simultaneously): 24.8%, 65.6%%, and 66.0%
Albrecht 2017 (USA)[12]	Medicare database	Retrospective cohort	- COPD - Depression	16075	PDC (COPD maintenance medications; antidepressants)	PDC < 0.8 (also report PDC = 0)	Person months: COPD: 78.8% Antidepressants: 45.2%
Albrecht 2016 (USA)[13]	Medicare database	Retrospective cohort	- COPD - Depression	31033; MM = 9593	PDC (COPD maintenance medications)	PDC < 0.8 (also report other categories)	7674 (80%)
An 2013 (USA)[14]	Administrative claims and electronic medical records	Retrospective observational	- Hypertension - Type 2 diabetes	2334	PDC (diabetes medications and hypertension medications simultaneously)	PDC < 0.8 for both drugs simultaneously	1671 (71.6%)
Applebaum 2009 (USA)[15]	HIV/AIDS primary care	Baseline data from longitudinal study	- HIV - Substance related disorder - Psychiatric disorder	67	Medication Adherence Questionnaire	Adherence < 95%	20 (29.85%)
Applebaum 2009 (USA)[16]	Healthcare clinics	Cross-sectional (data from RCT screening)	- HIV - History of injection drug use/dependence	108	MEMS (HAART); AIDS Clinical Trials Group Adherence Questionnaire (HAART)	Report mean	Mean MEMS = 68.21% Mean self-report = 93.82%
Axon 2016 (USA)[17]	Veterans health administrative databases	Retrospective cohort	- Type 2 diabetes - Depression	740197; MM = 88895	MPR (diabetes medications)	MPR < 0.8	28143 (31.7%)
Baca-Garcia 2009 (USA)[18]	University hospital	Cross-sectional	- Current bipolar major depressive episode - Alcohol use disorder	97; MM = 39; analysed n = 36	Self-report (antidepressants)	Report mean (SD) median global compliance	Mean (SD) = 90.7 (14.0)
Badiee 2012 (USA)[19]	Ongoing studies at university research centre	Cross-sectional	- HIV - Bipolar disorder I or II	74; MM = 43	MEMS (protease inhibitor or related) AIDS Clinical Trials Group (ACTG) 4-day Adherence to	MEMS < 90% ACTG < 95%	MEMS: 22 (51%) ACTG: 6 (13%)

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					Antiretrovirals Questionnaire (antiretroviral medications) Visual Analogue Scale (protease inhibitor)	Visual Analogue Scale < 95%	Visual Analogue Scale: 22 (52%)
Baik 2012 (USA)[20]	Medicare database	Retrospective cohort	- Depression - Heart Failure	552956; MM = 22118; analysed n = 13981	MPR (heart failure medications; antidepressants)	Report mean	No coverage group: Antidepressants: Mean MPR = 0.56 Heart failure medications: Mean MPR = 0.89 Generic coverage group: Antidepressants: Mean MPR = 0.60 Heart failure medications: Mean MPR = 0.90
Banegas 2018 (USA)[21]	Healthcare database	Retrospective cohort	- Treated with statin - Newly diagnosed cancer (breast / prostate / colorectal)	10177	PDC (statins)	Partial adherence = PDC 0.20-0.79 Non-adherence = PDC < 0.2	Year 1: PDC < 0.20 = 837 (8.2%) PDC 0.20-0.79 = 2280 (22.4%) Year 2: PDC < 0.20 = 887 (8.7%) PDC 0.20-0.79 = 2155 (21.2%)
Basak 2014 (USA)[22]	Healthcare database	Longitudinal observational	- Type 2 diabetes - Hypertension and/or dyslipidaemia or angina	6043	PDC	Report mean (SD)	Report according to medication combinations per quarter. Range = 24.43 (39.10) for antianginal medications to 86.40 (23.27) for hypertension medications.
Batchelder 2014 (USA)[23]	Community and hospital	Cross-sectional	- HIV - Type 2 diabetes	62	VAS (antiretrovirals; diabetes medications; insulin injections where appropriate)	VAS < 100%	ARV: 32 (51.6%) Antidiabetes: 39 (63%) Of 32 participants reporting <100 % to HIV medication, 97 % (31) reported < 100 % to diabetes medication
Bauer 2013 (USA)[24]	Healthcare database	Observational new prescription cohort follow-up	- Type 2 diabetes - Depression	1366	Pharmacy utilisation data (antidepressants)	Primary non-adherence: no dispensing of index prescription within 60 days of index prescription date. Non-persistence: dispensing of index	Primary non-adherence: 52 (3.8%) Early non-persistent: 432/1314 (32.9%) 2nd stage non-persistent: 83/1314 (6.3%)

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					prescription once, no additional dispensing within period defined by number of days' supply of medication dispensed plus a grace period equal to 100% of prior days' supply.	
Beadles 2015 (USA)[25]	Medicaid database	Retrospective cohort	≥ 2 of the following: major depressive disorder, schizophrenia, diabetes mellitus, hypertension, hyperlipidaemia, seizure disorder, asthma, chronic obstructive pulmonary disease	depression = 9303; hypertension = 12595; diabetes = 6409; hyperlipidaemia = 9263	PDC (target medication)	Report mean (SD)	Mean (SD): Depression = 40% (43%) Hypertension = 50% (43%) Diabetes = 49% (43%) Hyperlipidaemia = 47% (43%)
Benner 2009 (USA)[26]	Healthcare database	Retrospective claims analysis	- Treated with antihypertensives - Treated with lipid-lowering therapy	5759	PDC (for combination therapy, reported according to prescription burden)	PDC < 80%	By number of other medications: 0 = 59.0% 1 = 65.5% 2 = 69.9% 8 = 79.9% 9 = 74.5% 10/+ = 79.9%
Berg 2018 (Denmark)[27]	Hospital	Cross-sectional	- Cardiac Disease - Depression/anxiety (HADS)	14239; MM (anxiety) = 1978; MM (depression) = 982	Single item self-report (cardiac medication)	Forgetting to take prescribed medication more than once a week among patients who took medication prior to hospital admission	Anxiety group = 113 (7%) Depression group = 70 (8%)
Bifftu 2015 (Ethiopia)[28]	Hospital outpatient	Cross-sectional	- Epilepsy - Depression (BDI-II)	405; MM = 183	Single item self-report (antiepileptic medications)	Not defined (reported as number with difficulties of adherence to antiepileptic drugs)	81 (44.3%)

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Blashill 2015 (USA)[29]	HIV primary care clinics	Baseline data of RCT screening	- HIV - Depressive symptoms (PHQ-2)	333	MEMS (antiretrovirals)	MEMS < 80%	191 (57.4%)
Bogner 2014 (USA)[30]	Primary care	Longitudinal	- Type 2 diabetes - Depression	180	MEMS (hypoglycaemic agents and antidepressants) Brief Medication Questionnaire (BMQ) (hypoglycaemic agents and antidepressants)	MEMS < 80% BMQ < 80%	Oral hypoglycaemic agent: MEMS = 103 (57.22%) Self-report = 34 (18.89%) Antidepressants: MEMS = 111 (61.67%) Self-report = 47 (26.11%)
Cabrera-Marroquin 2014 (Mexico)[31]	Tertiary care (ambulatory)	Observational cross-sectional	- Rheumatoid arthritis; - Major depressive disorder	78; MM = 24	Concordance with Therapy self-report measure of adherence and persistence (RA medication)	Based on combinations of item scores	10 (41.7%)
Calderón-Larrañaga 2016 (Spain)[32]	Primary care	Cross-sectional	-Hypertension -Physical and/or mental comorbidity	113,397; MM with physical co-morbidity = 103913; MM with mental comorbidity = 20349	MPR (antihypertensives)	MPR < 80%	+ physical comorbidity = 20868 (20.08%) + mental comorbidity = 4387 (21.56%)
Calip 2013 (USA)[33]	Healthcare database	Retrospective cohort	- Breast cancer survivors - Treated with statins	1393	Adherence: MPR (statins) Persistence: discontinuation rate (statins)	MPR < 0.8 Discontinuation rate (DR): gap ≥ 90 days between end of previous days' supply and subsequent dispensing of statin medication	Cancer treatment period: MPR < 0.8 = 666/1384 (48.1%) Mean (SD) DR = 0.5 (0.6) Year +1: MPR < 0.8 = 572/1129 (50.7%) Mean (SD) DR = 1.2 (1.3) Year +2: MPR < 0.8 = 672/1040 (64.6%) Mean (SD) DR = 1.2 (1.2) Year +3: MPR < 0.8 = 477/959 (49.7%) Mean (SD) DR = 1.2 (1.2)
Calip 2015 (USA)[34]	Healthcare database	Retrospective cohort	- Diabetes mellitus - Breast cancer	509; analysed n = 499	MPR (oral diabetes medications)	MPR < 0.8	376 (75.4%)

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Calip 2017 (USA)[35]	Healthcare database	Retrospective cohort	- Breast cancer survivors - Treated with at least one of antihypertensives, oral antidiabetes medications or statins	2308	MPR (antihypertensive or statins or oral antidiabetic medications)	MPR < 0.8	Antihypertensives = 666/1779 (37.4%) Oral diabetes medications = 379/499 (76.0%) Statins = 416/1072 (38.8%)
Cantudo-Cuenca 2014 (Spain)[36]	Hospital pharmacy service	Prospective observational	- HIV - Comorbidity(s)	594; MM = 431	MMAS-4 Pharmacy dispensing records: Number of medications dispensed/number of medications prescribed x 100	MMAS-4 < 4 Pharmacy data < 90%	129 (29.9%) (results show pharmacy data; authors reported no difference in adherence between both measures)
Casaletto 2016 (USA)[37]	Not reported	Cross-sectional	- HIV - Bipolar I or II	50	MEMS (bipolar medications)	MEMS < 90%	28 (56.0%)
Caughey 2013 (Australia)[38]	Veterans' Affairs database	Retrospective cohort	- Diabetes - Depression	29710; MM = 7171	Pharmacy dispensing data – treatment discontinuation (antidiabetic medications)	Gap between two dispensings of more than three times the duration	2868 (40%) Median (95% CI) = 1.81 (1.72-1.94)
Cha 2012 (USA)[39]	Not reported	Cross-sectional using baseline data from intervention	- Heart failure - Diabetes mellitus	116; MM = 46	MMAS-8 (heart failure medications) MEMS (ACE inhibitor and diuretic)	Report mean (SD): MMAS-8 range = 1-12; higher score = more adherent	MMAS-8 mean (SD) = 10.6 (1.6) MEMS: correct doses: ACE-I = 88.0 (16.6) Diuretics = 89.0 (17.0) MEMS: correct days: ACE-I = 81.7 (22.8) Diuretics = 81.1 (22.6)
Chou 2017 (USA)[40]	Healthcare database	Retrospective cohort	- Depression - Breast cancer (newly diagnosed)	2284; MM = 1142	PDC (antidepressants)	PDC < 80%	494 (43.3%)
Cicolini 2016 (Italy)[41]	Primary care	Cross-sectional	≥ 2 chronic condition from cardiovascular, neurological, respiratory, musculoskeletal, oncologic diseases or diabetes mellitus	567	MMAS-8 (Italian Version)	MMAS-8 score between 1-5.75 (defines "low adherence")	263 (46.4%)

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Crowley 2013 (USA)[42]	Community practice	Cross-sectional (baseline data from RCT)	- Type 2 diabetes - Hypertension	377	MMAS-4 (hypertension medications; diabetes medications)	Score ≤ 2 on any item	Hypertension = 110 (29.2%) Diabetes = 162 (43.0%)
Crowley 2015 (USA)[43]	Tertiary care health system	Cross-sectional (baseline data from RCT)	- Acute myocardial infarction in previous 3 years - Hypertension	405	MMAS-4 (general adherence)	Response of "strongly agree," "agree," "don't know," or "refused" for any statements	173 (42.7%)
Cruess 2012 (USA)[44]	Physician outpatient clinics	Prospective cohort	- HIV - Treated with psychotropic medication	324; MM = 106	Monthly unannounced pill counts (HIV medication; psychiatric medication)	Report mean (SD)	HIV = 84.09% (19.26%) Psychiatric = 79.86% (20.88%)
Cziraky 2015 (USA)[45]	Healthcare database	Retrospective cohort	- Acute coronary syndrome or unstable angina - Type 2 diabetes	Year 1 = 85097; MM = 22588 Year 3 = 35540; MM = 9160	MPR (cardiovascular medications) Treatment discontinuation (cardiovascular medications)	MPR: not reported Treatment discontinuation: gap in treatment of 45 days or longer in each medication based on days of supply	Oral diabetes medication: Year 1 = 49.3% Year 3 = 46.8% Insulin Year 1 = 77.4% Year 3 = 78.9% Additional medications reported in original paper.
Daniali 2018 (Iran)[46]	Government healthcare centres	Cross-sectional	- Chronic condition - Depression (PHQ-9)	483; MM = 79	4-item self-report	Not reported	40 (50.6%)
Daniels 2016 (USA)[47]	Primary care	Pilot intervention	≥ 2 chronic conditions	20	MMAS-8	Report mean (SD) MMAS-8 score > 2 (defined as "low adherence"); lower score = lower non-adherence	Mean (SD) = 4.65 (2.37)
Davies 2015 (USA)[48]	Tertiary care Veterans Affairs Medical Centre (patients receiving primary care)	Retrospective chart review	- HIV/AIDS - Diabetes mellitus	56	MPR (ART)	MPR < 95%	34 (60.7%)
Davis 2011 (USA)[49]	Clinical centres	Data from RCT	- Hypertension	1065	Not reported (antihypertensives)	Report mean (SD)	Lower BP goal group: Achieved goal: 92.6% (7.7%)

			- Chronic kidney disease				Above goal: 83.0% (14.7%) Usual BP goal group: At goal: 91.4% (8.8%) Below goal: 90.8% (13.2%) Above goal: 78.5% (19.1%)
DeLeon 2015 (USA)[50]	Primary care	Retrospective cohort	- Diabetes - Hypertension	20085; MM = 3827	MPR (diabetes medications, hypertension medications, lipid controlling medications)	MPR < 80%	Diabetes: 1056/1679 (62.9%) Hypertension: 1084/1923 (56.4%) Lipid controlling: 928/1295 (71.7%)
Desai 2014 (USA)[51]	Medicaid database	Retrospective analysis	- Treatment by antipsychotic - Treatment by antidiabetic medication	3642; MM = 1821	PDC (oral antidiabetics; oral antipsychotics) Persistence (oral antidiabetics)	PDC < 0.8 Persistence: number of days of continuous access to oral antidiabetics, allowing for a 30-day gap period.	Antidiabetics: 1143 (62.8%) Mean (SD) persistence = 180.8 (124.8) Antipsychotics: 689 (37.8%)
Dhamane 2017 (USA)[52]	Healthcare database	Retrospective cohort	- COPD - Treatment by ≥ 1 of antianxiety, anticoagulants, antidepressants, antihyperglycaemics, antihypertensives, beta-blockers, bisphosphonates, calcium channel blockers, diuretics, insulin, NSAIDs, statins	14117	PDC (COPD medications; non-COPD medications)	PDC < 0.8 Report mean PDC for non-COPD medications	COPD medications: 11177 (79.2%) non-COPD medications mean: antianxiety = 0.39 anticoagulants = 0.55 antidepressants = 0.65 antihyperglycaemics = 0.70 antihypertensives = 0.71 beta-blockers = 0.70 bisphosphonates = 0.54 calcium channel blockers = 0.68 diuretics = 0.65 insulin = 0.53 NSAIDs = 0.33 statins = 0.67
Domino 2014 (USA)[53]	Administrative claims databases	Retrospective secondary data analysis	≥ 2 chronic conditions from asthma, COPD, diabetes, hypertension, hyperlipidaemia, seizure disorder, depression, schizophrenia	188531	PDC	Report mean PDC	Diabetes = 0.29 Hyperlipidaemia = 0.29 Hypertension = 0.43 Depression = 0.33 Schizophrenia = 0.46

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Druss 2010 (USA)[54]	Community mental health centre	Pilot RCT	- ≥ 1 chronic medical condition - Severe mental illness	80	MMAS-4 (chronic medical condition)	Report mean (SD) score from 0 (no problems with adherence) to 4 (more problems with adherence)	Mean (SD) = 1.5 (1.3)
DuChane 2014 (USA)[55]	Pharmacy claims database	Retrospective cohort	- HIV - Comorbidity (treated with angiotensin-converting enzyme (ACE) or angiotensin receptor blocker (ARB) or statin)	ACE/ARB group = 2968; MM = 1484; Statins group = 2744; MM = 1372	PDC	ART medications: PDC < 95% Other medications: PDC < 80%	HIV + ACE/ARBs = 1118 (37.7%) HIV + statins = 936 (34.1%) Mean (SD) ART = 83.9 (24.3)
Eton 2017 (USA)[56]	Primary care	Cross-sectional	≥ 2 chronic conditions requiring self-management	332 (analysed n = 311)	Single-item self-report	Response "usually /sometimes" vs "always" re adherence to physician-recommended medications	Usually/sometimes = 51 (16.4%)
Farley 2012 (USA)[57]	Medicaid database	Retrospective cohort	- Schizophrenia - Pre-existing cardiometabolic condition(s): hypertension, hyperlipidaemia and/or diabetes	1006	PDC (antipsychotic medications; cardiometabolic medications)	PDC < 80% (unweighted mean of 3 cardiometabolic classes) Report mean (SD)	Antipsychotic (acute phase) = 89 (18.8%) Diabetes medications: acute phase = 0.90; continuation phase = 0.63; maintenance phase = 0.59 Antihypertensives: acute phase = 0.86; continuation phase = 0.71; maintenance phase = 0.66 Statins: acute phase = 0.79; continuation phase = 0.65; maintenance phase = 0.57
Feldman 2016 (USA)[58]	Not reported (recruited from outpatients, emergency room, referrals and flyers)	RCT	- Panic disorder - Asthma	48	MARS (inhaled corticosteroids)	Report mean (SD) Range = 1-5; higher score = lower non-adherence. Score 4.5+ = good adherence.	Mean (SD) = 3.8 (0.4)

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Ferguson 2017 (Australia)[59]	Cardiology ward at academic medical centre	Prospective cohort	- Chronic heart failure; - Atrial fibrillation	137; analysed n = 83	MMAS-4	≥ 1 responses to non-adherence items	36 (43.4%)
Foguet-Boreu 2017 (Spain)[60]	Primary care	Cross-sectional	- Hypertension - Diabetes type 1 or type 2	27637	MPR (hypoglycaemic medications; antihypertensive medications; both combined)	MPR < 80% Report MPR ≥ 80%	Diabetes = 10898 (39.8%) Hypertension = 4322 (23.8%) Combined diseases (MPR < 80% for at least one condition) = 13593 (49.2%)
Gathright 2017 (USA)[61]	Hospital	Secondary analysis of data from longitudinal observational study	- Heart Failure - Depressive Symptoms (PHQ-9)	308; MM = 107	MEMS (up to 4 common heart failure medications)	Report mean (SD) percent of days a participant did not follow their prescribed regimen	Mean (SD) = 65.3% (27.5%)
Gaudiano 2017 (USA)[62]	Psychiatric hospital	Cross-sectional (study of baseline data from clinical trial)	- Bipolar I or II or NOS - Drug and/or alcohol use disorder	39	Medication Compliance Questionnaire (mean rating across all psychotropic medications)	Report mean (SD) Range = 0-4, higher score = higher non-adherence.	Mean (SD) = 1.2 (1.2)
Gaynes 2015 (USA)[63]	Infectious disease clinics	Data from RCT	- HIV - Depression (PHQ-9 and major depressive disorder (MINI)	304	Visual Analogue Scale (HIV/AIDS medications)	VAS score ≤ 80%	78 (25.7%) No psychiatric comorbidity = 13/54 (20.1%) ≥ 1 psychiatric comorbidity = 65/250 (26.0%)
Gentil 2015 (Canada)[64]	Community	Data from longitudinal study	- Type 2 diabetes - Depression/anxiety	301; MM = 47	MPR (oral antidiabetic medication)	MPR < 80%	18 (38.3%)
Gibson (USA) 2018[65]	Primary care	Cross-sectional	- Type 2 diabetes - Hypertension	60	Self-report during semi-structured interview	Self-report missing 1 or more days in previous week	Hypertension: 14 (23.3%) Type 2 diabetes: 16 (26.7%)
Gnanasan 2011 (Malaysia)[66]	Hospital outpatients	Feasibility study (action research)	- Diabetes mellitus - TB	35	Medication-related problems identified by clinical pharmacist (TB medications; non-TB medications)	Not reported	TB treatment: 3 (8.6%) Non-TB treatment: 10 (28.6%)
Gonzalez 2011 (USA)[67]	Methadone maintenance treatment	Cross-sectional	- HIV - History of injection drug use	91	MEMS (HAART)	MEMS < 100% doses as prescribed	73 (80.2%)

Grove 2017 (USA)[68]	Medicaid database	Retrospective cohort	- Schizophrenia - ≥ 1 chronic condition from hypertension, hyperlipidaemia, seizure disorder, chronic obstructive pulmonary disease, diabetes, asthma, or major depression	12662	PDC (antipsychotic medications)	Report mean (SD)	Mean (SD) = 45.4% (37.8%)
Gundling 2013 (Germany)[69]	Hospital	Retrospective analysis	- Diabetes mellitus - Cirrhosis	285; MM = 87; analysed n = 69	Information about compliance based on doctors', patients' and relatives' reports (report general adherence)	"e.g. concerning regular intake of medication and following the doctor's advice"	36 (52.2%)
Gupte-Singh 2015 (USA)[70]	Medicaid database	Retrospective analysis	- Asthma - Depression	3626; MM = 226	PDC (asthma controller; antidepressant medications)	Asthma controller: PDC < 50% Antidepressant: PDC < 80%	Asthma controller medications: 140 (62.0%) Antidepressants: 168 (74.3%)
Halava 2014 (Finland)[71]	Data from public sector employees	Prospective cohort	- Prescribed statins - Cardiovascular comorbidities	9285; MM = 2827	PDC (statins)	PDC < 80%	1155 (40.9%)
Han 2018 (USA)[72]	Medicare database	Retrospective cohort	- Hypertension - Chronic kidney disease	70201	PDC (antihypertensive medications)	PDC < 80%	27468 (39.1%)
Hansen 2009 (USA)[73]	Ambulatory care practice	Secondary analysis of RCT data	- Heart failure - Depression (GDS)	314; MM = 117	MEMS (heart failure medication) MMAS-4 + one-item screening questionnaire (heart failure medication)	Report mean (95% CI) MMAS-4: Range = 0-1; higher score = better adherence	MMAS-4 = 79% (71-87%) MEMS not reported
Hansen 2018 (USA)[74]	Medicaid database	Retrospective cohort	- Schizophrenia - Cardiometabolic condition (hypertension, hyperlipidaemia and/or diabetes)	4223	PDC (antihypertensive medications; statins; oral antidiabetics; antipsychotic medications)	PDC < 80%	Range of non-adherence reported: Antipsychotic = 24.4% - 44.5% Antidiabetic = 39.4% - 61.5% Antihypertensive = 32.5% - 52.6% Statins = 44.5 – 63.5% Full results reported according to number of primary care prescribers

							and psychiatric specialist prescribers in original paper.
Hansen 2012 (USA)[75]	Medicaid database	Cross-sectional time-series	- Schizophrenia - Cardiometabolic condition (hypertension, hyperlipidaemia and/or diabetes)	33230	PDC (oral second-generation antipsychotics; oral antidiabetics; statins; antihypertensive medications)	PDC < 0.8	Antipsychotic: 11565/33230 (34.8%) Antihypertensives: 10069/24843 (40.5%) Antidiabetics: 4869/12349 (39.4%) Antihyperlipidemics: 4192/10909 (38.4%)
Hansen 2014 (USA)[76]	Veterans' Administration Medical Centre	Retrospective cohort	- Cardiometabolic condition (hypertension, hyperlipidaemia and/or diabetes) - ≥ 1 additional chronic condition(s)	7933; Diabetes MM = 1417; Hypertension MM = 3318; Dyslipidaemia MM = 3009	CMG (diabetes medications; hypertension medications; dyslipidaemia medications)	CMG ≥ 20%	Diabetes (+ ≥ 1 other): 392 (27.7%) Hypertension (+ ≥ 1 other): 766 (23.1%) Dyslipidaemia (+ ≥ 1 other): 948 (31.5%)
Hapangama 2013 (Sri Lanka)[77]	Tertiary care psychiatric unit (inpatient and outpatient)	Cross-sectional	- Mental disorder - Substance use disorder	325; MM = 141	Interview schedule administered by senior registrar in psychiatry	Ceased medication for ≥ 2 days for any reason, without medical advice, during treatment for mental illness	91 (64.5%)
Hawke 2013[78]	Academic medical centres primarily	RCT (secondary data analysis)	- Bipolar I or II - Comorbid anxiety disorder	204; MM = 85	Single-item self-report	Report mean (SD) (possible range not reported)	Mean (SD) = 1.36 (0.53)
Hussein 2010 (USA)[79]	Healthcare database	Retrospective cohort	- Hypertension - Using a two-pill amlodipine + atorvastatin; using a two-pill CCB + statin; or using a single pill amlodipine/ atorvastatin	35430	PDC (statins; calcium channel blocker)	PDC < 80% (both medications together)	Range of non-adherence reported: 39.1 - 73.4% Full results according to status of experienced/naïve with respective medications reported in original paper.
Ihle 2017 (Switzerland)[80]	Primarily hospital	Cross-sectional	≥ 2 chronic conditions	84	Self-report items adapted from Moriskey, Green, Levine measure (specifically, assesses	Report mean (SD) from 1-5 (higher score = higher non-adherence)	Omission errors: mean (SD) = 1.39 (0.88)

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					omission errors and commission errors)		Commission errors: mean (SD) = 1.12 (0.57)
Inauen 2017 (Switzerland) [81]	University hospital (outpatient) and other channels, e.g. online, pharmacy	3-wave panel study	≥ 2 chronic conditions and ≥ 2 prescribed medications	84	Non-Adherence Report, Medication Based (NAR-M) Non-Adherence Report, Diary (NARD) MARS MEMS	Report mean (SD) (lower mean = higher non-adherence)	NAR-M: T1 = 92.8 (11.1); T2 = 92.1 (12.4); T3 = 94.5 (10.5) NARM-D (no MEMS group) = 96.9 (7.9) MEMS group = 98.5 (4.1) MARS T1 = 89.4 (18.3) T2 = 90.1 (18.1) T3 = 94.5 (13.3) MEMS T1-T2 = 95.7 (3.8) T2 – T3 = 93.7 (10.4)
Iyengar 2016 (USA)[82]	Medicare database	Retrospective claims analysis	- Diabetes - Hypertension - High blood cholesterol	40632	PDC (diabetes medications, cholesterol medications, hypertension medications)	PDC < 80%	Antidiabetics = 13898 (34.2%) Antihypertensives = 13462 (33.1%) Antihyperlipidemics = 10584 (26.1%)
James 2018 (USA)[83]	Hospital	Longitudinal	- Type 2 diabetes - Early stage cancer (recently diagnosed)	56	MMAS-8 (diabetes medications)	MMAS: positive response to ≥ 2 questions	Baseline: 25 (44.6%); 3-months: 20 (35%)
Jaworski 2011 (France)[84]	University hospital psychiatric department (outpatient)	Cross-sectional	- Schizophrenia or bipolar disorder - Substance use disorder	80; MM = 40: schizophrenia + SUD= 20; bipolar + SUD = 20	Self-report number of times forgot to take medication in preceding month (psychotropic medications)	Score < 75%	Bipolar + SUD = 15 (75%); Schizophrenia + SUD = 14 (70%)
Jing 2011 (USA)[85]	Medicare and healthcare databases	Retrospective analysis	- Treatment with antihypertensive - Treatment with antidiabetic	30132	MPR	MPR < 77% for ≥ 1 antihypertensives and/or antidiabetic medications for each standalone calendar quarter	10722 (35.6%)
Joseph 2017 (India [86]	Medical college hospital & dialysis centre	Prospective observational	- Hypertension - Chronic kidney disease	120	MMAS-8 (antihypertensives)	Not reported	75 (62.5%)

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	(inpatient and outpatient)						
Kacanek 2010 (USA)[87]	Physician network, newspapers, radio advertisement	Secondary analysis of data from longitudinal cohort	- HIV - Depression symptoms (CES-D and Diagnostic Interview Schedule)	225; MM= 51	For each medication, participants reported the number of days over the last seven days they had missed a dose of that antiretroviral medication.	Missing at least 5% of HAART doses in the seven days prior to study	23 (45.1%)
Kalichman 2015 (USA)[88]	AIDS community services and infectious disease clinics	Prospective	- HIV - Mental health condition	123	Unannounced pill-counts (antiretroviral medications; psychiatric medications)	Antiretrovirals: < 85% Psychiatric medications: report mean (SD)	ART: 57 (46.3%) ART mean (SD) = 77.2 (25.3) Psychiatric mean (SD) = 64.6 (36.4)
Kamal 2018 (Switzerland)[89]	University hospital infectious disease service	Prospective observational cross-sectional	- HIV - Comorbidity (≥ 1 concomitant oral medication from any therapeutic class for at least 6 months)	105	Swiss HIV Cohort Study Adherence Questionnaire (SHCS-AQ) (ART; co-treatments)	Missing ≥ 1 dose of treatment in previous 4 weeks	ART = 18 (17.1%) Co-treatments = 30 (28.6%)
Katerndahl 2012 (USA)[90]	Primary care	Cross-sectional	- Diabetes - Mild or moderate depression (PHQ-9)	106; MM = 37	Medication subscale from Diabetes Self-Care Activities Measure	Report mean (possible range = 0 – 7; lower score = lower compliance	Mean = 4.89
Katon 2009 (USA)[91]	Primary care	Prospective	- Diabetes type 1 or type 2 - Probably major depression (PHQ-9)	4117; MM = 495	CMG (hyperlipidaemia medications; hypertension medications; hyperglycaemia medications)	CMG ≥ 20%	Only reported in patients with poor glycaemic, blood pressure and lipid control: Poor diabetes control = 47/127 (37.0%) Poor hypertension control = 67/232 (28.9%) Poor lipid control = 17/44 (38.6%)
Kebede 2012 (Ethiopia)[92]	Hospital	Cross-sectional	- HIV/AIDS - TB	24	Self-report tool	Missing a dose for ≥ 1 day per month (ART; TB)	ART: 16 (66.7%) TB: 11 (45.8%)
Kenning 2015 (UK)[93]	General practice	Prospective	≥ 2 from chronic obstructive pulmonary disease, coronary heart disease, diabetes, osteoarthritis and	410	Modified Moriskey Scale (6-items)	Not reported	Not reported

			depression (self-reported and/or HADS).				
Kim 2010 (South Korea)[94]	Pharmacy claims database	Retrospective cohort	- Hypertension - Psychiatric disorder	2454844; MM = 158982	Cumulative Medication Adherence (CMA) (antihypertensives)	CMA < 80%	67297 (42.3%)
Kong 2012 (USA)[95]	Medicaid database	Retrospective cohort	- HIV - Depression	7034; MM = 3286; analysed n = 3202	MPR (antiretrovirals)	Adherence < 90%	2386 (74.5%)
Kostev 2018 (Germany)[96]	General practice	Retrospective cohort	- Type 2 diabetes - Depression	12898; MM = 6449	Persistence (oral antihyperglycaemic medications)	At least 3 months without drug	2096 (32.5%)
Kreyenbuhl 2010 (USA)[97]	Veterans' Affairs administrative database	Retrospective cohort	- Type 2 diabetes - Schizophrenia or schizoaffective disorder	22014; MM = 11454	MPR (oral hyperglycaemic medications; antipsychotic medications)	MPR < 0.8	Oral antihyperglycaemics = 4925 (43%) Antipsychotic medications = 4009 (35%)
Kreyenbuhl 2011 (USA)[98]	Mental health clinics (outpatient)	Cross-sectional	- Type 2 diabetes - Serious mental illness	74; MM = 44	Brief Medication Questionnaire (oral hypoglycaemic medications)	Not taking any hypoglycaemic medications as prescribed over the past week	12 (27%)
Krivoy 2015 (Isreal)[99]	Healthcare database	Retrospective analysis	- Ischaemic heart disease - Depression	63437	MPR with addition of physician prescription data from electronic health record (antidepressants)	MPR < 20% = non-adherent MPR 20% to <50% = poorly adherent MPR 50% - <80% = moderate adherence	Non-adherence = 18212 (28.7%) Poor adherence = 10094 (15.9%) Moderate adherence = 11041 (17.4%) Total < 80% = 39347 (62.03%)
Krivoy 2017 (Israel)[100]	Healthcare database	Retrospective analysis	- Depression - Cerebrovascular accident	32361	MPR plus physician prescription data (antidepressants)	MPR < 20 % = non-adherent MPR 20-50% = poorly adherent MPR 50-80% = moderately adherent	< 20% = 8619 (26.6%) 20-50% = 5108 (15.8%) 50-80% = 5656 (17.5%) Total < 80% = 19383 (59.9%)
Krousel-Wood 2010 (USA)[101]	Managed care organisation	Prospective cohort	- Hypertension - Depression (CEDS)	2180; MM = 284	MMAS-8 (antihypertensives); MPR (antihypertensives)	MMAS-8 score < 6 ("low adherence"); MPR < 0.8	MMAS: 67 (23.6%); MPR = 104 (36.5%)

Krumme 2018 (USA)[102]	Medicare database	Retrospective cohort	≥ 2 chronic conditions, at least one of which is hypertension, hyperlipidaemia or diabetes	22805; MM = 6519	PDC (cardiovascular medication)	PDC < .80	PDC ≤ 70 = 1303 (20.0%) PDC ≤ 85 = 2865 (44.0%)
Kumar 2009 (USA)[103]	Data from longitudinal study	Cross-sectional	- HIV - Depression	1192; MM = 210	4-item self-report for past week (antiretroviral medication)	Report categories of 100%, 85%, 57-85%, <57.	Adherence < 100% = 88 (41.91%) Adherence < 85% = 67 (31.91%)
Langness 2014 (USA)[104]	Pharmacy records	Retrospective review	- HIV - Treatment with mental health medication and/or blood pressure medication	Blood pressure group = 269; Mental health group = 295	PDC (blood pressure medication; mental health medication)	PDC < 80%	Blood pressure: 92 (34.2%) Mental health: 108 (36.6%)
Lee 2011 (USA)[105]	Medicaid database	Retrospective cohort	- Serious mental illness (schizophrenia or major affective disorder) - HIV	23413; MM = 788	Continuity of medication use measures (ART and psychotropic medication)	Monthly use of any antiretroviral or psychotropic medications (divide total months patients had prescription filled by follow-up months after indexed prescription)	Percent of continuous time on medication: Psychotropic medication: 72.5% ART: 71.6%
Li 2016 (Hong Kong)[106]	Primary care	Cross-sectional	- Hypertension - Co-occurring conditions	2445; MM = 1159	MMAS-8 (Chinese version) (antihypertensives)	MMAS-8 ≤ 6	549 (47.4%) + 1 comorbidity = 339 (47.7%) + ≥ 2 comorbidities = 210 (46.9%)
Lin 2012 (USA)[107]	Primary care	RCT	- Depression (PHQ-9) - Diabetes or heart disease	214	CMG (diabetes/heart disease)	Report mean (SD)	Mean (SD): Oral hypoglycaemic = 0.83 (0.19) Antihypertensive = 0.85 (0.18) Lipid lowering = 0.83 (0.20) Antidepressant = 0.80 (0.21)
LoRe 2013 (USA)[108]	Veterans Affairs (VA) database	Retrospective cohort	- HIV - Hepatitis C virus	333	Sum of days' supply of medication dispensed over 12-week interval divided by number of days between first final fills of that interval (antiviral medications)	Pharmacy data < 70% (also report additional intervals above 70%)	Interferon: Interval 1 = 28/267 (10.5%) Interval 2 = 38/189 (20.1%) Interval 3 = 22-131 (16.8%) Interval 4 = 28-94 (29.8%) Ribavirin: Interval 1 = 46/267 (17.2%)

							Interval 2 = 55/185 (29.7%) Interval 3 = 46/127 (36.2% Interval 4 = 49/91 (53.8%)
Long 2014 (USA)[109]	Veterans Affairs medical facilities (primary care)	Cross-sectional	- Type 2 diabetes - Serious mental illness	363	MPR (diabetes medications including insulin) MMAS-4 (diabetes specific)	MPR: report mean (SD) MMAS-4: report as perfect vs non-perfect adherence	MPR mean (SD) = 0.8 (0.3) MMAS-4 = 178 (49%)
Lunghi 2017 (Canada)[110]	Healthcare database	Retrospective cohort	- Type 2 diabetes - Depression	3106	PDC (antidiabetic medication)	PDC < 90%	1614 (52.0%)
Magidson 2012 (USA)[111]	Substance use treatment centre	Cross-sectional (RCT baseline data)	- HIV/AIDS - Substance use	73	ACTG Adherence to Antiretroviral Medication Questionnaire (ART and other medications)	Report mean adherence over previous 4 days	ART: Yesterday = 100%; 2 days ago = 94.6%; 3 days ago = 97.8% 4 days ago = 97.8% Other medications: Yesterday = 97.3% 2 days ago = 94.9% 3 days ago = 96.5% 4 days ago = 97.4%
Manuel 2011 (USA)[112]	Outpatient community mental health centres	Data from RCT	- Psychotic disorder - Active substance use or dependence	198	Self-report missed medication days in past 6 months (psychiatric medications)	≥ 20% missed medication days	30 (15.2%)
Mathew 2013 (India)[113]	Hospital	Non-randomised non-controlled intervention	- Type 2 diabetes - Comorbidity(s) (receiving ≥ 5 drugs)	240	14-item self-report questionnaire (diabetes medication)	Patients were categorised into adherent and non-adherent (criteria for sorting not reported)	116 (48.3%)
May 2010 (USA)[114]	Hospital	Longitudinal observational	- Coronary artery disease - Depression	585; MM = 73	MPR across multiple refills (continuous measure of adherence (CMA) for lipid-lowering medications)	Report mean CMA	Mean adherence: 6 months = 90.9% 1 year = 89.3% 2 years = 85.7%
McCusker 2016 (Canada)[115]	Primary care	Data from RCT	- ≥ 1 chronic physical condition(s) or chronic pain ≥ 6 months - ≥ mild depressive symptoms (PHQ-9)	215; analysed n = 202	MMAS-4 (any medication)	Report mean (SD) MMAS-4 higher score = higher non-adherence (range = 1-4).	Mean (SD) = 0.7 (0.9)

Mellins 2009 (USA)[116]	Community	Cross-sectional (baseline data from cohort study	- HIV - Psychiatric disorder - Substance use disorder	1138 (n prescribed ARV = 542; n prescribed antipsychotics = 501)	Antiretrovirals: Modified version of AACTG Adherence to Antiretroviral Medication Questionnaire Psychiatric medications: 1-item self-report	Self-report < 100% adherence in past 3 days	ARV = 244 (45.0%) Psychiatric medication = 190 (38%)
Miketic 2012 (USA)[117]	Hospital	Prospective descriptive pilot study	- Coronary artery disease - Diabetes mellitus	51; MM = 22	MMAS-4	Report mean (SD)	Mean (SD) = 0.86 (0.94)
Mira 2014 (Spain)[118]	Health centres	RCT	- Multimorbid taking multiple medications	99	MMAS-4	Report mean (SD)	Mean (SD) = 6.9 (1.1)
Monroe 2018 (USA)[119]	HIV clinic	RCT	- HIV - Diabetes and/or hypertension	46	MPR	Report median (IQR)	ART: 92% (71-100%) Other medications: 79% (63-96%) (if patient had hypertension + diabetes, only hypertension adherence recorded)
Moore 2015 (USA)[120]	Ongoing studies at research centre	RCT	- HIV - Bipolar type I or II	50	Visual analogue scale (HIV medication and psychiatric medication)	VAS < 100% = some adherence difficulty in previous 30 days	HIV medication = 12 (47.8%) Psychiatric medication = 14 (54.3%)
Moore 2012 (USA)[121]	Ongoing studies at research centre	Cross-sectional	- HIV - Bipolar type I or II	44 (n with psychiatric medication assessed = 39)	MEMS (psychiatric medication and HIV medication)	MEMS <90%	HIV medication = 23/44 (52.3%) Psychiatric medication = 25/39 (64.1%)
Moore 2012 (USA)[122]	Ongoing studies at research centre	Cross-sectional	- HIV - Current or lifetime history of methamphetamine use/dependence	127; MM = 75	AIDS Clinical Trials Group (ACTG) four day adherence self-report questionnaire (HIV medications)	Report of any skipped dose in previous 4 days	Total = 16 (21.3%) 12 (18%) with lifetime history METH+ use 4 (50%) with current METH+ use
Muntner 2010 (USA)[123]	Community	Cross-sectional	- Chronic kidney disease - Hypertension	13065; MM = 3936	MMAS-4 (antihypertensive medications)	MMAS-4 = 0: 'better adherence' MMAS-4 ≥ 2: worse adherence	MMAS-4 = 0: 2715 (69.0%) MMAS-4 = 1: 916 (23.3%) MMAS-4 ≥ 2: 305 (7.8%)
Muth 2016 (Germany)[124]	General practice	Pilot cluster-RCT	≥ 3 chronic conditions	100	MMAS-4 Medication Adherence Rating Scale (MARS)	MMAS-4: scores 0–4, higher scores = lower adherence	MMAS-4 mean (SD) =0.3 (0.6) MARS mean (SD) = 24.0 (1.4)

					Observed medication adherence: (1) drug score (2) dose score (3) the regimen score	MARS: scores 5–25, lower scores = lower adherence Observed medication adherence: Scores outside 0.8–1.2 are divergent	Observed adherence: Drug score = 43/97 (44.3%) divergent Dose score = 93/94 (98.9%) divergent Regimen score = 93/93 (100%) divergent
Muth 2018 (Germany)[125]	General practice	Pragmatic cluster-RCT	≥ 3 chronic conditions	Total = 505; intervention = 253; control = 252	MMAS-4 Observed medication adherence: (1) drug score (2) dose score (3) the regimen score	Report mean (SD) MMAS-4: scores 0–4, higher scores = lower adherence Observed medication adherence: scores outside 0.8–1.2 are divergent	MMAS-4 =3.7 (0.6) Observed adherence: Drug score = 188/501 (37.5%) divergent Dose score = 259/499 (50.9%) divergent Regimen score = 255/500 (51.0%) divergent
Naidoo 2013 (South Africa)[126]	Primary care	Cross-sectional	- TB - HIV	3107; MM = 1729; analysed n = 757	Anti-TB medication: 1-item self-report of percent medication taken in prior 3–4 weeks HIV medication: 1-item self-report of percent medication taken in prior 4 weeks	Self-report adherence < 90%	TB medication: 315 (41.6%) HIV medication: 321 (42.4%)
Nakimuli-Mpungu 2019 (Uganda)[127]	Primary care	Baseline data from cluster-RCT	- Depression (MINI) - HIV	1140	Missed-dose method: 1-item self-report of days missed taking all doses in prior week (ART)	Self-report adherence< 100%	Not reported
Natarajan 2013 (Canada)[128]	General practice	Cross-sectional	- Diabetes mellitus - Hypertension	527; analysed n = 484	MMAS-4 (antihypertensives)	Answer yes to 1 or more questions = medium or low adherence	Low adherence = 8 (1.7%) Medium adherence = 101 (20.9%) Combined = 109 (22.5%)
Nelson 2011 (USA)[129]	Veterans Affairs medical centre	Retrospective record review	- Hyperlipidaemia - Diabetes - Psychotic disorder (not all sample has psychotic disorder)	total = 124; +psychotic disorder = 62; -psychotic	Cumulative mean gap ratio (CMGR): (both medication classes; for combination therapy, the medication with the greatest non-adherence was used)	Report mean (SD)	Antihyperlipidemic medication: +psychotic disorder mean (SD) = 12.0% (12.2) reflecting approx. 44 days without therapy -psychotic disorder mean (SD) = 16.9% (13.0%) reflecting approx. 62 days without therapy

				disorder = 62			Antipsychotic medication: mean (SD) = 18.3% (16.4%) reflecting approx. 68 days without therapy
Nelson 2011 (USA)[130]	Veterans Affairs medical centre	Retrospective record review	- Diabetes - Schizophrenia or related psychotic disorder	124; MM = 62	Cumulative mean gap ratio (CMGR) (antihyperglycaemic medication; antipsychotic medication)	Report mean (SD)	Antihyperglycaemics = 15.8% (16.8%) Antipsychotics = 18.3% (16.4%)
Neuget 2016 (USA)[131]	Administrative claims database	Retrospective cohort	- Non-metastatic breast cancer - Chronic condition from hypertension, hyperlipidaemia, GERD, thyroid, diabetes, osteoporosis	21255; MM = 13427	MPR (tamoxifen and/or aromatase inhibitor) Non-adherence after breast cancer diagnosis reported per disease group	MPR < 80%	Hypertension group: 1233/8683 (14.2%) Hyperlipidaemia group: 671/5829 (11.51%) GERD group: 567/4264 (13.97%) Thyroid group: 484/3542 (13.67%) Diabetes group: 267/1742 (15.33%) Osteoporosis group: 147/1384 (10.62%)
Newville 2014 (USA)[132]	Methadone maintenance clinic	Cross-sectional	- HIV - Treated with methadone maintenance therapy	100	Self-report using questions from AIDS Clinical Trials Group (HAART)	Self-report < 100%	30 (30%)
Nichol 2009 (USA)[133]	Administrative claims database	Retrospective record review	- Hypertension - Treated with lipid-lowering medication	5943	PDC (antihypertensives; lipid-lowering medication)		Year 1 Non-adherent to at least one = 1817 (30.6%) Year 2 Non-adherent to at least one = 2955 (49.7%)
Otenyo 2018 (Kenya)[134]	Hospital renal unit	Cross-sectional	- Chronic kidney disease - Hypertension	144	MMAS-8 (antihypertensives)	MMAS-8 < 6	61 (42.4%)
Owen-Smith 2016 (USA)[135]	Healthcare database	Retrospective analysis	- Schizophrenia or schizoaffective disorder - Treatment by any statin or ACEI/ARB	1420; MM = 710	MPR (cardiovascular medications; if taking both statin and ACEI/ARB, adherence calculated separately for each)	MPR < 0.80	Statin: 350/583 (60.0%) ACEI/ARB: 232/383 (60.6%)
Padman 2017 (India)[136]	Hospital general medicine department; Dialysis centre	Prospective observational	- Chronic renal failure - Diabetes mellitus	107	MMAS-8 (diabetes medication)	Rated MMAS-8 scores as low, moderate and high adherence (criteria for categorising not reported)	Low = 61 (57.0%) Moderate = 36 (33.6%)

Pantalone 2012 (USA)[137]	HIV/AIDS clinics	Cross-sectional	- HIV - Hepatitis C virus	178; MM = 54; analysed n = 45	1-item self-report number of doses missed in prior 30 days (HIV medications)	Self-report adherence < 100%	17 (37.8%)
Parpouchi 2017 (Canada)[138]	Administrative database	Longitudinal (using baseline data, longitudinal data and retrospective data)	- Mental illness - Treated with methadone maintenance	497; MM with adherence to MMT reported = 78	MPR (methadone)	MRP < 0.8	62 (79.5%)
Patel 2016 (USA)[139]	Medicaid database	Retrospective cohort	- Type 2 diabetes - Developmental disability (cerebral palsy, autism, Down's syndrome, cognitive disabilities)	1529	Adherence: MPRm (anti-diabetes medications) Persistence: period from start of medication therapy to discontinuation (anti-diabetes medications)	Adherence: MPRm < 0.8. Report mean SD. Persistence: gap > 120 days	Pooled mean (SD) MPRm = 0.90 (0.19) Non-persistence: 464 (30.4%)
Pence 2015 (USA)[140]	Infectious disease clinics	Pseudo-cluster RCT	- HIV - Major depression (MINI confirmed diagnosis & PHQ-9)	304	Visual analogue scale (ART adherence in prior 30 days)	Report mean (SD)	Mean (SD) = 86.5% (22.7%)
Pentakota 2013 (USA)[141]	Administrative research database	Retrospective cohort	- Diabetes - Comorbidity(s)	79249; MM at some stage (baseline, end, both) = 71334	Adherence: PDC (diabetes medications) Persistence: time from initiation of treatment to first discontinuation in treatment during first 2 years of treatment (diabetes medications)	PDC < 0.8 Persistence: discontinuity > 60 days	Reported according to concordant and discordant comorbidities: Concordant_Concordant = 34442 (44.5%) Discordant_Discordant = 39469 (50.95%) Additional combinations reported in original paper.
Poon 2009 (USA)[142]	Veterans Health Administration database	Retrospective cohort	- Dementia - Hypertension	56561	MPR (antihypertensive medications)	MPR < 0.8	Reported by race and drug class: Min = 13.0%; Max = 39.6%
Puyat 2017 (Canada)[143]	Healthcare databases	Retrospective cohort	- Depression - Additional chronic health condition	105476; MM = 13329	PDC (antidepressants)	Report mean (SD) in 180-day period and in 365-day period	Mean (SD) = 0.84 (0.24) Mean (SD) = 0.80 (0.27)
Qian 2014 (USA)[144]	Administrative database	Cross-sectional	- COPD - Depression	74863; MM = 25130; analysed n = 12574	PDC (COPD medications) Discontinuation (COPD medications)	PDC: < 0.80 Discontinuation: binary variable of medication stopped	PDC: 8621 (68.6%) Discontinuation: 3009 (23.9%)

						3 months or more prior to end of observation period or death, whichever first	
Rattinger 2012 (USA)[145]	Administrative database	Cross-sectional	- Chronic heart failure - Alzheimer's disease or related disorders	9827; MM = 2379	MPR (chronic heart failure medication)	Report mean (SD)	EBM = 0.86 (0.15) Diuretic = 0.81 (0.20) Cardiac glycoside = 0.86 (0.18) Calcium channel blocker = 0.88 (0.17)
Reed 2018 (Australia)[146]	General practice	RCT	≥ 2 chronic conditions from cardiovascular, respiratory, musculoskeletal, psychological, digestive, kidney disease, diabetes (type 1 or 2), cancer	254	5-item self-report	Higher scores indicate poorer levels of medication adherence (range of possible scores not reported)	Mean (SD) = 24.14 (0.14)
Rosenblum 2014 (USA)[147]	Residential and outpatient psychiatric and substance use facilities	RCT	- Mental illness - History of substance use	203	Medication Adherence Rating Scale (MARS) (psychiatric medication)	Report mean (SD) Lower score = higher medication adherence (range of possible scores not reported)	Mean (SD) = 1.55 (0.48)
Roux 2012 (France)[148]	Hospital	Prospective cohort	- HIV - Hepatitis C virus	593 at enrolment; 246 at follow-up	7-item self-report adherence questionnaire (HIV medications)	Self-report adherence < 100% during 4-days prior	Enrolment: 214 (36%) Follow-up: 72 (29.3%)
Saadat 2015 (Iran)[149]	Healthcare centres	Cross-sectional	- Hypertension - ≥ 1 comorbidity	280; MM = 160	MMAS-8 (Persian version) (antihypertensive medication)	MMAS-8 < 6	77 (48.1%)
Safren 2009 (USA)[150]	Community	RCT	- HIV - Depressive mood disorder	45	MEMS (antiretroviral medication)	Report mean (SD)	Control group = 54.41 (30.94) Intervention group = 62.26 (26.82)
Safren 2012 (USA)[151]	Community outreach, hospital HIV clinics, substance use clinics	RCT	- HIV - History of injection drug use (self-endorsed + enrolled in opioid treatment for ≥ 1 month(s)) - Criteria for diagnosis of current or subsyndromal	89	MEMS (antiretroviral medication)	Report mean (SD) percent MEMS-based adherence since last visit	Mean (SD) = 69.0% (28.3%)

			depressive mood disorder				
Safren 2014 (USA)[152]	Primary care	RCT	- Unipolar depression - Type 2 diabetes	87	MEMS (antidiabetic oral medication, insulin or both)	Report mean (SD)	Control group = 85.7 (24.6%) Intervention group = 76.6% (26.0%)
Santorelli 2016 (USA)[153]	Administrative databases	Retrospective cohort	- Breast cancer - Diabetes or hypertension or lipid disorder	9340; MM groups: diabetes = 298; lipid disorders = 508; hypertension = 1062	Adherence: PDC (hypertension; diabetes; lipid disorder medications) Persistence: largest number of consecutive days without coverage by relevant medication class during first 18 months (hypertension; diabetes; lipid disorder medications)	Adherence: PDC < 80% Persistence: Diabetes = 26 days without medications Hypertension = 26 days without medication Lipid-lowering = 62 days without medications	PDC: Diabetes = 78 (26.2%) Hypertension = 151 (14.2%) Lipid disorder = 147 (28.9%) Persistence: Diabetes = 94 (31.5%) Hypertension = 200 (18.8%) Lipid disorder = 123 (24.2%)
Schmitt 2010 (USA)[154]	Veterans Affairs medical centre	Retrospective observational	- Chronic kidney disease - Treatment with antihypertensive	7227	MPR (antihypertensive medication)	MPR < 80%	2360 (32.7%)
Schüz 2011 (a) (Germany)[155]	Community	Longitudinal	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	309	2 self-report items from Reported Adherence to Medication Scale (RAM) measuring: (1) intentional non-adherence (2) unintentional non-adherence	Report mean (SD) Higher values indicate higher degrees of non-adherence (range of possible scores not reported)	Intentional mean (SD) = 1.32 (0.68) Unintentional mean (SD) = 1.40 (0.67)
Schüz 2013 (Germany)[156]	Community	Longitudinal	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	215	1-item self-report intentional non-adherence from Reported Adherence to Medications Scale (RAM)	Report mean (SD) Higher scores indicate better adherence (range of possible scores not reported)	Mean (SD) = 4.68 (0.68)
Schüz 2011 (b) (Germany) [157]	Community	Longitudinal	≥ 2 chronic conditions from Charlson Comorbidity Index or	309	2 self-report items from Reported Adherence to Medication Scale (RAM) measuring:	Not reported	Not reported

			Functional Comorbidity Index		(1) intentional non-adherence (2) unintentional non-adherence		
Shi 2010 (USA)[158]	Veterans Health administration database	Retrospective analysis	- Schizophrenia - Metabolic syndrome	593; MM = 209	Adherence: MPR (antipsychotics) Discontinuation: medication gap (antipsychotics)	Adherence: report mean (SD) MPR Discontinuation: medication gap > 30 consecutive days	Mean (SD) MPR = 0.33 (0.47) Median time to discontinuation = 50 days
Shin 2010 (South Korea) [159]	Pharmacy claims data	Retrospective record review	- Hypertension - Cancer survivor	2455193; MM = 12636	CMA (antihypertensive medication)	CMA < 80%	5762 (45.6%)
Shoval 2017 (Israel)[160]	Healthcare database	Retrospective analysis	- Parkinson's disease - Depression	8553	MPR (antidepressant medication)	MPR < 20% = non-adherent MPR 20% - 50% = poorly adherent MPR 50% - 80% = moderately adherent	< 20% = 1566 (18.4%) 20-50% = 1184 (13.8%) 50-80% = 1584 (18.5%) < 80% = 4334 (50.7%)
Siika 2013 (Kenya)[161]	Ambulatory HIV care program	Retrospective cohort	- HIV - Tuberculosis	21242; MM = 5186; analysis n = 4838	1-item self-report number of pills taken during prior week (ART)	If patient ever answers 'most', 'half', 'few', or 'none' (defined as non-perfect adherence to ART)	986 (20.4%)
Slabbert 2015 (South Africa)[162]	Administrative database	Prospective cohort	- Major depressive disorder - HIV/AIDS	127	MPR (antidepressant medications)	Report mean (SD)	74.43 (32.03)
Soto 2015 (Chile) [163]	Hospital	Prospective	- Type 2 diabetes - Hypertension	50	Pill count (seems like both medications together)	Pill count < 80% or > 110%	35 (70.0%)
Stack 2010 (UK)[164]	Healthcare database (secondary care)	Cross-sectional	- Type 2 diabetes - Hypertension	480	MMAS-4 (separately for antidiabetic medication, antihypertensive medication and statin)	0 = total adherence 4 = non-adherence: presented as no unintentional non-adherence; no intentional non-adherence	Unintentional non-adherence: Diabetes medication = 302 (62.9%) Hypertension medication = 226 (47.1%) Statins = 225 (46.9%) Intentional non-adherence: Diabetes medication = 63 (13.1%)

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							Hypertension medication = 92 (19.2%) Statins = 87 (18.1%)
Stilley 2010 (USA)[165]	Primary care	RCT	≥ 2 from type 2 diabetes, hypertension, hyperlipidaemia	373	MEMS (not clear to what medications)	Report mean (SD): (1) prescribed number of doses (2) days with prescribed dose (3) days with correct inter-dose interval	Prescribed number of doses = 92.6 (17.6) Days with prescribed dose = 82.2 (22.4) Days with correct inter-dose interval = 64.2 (30.1)
Stuart 2015 (USA)[166]	Medicare database	Retrospective cohort	- Diabetes - Cancer	32855; MM = 4348	PDC (diabetes medication)	Report mean PDC	OHA (n = 2248) mean PDC = 73.8% RAAS-I (n = 2618) mean PDC = 70.5% Statins (n = 2272) mean PDC = 69.0%
Taira 2017 (USA)[167]	Healthcare database	Retrospective observational	- Type 2 diabetes - Treatment with antihypertensive, oral antidiabetic and cholesterol lowering medication	5163	MPR (each medication)	MPR < 0.8	Range of non-adherence reported: 16% - 48.2% Non-adherence for each medication according to ethnicity reported in original paper.
Tang 2014 (USA)[168]	Outpatient sites	Secondary analysis of prospective descriptive cohort	- Heart failure - Depression (GDS)	280; MM = 103	MEMS (heart failure medication); Basel Assessment of Adherence Scale (BAAS) (heart failure medication)	MEMS: < 80% BAAS: a positive answer on any question	Self-report = 77 (75%) MEMS = 29 (28%)
Thaler 2015 (USA)[169]	HIV treatment facilities	Not reported	- HIV - neurocognitive impairment and/or substance dependence/use and/or psychiatric illness	204; MM not reported	MEMS (HAART) Self-report (not described) (HAART)	Report mean (SD) per additional risk factor	One risk factor: 64.4% (27.0%) Two risk factors: 62.8% (28.0%)
Vacek 2013 (USA)[170]	Medicaid database	Retrospective analysis	- Developmental disability - Hypertension	280; analysed n = 201	PDC (antihypertensive medication)	PDC < 0.8	90 (45%)
Vatcharavongvan 2017 (Thailand)[171]	Primary care	Cross-sectional	≥ 2 chronic conditions from type 2 diabetes, hyperlipidaemia,	397; analysed n (polyphar	Structured questionnaire during face-to-face interview	At least one medication that was (a) taken at wrong	74 (60.7%)

			hypertension, ischemic heart disease, chronic heart failure, cerebrovascular disease, epilepsy	macy group) = 122		time, (b) missed (c) intentionally stopped by patient during 2 weeks prior	
Vega 2017 (USA)[172]	Healthcare database	Retrospective cohort	- Major depressive disorder - Type 2 diabetes	1361	Adherence: MPR (depression medication; diabetes medication) Persistence: Length of treatment (LOT) (depression medication)	MPR < 80% LOT ≥ 15 day gap in first 180 days of therapy	Depression medication: Non-adherent = 872 (64.1%) Non-persistent = 926 (68.0%) Non-adherent/non-persistent = 936 (68.8%) Diabetes medication: Non-adherent = 525 (38.6%)
Villar 2009 (Spain)[173]	Ongoing studies, administrative database	Prospective cohort	- Severe obstructive sleep apnoea - Cardiovascular disease	2158; antihypertensives = 659; statins = 175; antiplatelets = 109	MPR (cardiovascular disease medications)	MPR < 80%	Antihypertensives: 118/659 (17.91%) Statins: 59/175 (33.71%) Antiplatelets: 9/109 (8.26%)
Wagner 2011 (USA)[174]	Pooled data from longitudinal studies	Cross-sectional and longitudinal	- HIV - Depression (BDI-II; BDI (original version); CESD; depression subscale of Brief Symptom Inventory)	1374; MM = 489	MEMS (HIV medication)	MEMS < 90%	Severe depression = 72% (148/205) Mild to moderate depression = 56% (159/284) Combined = 307/489 (62.8%)
Wakefield 2012 (USA)[175]	Primary care	RCT	- Type 2 diabetes - Hypertension	302 (control = 107; low intensity = 102; high-intensity = 93)	1-item self-report adherence	Report mean (SD)	Mean (SD): Control group = 99.5 (4.4) Low-intensity group = 99.7 (1.6) High-intensity group = 99.7 (2.7)
Warner 2013 (Germany)[176]	Community	Longitudinal	≥ 2 chronic physical conditions from Charlson Comorbidity Index or Functional Comorbidity Index	309	1-item self-report unintentional non-adherence from Reported Adherence to Medication Scale (RAM)	1 < RAM < 4	T1 = 83 (27%) T2 = 99 (32%)
Wei 2018 (USA)[177]	Medicare database	Retrospective cohort	- Major depression - Chronic obstructive pulmonary disease	25458; analysed n = 11572	PDC (COPD medications)	PDC < 0.8	7059 (61.0%)

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Weiss 2016 (USA)[178]	Ambulatory practices	Longitudinal	- HIV - Hypertension or chronic kidney disease	152; +hypertension = 115 (analysed n = 102); +CKD = 37 (analysed n = 20)	MEMS (1) taking adherence (2) dosing adherence (3) timing adherence (HIV medications and CKD/hypertension medications separately)	MEMS < 95% (HIV medications) MEMS < 80% (other medications)	+hypertension: HIV medication = 40 (39%) Hypertension medication = 20 (20%) +CKD: HIV medication = 7 (35%) CKD medication = 4 (20%)
Wenze 2015 (USA)[179]	Hospital	Pilot RCT	- Bipolar I or II - Substance use disorder	30	Adapted version of Medication Compliance Questionnaire (MCQ) (prescribed mood-stabilizing agent)	MCQ ≥ 2	12 (40.0%)
Wong 2014 (China)[180]	Administrative database	Cross-sectional	≥ 2 chronic conditions	3866; MM = 523	MMAS-8 (antihypertensives, oral hypoglycaemic agents, lipid lowering agents separately)	MMAS-8 ≤ 6 (MMAS-8 sections for each drug should score >6 to be defined as having optimal adherence)	321 (45.1%)
Wroe 2014 (Rwanda)[181]	Rural health centres	Cross-sectional	- HIV - Depression (Hopkin's Symptom Checklist)	292; MM = 32	Self-report 4-day recall (HIV medications) Self-report 2-day and 7-day treatment interruptions (HIV medications) CASE index (HIV medications)	Self-report 4-day recall: ≥ 1 dose(s) in four days prior Self-report 2-day and 7-day treatment interruptions: non-adherence on sequential days CASE index ≤ 9	4-day recall = 5 (17.0%) 2-day interruptions = 7 (22.2%) CASE index = 5 (15.8%)
Wu 2011 (USA)[182]	Medicaid database	Retrospective observational	- Hyperlipidaemia - Type 2 diabetes	1705	MPR (statins; antidiabetic medication)	MPR < 0.8	Statins = 1079 (63.3%) Mean (SD) MPR (statins) = 0.61 (0.32) Mean (SD) MPR (antidiabetics) = 0.71 (0.30)
Wu 2010 (USA)[183]	Medicaid database	Retrospective cohort	- Major depressive disorder - Anxiety disorder	3083; MM = 776	Adherence: MPR modified (MPRm) (antidepressants) Persistence: Refill gap (antidepressants)	MPRm: report mean (SD) Refill gap: median (IQR)	Mean (SD) MPRm = 0.71 (0.37) Median (IQR) persistence (days) = 237 (216-256)

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Wu 2013 (USA)[184]	Inpatient and outpatient cardiology clinics	Secondary analysis of two prospective studies	- Heart failure - Depression (PHQ-9)	216; MM = 55	MEMS (heart failure medication)	MEMS < 88%	30 (54.55%)
Wulaningsih 2018 (Sweden)[185]	Administrative database	Retrospective analysis	- ER-positive breast cancer - Charlson Comorbidity Index score ≥ 1	4645; MM baseline = 715; MM follow-up = 1348	MPR (endocrine therapy)	MPR < 80%	Baseline: 168/715 (23.5%) 5-years: 330/1348 (24.5%)
Xia 2015 (China)[186]	Hospital	Cross-sectional	- ≥ 1 chronic condition from chronic obstructive pulmonary disease, diabetes, hyperlipidaemia, hypertension - Depression (PHQ-9) and/or anxiety (GAD-7)	325; +anxiety = 276; +depression = 266	Chinese Medication Adherence Scale (CMAS) (chronic condition)	CMAS <6 to	+depression: 222 (83.5%) +anxiety: 226 (81.9%)
Xue 2018 (China)[187]	Primary care	Baseline data from RCT	- Hypertension - Depression (PHQ-9)	2362	1-item self-report adherence (antihypertensive medications)	Answer yes to "do you sometimes forget to take your antihypertensive pills"	954 (40.4%)
Yang 2010 (USA)[188]	Medicare database	Retrospective cohort	- Diabetes - Hypertension and/or renal disease	599141	PDC (antihypertensive medication)	PDC < 0.8	276945 (46.22%)

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References to studies included in analyses

Multimorbidity sample	
≥ 2 specific conditions	[1–134]
1 specified condition and ≥ 1 other condition(s)	[135–160]
Inclusive definition of multimorbidity, e.g. ≥ 2 conditions	[161–178]

Adherence measurement	
Pharmacy data	[1–4,7,10,11,13,14,20,21,23,25,29,30,32,33,36,40,45–47,51,54,59,61–64,66–68,70–75,78,81,89,90,92,94,97,98,100–102,108–111,113,117,118,121–123,126,131,132,135–151,161,133,160,164,50]
Self-report	[5,6,8,9,12,15,16,18,19,24,27,28,34,35,38,39,41,47–49,52,53,55,56,58,60,65,68,69,74,76,77,79,80,82,83,85–88,91,93,93,95,96,99,103,104,112,115,119,120,125,128,129,134,137,152–159,162,163,165–169,171,170,172–174,178,176,177,175,43]
Electronic monitoring	[5,9,17,18,22,24,37,43,47,84,105–107,116,119,120,124,127,130,167]
Pill counts	[26,57,114]
Healthcare provider report	[42,44]

Reporting of non-adherence	
Prevalence of non-adherence	[1–4,6,7,9,11–23,25,27–30,32,33,35,36,39–46,48,50–57,59–70,73–75,77,79,82–86,88,91–98,100,101,104,108,110–112,114,115,118,119,121–124,126–131,133,134,136–140,142–148,150–152,154–156,158–160,162,165,170,171,176–178]
Average non-adherence	[5,8,10,24,26,31,34,37,38,47,49,58,71,72,76,78,80,81,89,90,99,102,103,105–107,109,113,116,117,120,125,132,135,141,149,153,157,161,163,164,166,167,169,172,173,175]

Study setting	
Primary care / community / general practice	[6,17,18,26,27,30,32,36,40,41,57–59,63,72,74,77,79,85–88,99,105,107,116,125,129,134,153,155,157,160,162,163,165,168,170–177]
Secondary care / hospital /tertiary care	[8,15,16,19,28,35,37,38,42,44,48,52,53,55,60,65,76,78,80,93,95,103,104,114,115,128,130,137,144,152,154,156,159,166]
Administrative database	[1–4,7,10,11,13,14,20,21,23,25,29,33,45,46,50,51,54,61,62,64,66,67,70,71,73,75,92,94,97,98,100–102,109–111,113,117,118,121–123,126,131–133,135,136,138–143,145–151,161,164,178]
Other / not clearly reported / multiple sources	[152,5,9,12,22,24,31,34,39,43,47,49,167,56,68,69,169,83,84,89–91,96,158,106,108,112,119,120,124,127,81,82]

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Supplementary file 7 - Predictors of medication (non-)adherence 1

Study (country)	Population	Predictors assessed	Dependent variable	Significant predictors of and/or associations with medication adherence
Beadles 2015 (USA) [1]	≥ 2 of the following: major depressive disorder, schizophrenia, diabetes mellitus, hypertension, hyperlipidaemia, seizure disorder, asthma, chronic obstructive pulmonary disease	• Medical home enrolment	Proportion Days Covered (PDC) ≥ 0.8	• Medical home enrolment was positively associated with PDC ($p < .001$) across all cohorts (see original paper for breakdown by cohort)
Cicolini 2016 (Italy) [2]	≥ 2 of the following: cardiovascular, neurological, respiratory, musculoskeletal, oncologic diseases or diabetes mellitus	• Age • Sex • Education • Marital status • Number of prescribed drugs • Presence of diabetes • Presence of other disorders (cancer, musculoskeletal, respiratory, neurological, other) • Beliefs about medications (necessity, concerns) • Beliefs about medications (ambivalent, accepting)	8-item Moriskey Medication Adherence Scale (MMAS-8); "medium adherence" vs "low adherence"	• Necessity beliefs: OR (95% CI) = 1.61 (1.21-2.14), $p < .001$ • Concern beliefs: OR (95% CI) = 2.02 (1.64-2.49), $p < .001$ • Ambivalent vs accepting: OR (95% CI) = 0.24 (0.16-0.37), $p < .001$
Daniels 2016 (USA) [3]	≥ 2 chronic conditions	• Health literacy • Medication assistance • Medication adherence self-efficacy • Health-related problem solving • Functional health literacy • Chronic disease self-efficacy • Quality of life • Number of chronic conditions • Number of medications	MMAS-8; higher score = lower adherence	• Health-related problem solving (total score): $r = -.489$, $p = 0.024$) • Health-related problem solving (impulsive subscale): $r = -.521$, $p = 0.0015$ • Quality of life (physical subscale): $r = -.44$, $p < .05$ • Quality of life (environment subscale): $r = -.566$, $p = 0.009$

Domino 2014 (USA) [4]	≥ 2 of the following: asthma, COPD, diabetes, hypertension, hyperlipidaemia, seizure disorder, depression, schizophrenia	Adherence in medical index condition: • Number of comorbid medical conditions • Mental health conditions Adherence in mental health condition: • Number of comorbidities	PDC (results show percentage point change (standard error) in adherence associated with clinical diagnosis stated)	Persons with diabetes: • Depression: -0.0427 (0.0030), $p < .01$ • Schizophrenia: 0.0458 (0.0061), $p < .01$ • 2 medical comorbidities: 0.0201 (0.0037), $p < .01$ • ≥ 3 medical comorbidities: 0.0678 (0.0037), $p < .01$ Persons with hyperlipidaemia: • Depression: -0.0200 (0.0025), $p < .01$ • Schizophrenia: 0.0359 (0.0050), $p < .01$ • 2 medical comorbidities: 0.0447 (0.0037), $p < .01$ • ≥ 3 medical comorbidities: 0.0979 (0.0037), $p < .01$ Persons with hypertension: • Depression: -0.0276 (0.0021), $p < .01$ • Schizophrenia: -0.0105 (0.0041), $p < .05$ • 2 medical comorbidities: 0.0365 (0.0023), $p < .01$ • ≥ 3 medical comorbidities: 0.1039 (0.0024), $p < .01$ Persons with depression: • 1 medical comorbidity: 0.0193 (0.0029), $p < .01$ • 2 medical comorbidities: 0.0462 (0.0035), $p < 0.01$ • ≥ 3 medical comorbidities: 0.0894 (0.0037), $p < .01$ Persons with schizophrenia: • 1 medical comorbidity: 0.0362 (0.0067), $p < .01$ • 2 medical comorbidities: 0.0752 (0.0078), $p < 0.01$ • ≥ 3 medical comorbidities: 0.1082 (0.0079), $p < .01$
Eton 2017 (USA) [5]	≥ 2 chronic conditions requiring self-management	• Health provider relational quality (HPRQ)	Single-item self-report; “always” vs “usually/sometimes” taking medications	• Health provider relational quality: $t(294) = 3.79$, $p < .001$ (converted to $r = 0.22$, $p < .001$ for consistency with other studies)

Ihle 2017 (Switzerland) [6]	≥ 2 chronic conditions	• Prospective memory • Retrospective memory	Omission errors of adherence = 1-item self-report Commission errors of adherence = 1-item self-report	• Omission errors of adherence: prospective memory performance: r = -.29, p < .05 • Commission errors of adherence: retrospective memory performance: r = -.24, p < .05
Kenning 2015 (UK) [7]	Any combination of the following: chronic obstructive pulmonary disease, coronary heart disease, diabetes, osteoarthritis and depression.	• Gender • Age • Deprivation (IMD) • Number of conditions • Health status • Anxiety/depression (HADS) • Illness perceptions – consequences (IPQ) • Illness perceptions – treatment control (IPQ) • Self-help behaviours • Self-monitoring and insight (heiQ) • Baseline Moriskey medication motivation • Baseline Moriskey medication knowledge • Hassles with healthcare • Patient experience of multimorbidity (MULTIPleS)	Modified Moriskey Scale – Medication Motivation at follow-up	• Health status (B = -0.119, SE = 0.051; β= -0.132, p < .05) • Self-monitoring and insight (B = 0.093, SE = 0.045; β = 0.100, p < .05) • Moriskey medication motivation at baseline (B = 0.522, SE = 0.044; β = 0.552, p < .01). • Hassles with health care (B = -0.107, SE = 0.050; β = -0.108, p < .05)
Schüz 2011 (Germany) [8]	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	• Unintentional non-adherence Time 1 (T1) • Intentional non-adherence T1 • Sex • Age • Education • Number of illnesses • Number of medicines • Change in specific necessity (T1 to T2) • Change in specific concerns (T1 to T2) • Change in general harm (T1 to T2) • Change in general overuse (T1 to T2)	Two items from Reported Adherence to Medication Scale; Higher score = higher non-adherence.	Intentional non-adherence as dependent variable: • Intentional non-adherence time 1: B = .58, SE = .09, p < .01 • Change in specific necessity beliefs: B = −.19, SE = .06, p < .01 Unintentional non-adherence as dependent variable: • Unintentional non-adherence time 1: B = .70, SE = .09, p < .01 • Change in general overuse beliefs: B step 3 = .26, SE = .10, p < .01

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Supplementary file 7 - Predictors of medication (non-)adherence 4

Schüz 2011 (Germany) [9]	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	• Unintentional non-adherence Time 1 • Intentional non-adherence Time 1 • Sex • Age • Education • Functional health status • Number of medicines • Number of illnesses • General harm beliefs • General overuse beliefs • Specific necessity beliefs • Specific concern beliefs	Two items from Reported Adherence to Medication Scale; Higher score = higher non-adherence	Unintentional non-adherence as dependent variable: • Unintentional non-adherence time 1: β step 3 = .29, $p < 0.01$ • General overuse beliefs: β step 3 = .17, $p < .01$ Intentional non-adherence as dependent variable: • Intentional non-adherence time 1: β = .26, $p < .01$ • Number of medicines: β = -.15, $p < .05$ • Number of illnesses: β = .16, $p < .05$ • General harm beliefs: β = .15, $p < .05$ • Specific necessity beliefs: β = -.15, $p < .05$
Schüz 2014 (Germany) [10]	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	• Illness perceptions (illness consequences, timeline, control)	One item from Reported Adherence to Medication Scale. Higher score = higher adherence.	Main effects and interaction model: • Timeline of illness 2: β = -.18, $p < .05$ • Control of illness 2: β = .21, $p < .05$ Peak model: • No significant predictors Combined model: • Higher control: β = .20, $p < .01$ • More negative consequences: β = .16, $p < .05$
Warner 2013 (Germany) [11]	≥ 2 chronic physical conditions from Charlson Comorbidity Index or Functional Comorbidity Index	• Sex • Age • Education • Number of medicines • Physical fitness • Attitudes towards medication • General received support • Social conflict • Medication specific social support Time 1 • Medication specific social support Time 2 • Medication adherence Time 1 • Medication adherence Time 2	One item from Reported Adherence to Medication Scale assessing unintentional non-adherence. Higher score = higher non-adherence	Step 1: OR (95% CI) • Adherence Time 1 = 5.99 (3.18-11.29), $p < .001$ Step 2: OR (95% CI) • Adherence Time 1 = 5.37 (2.82-10.24), $p < .001$ • Medication specific social support = 0.69 (0.49-0.98), $p < .05$. Step 3: OR (95% CI) • Adherence Time 1 = 5.62 (2.91-10.85), $p < .001$ • Interaction medication support x social conflict = 0.42 (0.17-0.99), $p < .05$

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	(i)
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	(ii)
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	1
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	1
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	2
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	2
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	2
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	2+SM
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	2+3
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	3
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	3
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	3
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	2
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	3

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	n/r
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	n/r
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	4
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	4+SM
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	4
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	5
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	5
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	n/r
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	n/r
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	14-16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	17
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	17

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

SM = supplementary material

n/r = not reported

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